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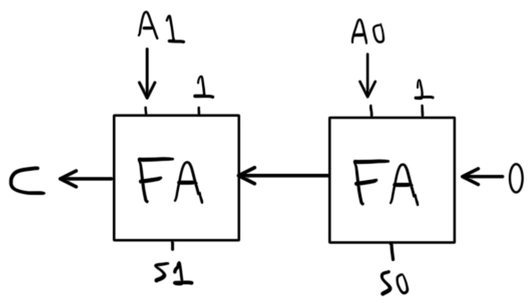
CHAPTER FOUR

Register Transfer and Microoperations



IN THIS CHAPTER

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4-2 Register Transfer
4-3 Bus and Memory Transfers
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4-7 Arithmetic Logic Shift Unit
4-8 Hardware Description Languages

a sign based way to describe the oberations that're done inside the Computer sys.

the simplest oberation

4-1 Register Transfer Language

تناغم بين الدوائر الالكترونية

A digital system is an interconnection of digital hardware modules that accomplish a specific information-processing task. Digital systems vary in size and complexity from a few integrated circuits to a complex of interconnected and interacting digital computers. Digital system design invariably uses a modular approach. The modules are constructed from such digital components as registers, decoders, arithmetic elements, and control logic. The various modules are interconnected with common data and control paths to form a digital computer system.

microoperation

Digital modules are best defined by the registers they contain and the operations that are performed on the data stored in them. The operations executed on data stored in registers are called microoperations. A microoperation is an elementary operation performed on the information stored in one or more registers. The result of the operation may replace the previous binary information of a register or may be transferred to another register. Examples of microoperations are shift, count, clear, and load. Some of the digital components introduced in Chap. 2 are registers that implement

For any function of the computer, a sequence of microoperations is used to describe it

Text

control signal تتم عن طريق

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Register + microoperation hardware + Control func = digital Computer

microoperations. For example, a counter with parallel load is capable of performing the microoperations increment and load. A bidirectional shift register is capable of performing the shift right and shift left microoperations.

The internal hardware organization of a digital computer is best defined by specifying:

حجرة تخزينية داخل المعالج

1. The set of registers it contains and their function.

لتخزين العمليات

2. The sequence of microoperations performed on the binary information stored in the registers.

المسؤول عن تنفيذ اشارات النظام

3. The control that initiates the sequence of microoperations.

Control unit

It is possible to specify the sequence of microoperations in a computer by explaining every operation in words, but this procedure usually involves a lengthy descriptive explanation. It is more convenient to adopt a suitable symbology to describe the sequence of transfers between registers and the various arithmetic and logic microoperations associated with the transfers. The use of symbols instead of a narrative explanation provides an organized and concise manner for listing the microoperation sequences in registers and the control functions that initiate them.

register transfer

The symbolic notation used to describe the microoperation transfers among registers is called a register transfer language. The term "register transfer" implies the availability of hardware logic circuits that can perform a stated microoperation and transfer the result of the operation to the same or another register.

language

The word "language" is borrowed from programmers, who apply this term to programming languages. A programming language is a procedure for writing symbols to specify a given computational process. Similarly, a natural language such as English is a system for writing symbols and combining them into words and sentences for the purpose of communication between people. A register transfer language is a system for expressing in symbolic form the microoperation sequences among the registers of a digital module. It is a convenient tool for describing the internal organization of digital computers in concise and precise manner. It can also be used to facilitate the design process of digital systems.

The register transfer language adopted here is believed to be as simple as possible, so it should not take very long to memorize. We will proceed to define symbols for various types of microoperations, and at the same time, describe associated hardware that can implement the stated microoperations. The symbolic designation introduced in this chapter will be utilized in subsequent chapters to specify the register transfers, the microoperations, and the control functions that describe the internal hardware organization of digital computers. Other symbology in use can easily be learned once this language has become familiar, for most of the differences between register transfer languages consist of variations in detail rather than in overall purpose.

next step

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4-2 Register Transfer

registers

Computer registers are designated by capital letters (sometimes followed by numerals) to denote the function of the register. For example, the register that holds an address for the memory unit is usually called a memory address register and is designated by the name *MAR*. Other designations for registers are *PC* (for program counter), *IR* (for instruction register, and *R*1 (for processor register). The individual flip-flops in an *n*-bit register are numbered in sequence from 0 through $n - 1$, starting from 0 in the rightmost position and increasing the numbers toward the left. Figure 4-1 shows the representation of registers in block diagram form. The most common way to represent a register is by a rectangular box with the name of the register inside, as in Fig. 4-l(a). The individual bits can be distinguished as in (b). The numbering of bits in a 16-bit register can be marked on top of the box as shown in (c). A 16-bit register is partitioned into two parts in (d). Bits 0 through 7 are assigned the symbol *L* (for low byte) and bits 8 through 15 are assigned the symbol *H* (for high byte). The name of the 16-bit register is *PC*. The symbol *PC*(0–7) or *PC*(*L*) refers to the low-order byte and *PC*(8–15) or *PC*(*H*) to the high-order byte.

register transfer

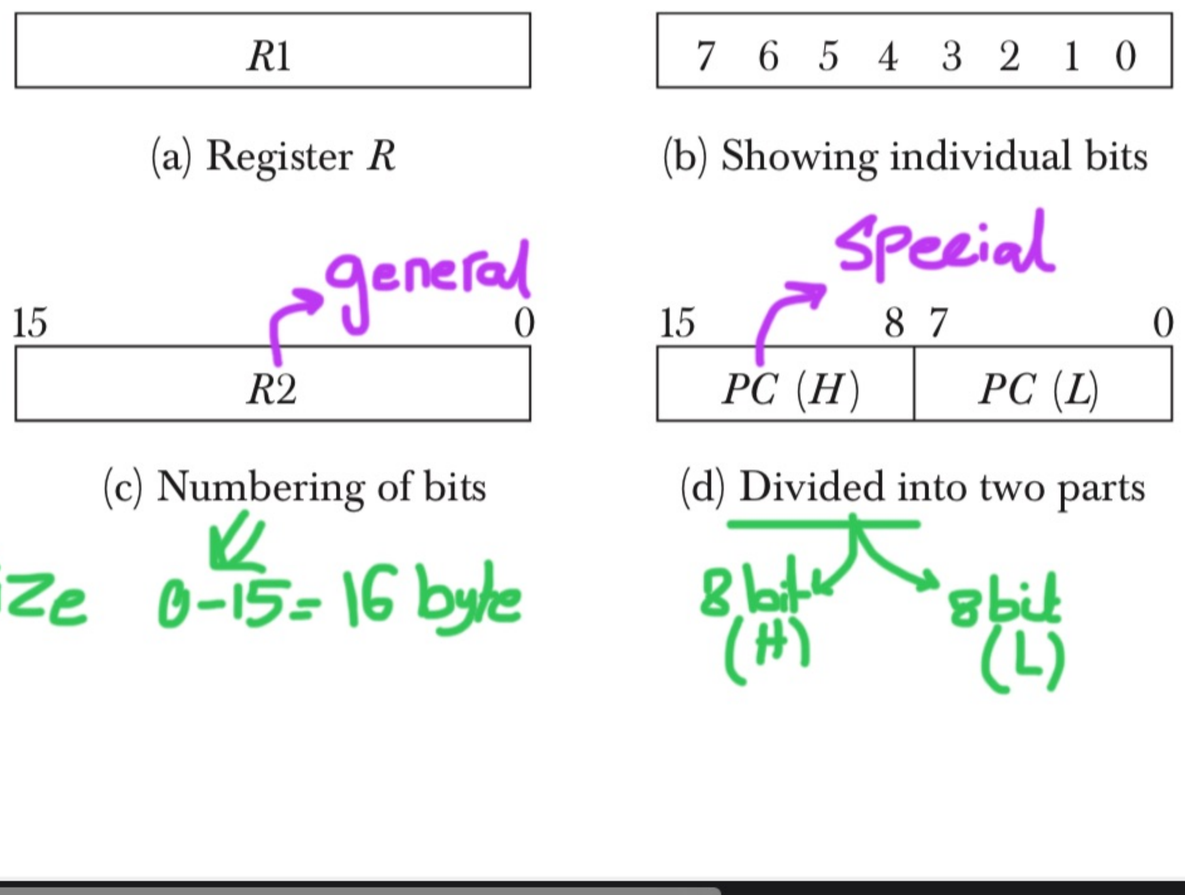
Information transfer from one register to another is designated in symbolic form by means of a replacement operator. The statement

$$R2 \leftarrow R1$$

denotes a transfer of the content of register *R*1 into register *R*2. It designates a replacement of the content of *R*2 by the content of *R*1. By definition, the content of the source register *R*1 does not change after the transfer.

A statement that specifies a register transfer implies that circuits are available from the outputs of the source register to the inputs of the destination register and that the destination register has a parallel load capability. Normally,

Figure 4-1 Block diagram of register.



(a) Register *R*
(b) Showing individual bits
(c) Numbering of bits
(d) Divided into two parts

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we want the transfer to occur only under a predetermined control condition. This can be shown by means of an *if-then* statement.

$$If\ (P = 1)\ then\ (R2 \leftarrow R1)$$

control function

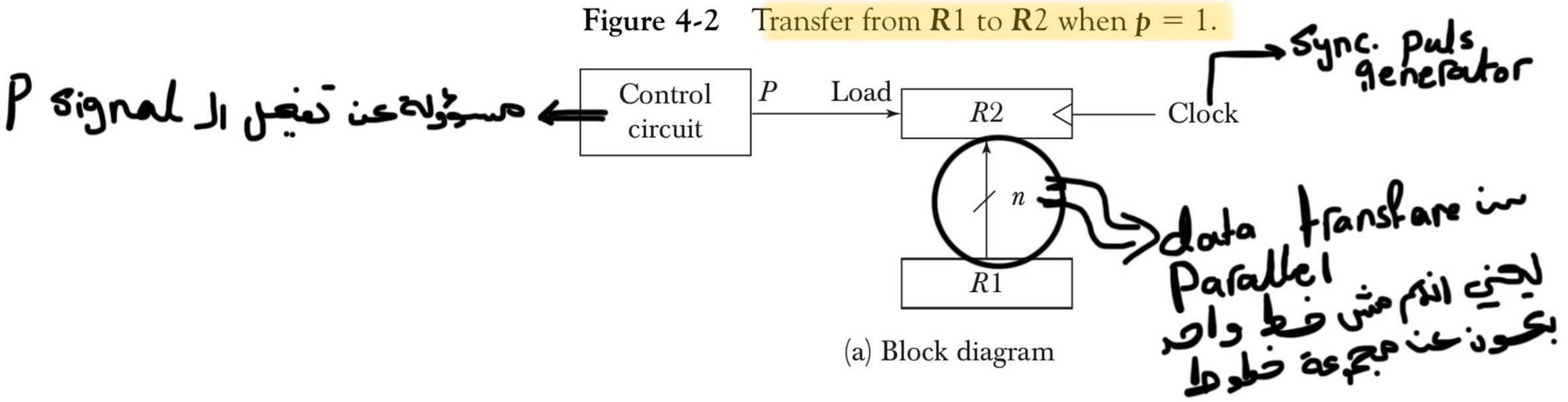
where P is a control signal generated in the control section. It is sometimes convenient to separate the control variables from the register transfer operation by specifying a *control function*. A control function is a Boolean variable that is equal to 1 or 0. The control function is included in the statement as follows:

$$P: \quad R2 \leftarrow R1$$

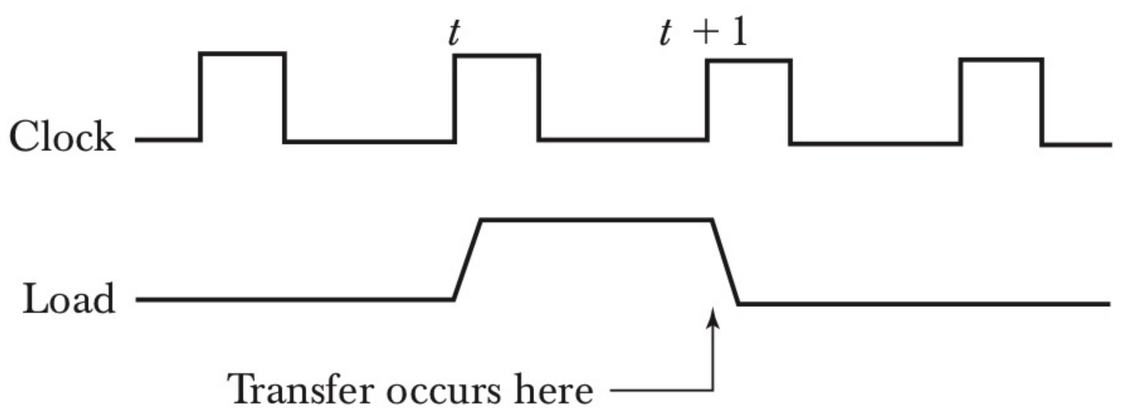
The control condition is terminated with a colon. It symbolizes the requirement that the transfer operation be executed by the hardware only if $P = 1$.

Every statement written in a register transfer notation implies a hardware construction for implementing the transfer. Figure 4-2 shows the block diagram that depicts the transfer from $R1$ to $R2$. The n outputs of register $R1$ are connected to the n inputs of register $R2$. The letter n will be used to indicate any number of bits for the register. It will be replaced by an actual number when the length of the register is known. Register $R2$ has a load input that is activated by the control variable P. It is assumed that the control variable is synchronized with the same clock as the one applied to the register. As shown

Figure 4-2 Transfer from **R**1 to **R**2 when $p = 1$.



(a) Block diagram



(b) Timing diagram

in the timing diagram, P is activated in the control section by the rising edge of a clock pulse at time t. The next positive transition of the clock at time $t + 1$ finds the load input active and the data inputs of $R2$ are then loaded into the register in parallel. P may go back to 0 at time $t + 1$; otherwise, the transfer will occur with every clock pulse transition while P remains active.

Note that the clock is not included as a variable in the register transfer statements. It is assumed that all transfers occur during a clock edge transition. Even though the control condition such as P becomes active just after time t, the actual transfer does not occur until the register is triggered by the next positive transition of the clock at time $t + 1$.

The basic symbols of the register transfer notation are listed in Table 4-1. Registers are denoted by capital letters, and numerals may follow the letters. Parentheses are used to denote a part of a register by specifying the range of bits or by giving a symbol name to a portion of a register. The arrow denotes a transfer of information and the direction of transfer. A comma is used to separate two or more operations that are executed at the same time. The statement

$$T: \quad R2 \leftarrow R1, \quad R1 \leftarrow R2$$

denotes an operation that exchanges the contents of two registers during one common clock pulse provided that $T = 1$. This simultaneous operation is possible with registers that have edge-triggered flip-flops.

TABLE 4-1 Basic Symbols for Register Transfers

Symbol	Description	Examples
Letters (and numerals)	Denotes a register	MAR, $R2$
Parentheses ( )	Denotes a part of a register	$R2(0\text{–}7)$, $R2(L)$
Arrow $\leftarrow$	Denotes transfer of information	$R2 \leftarrow R1$
Comma ,	Separates two microoperations	$R2 \leftarrow R1$, $R1 \leftarrow R2$

4-3 Bus and Memory Transfers

A typical digital computer has many registers, and paths must be provided to transfer information from one register to another. The number of wires will be excessive if separate lines are used between each register and all other registers in the system. A more efficient scheme for transferring information between registers in a multiple-register configuration is a common bus system. A bus structure consists of a set of common lines, one for each bit of a register, through which binary information is transferred one at a time. Control

common bus

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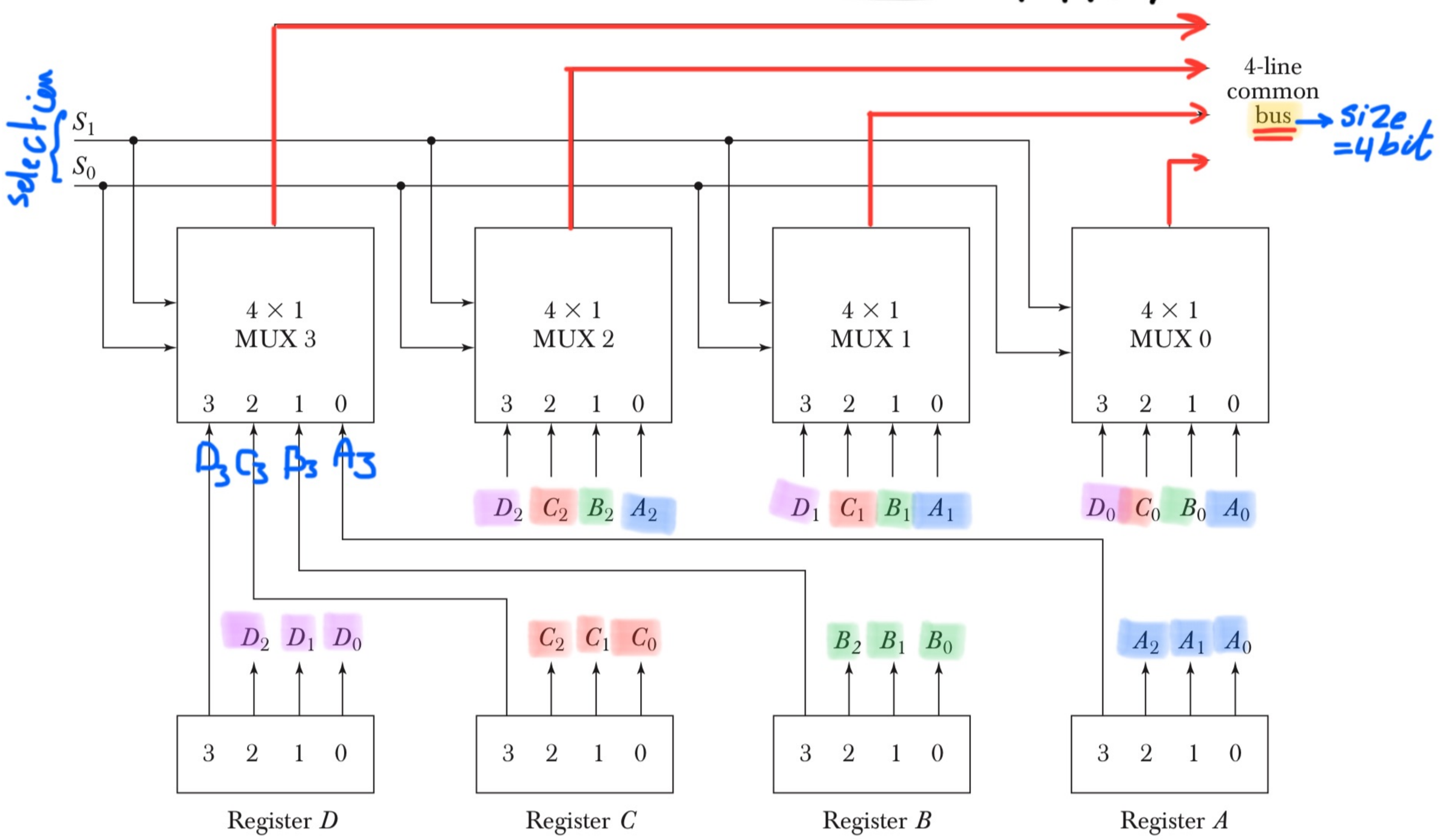
signals determine which register is selected by the bus during each particular register transfer.

One way of constructing a common bus system is with multiplexers. The multiplexers select the source register whose binary information is then placed on the bus. The construction of a bus system for four registers is shown in Fig. 4-3. Each register has four bits, numbered 0 through 3. The bus consists of four 4×1 multiplexers each having four data inputs, 0 through 3, and two selection inputs, S_1 and S_0. In order not to complicate the diagram with 16 lines crossing each other, we use labels to show the connections from the outputs of the registers to the inputs of the multiplexers. For example, output 1 of register A is connected to input 0 of MUX 1 because this input is labeled A_1. The diagram shows that the bits in the same significant position in each register are connected to the data inputs of one multiplexer to form one line of the bus. Thus MUX 0 multiplexes the four 0 bits of the registers, MUX 1 multiplexes the four 1 bits of the registers, and similarly for the other two bits.

each one Size = 4 bit

Computer sys : 4 registers

Figure 4-3 Bus system for four registers → 4 MUX



Selections ⇒ 0 0 → A_s, 0 1 → B_s, 1 0 → C_s, 1 1 → D_s } المسؤول الذي يختارهم CU (Control Unit)

if I have 4 registers and the size for each one is 8bit how many MUX we need and

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if I have 4 registers and the size for each one is 8bit how many MUX we need and what is the size for each one and what is the bus size

ans: 8 MUX (4 * 1) and the bus size = 4

bus selection

The two selection lines S_1 and S_0 are connected to the selection inputs of all four multiplexers. The selection lines choose the four bits of one register and transfer them into the four-line common bus. When $S_1S_0 = 00$, the 0 data inputs of all four multiplexers are selected and applied to the outputs that form the bus. This causes the bus lines to receive the content of register A since the outputs of this register are connected to the 0 data inputs of the multiplexers. Similarly, register B is selected if $S_1S_0 = 01$, and so on. Table 4-2 shows the register that is selected by the bus for each of the four possible binary value of the selection lines.

TABLE 4-2 Function Table for Bus of Fig. 4-3

S_1	S_0	Register selected
0	0	A
0	1	B
1	0	C
1	1	D

bus size (lines)

In general, a bus system will multiplex k registers of n bits each to produce an n-line common bus. The number of multiplexers needed to construct the bus is equal to n, the number of bits in each register. The size of each multiplexer must be $k \times 1$ since it multiplexes k data lines. For example, a common bus for eight registers of 16 bits each requires 16 multiplexers, one for each line in the bus. Each multiplexer must have eight data input lines and three selection lines to multiplex one significant bit in the eight registers.

The transfer of information from a bus into one of many destination registers can be accomplished by connecting the bus lines to the inputs of all destination registers and activating the load control of the particular destination register selected. The symbolic statement for a bus transfer may mention the bus or its presence may be implied in the statement. When the bus is includes in the statement, the register transfer is symbolized as follows:

$$BUS \leftarrow C, \qquad R1 \leftarrow BUS$$

The content of register C is placed on the bus, and the content of the bus is loaded into register $R1$ by activating its load control input. If the bus is known to exist in the system, it may be convenient just to show the direct transfer.

$$R1 \leftarrow C$$

From this statement the designer knows which control signals must be activated to produce the transfer through the bus.

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Three-State Bus Buffers

three-state gate

A bus system can be constructed with three-state gates instead of multiplexers. A three-state gate is a digital circuit that exhibits three states. Two of the states are signals equivalent to logic 1 and 0 as in a conventional gate. The third state is a *high-impedance state.* The high-impedance state behaves like an open circuit, which means that the output is disconnected and does not have a logic significance. Three-state gates may perform any conventional logic, such as AND or NAND. However, the one most commonly used in the design of a bus system is the buffer gate.

high-impedance

buffer

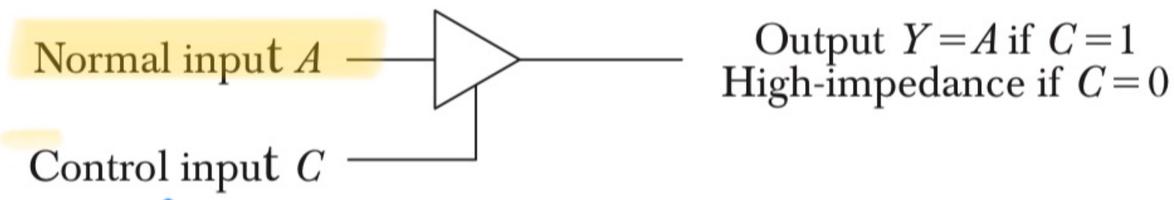
The graphic symbol of a three-state buffer gate is shown in Fig. 4-4. It is distinguished from a normal buffer by having both a normal input and a control input. The control input determines the output state. When the control input is equal to 1, the output is enabled and the gate behaves like any conventional buffer, with the output equal to the normal input. When the control input is 0, the output is disabled and the gate goes to a high-impedance state, regardless of the value in the normal input. The high-impedance state of a three-state gate provides a special feature not available in other gates. Because of this feature, a large number of three-state gate outputs can be connected with wires to form a common bus line without endangering loading effects.

bus system

The construction of a bus system with three-state buffers is demonstrated in Fig. 4-5. The outputs of four buffers are connected together to form a single bus line. (It must be realized that this type of connection cannot be done with gates that do not have three-state outputs.) The control inputs to the buffers determine which of the four normal inputs will communicate with the bus line. No more than one buffer may be in the active state at any given time. The connected buffers must be controlled so that only one three-state buffer has access to the bus line while all other buffers are maintained in a high-impedance state.

One way to ensure that no more than one control input is active at any given time is to use a decoder, as shown in the diagram. When the enable input of the decoder is 0, all of its four outputs are 0, and the bus line is in a high-impedance state because all four buffers are disabled. When the enable input is active, one of the three-state buffers will be active, depending on the binary value in the select inputs of the decoder. Careful investigation will reveal that Fig. 4-5 is another way of constructing a 4×1 multiplexer since the circuit can replace the multiplexer in Fig. 4-3.

Figure 4-4 Graphic symbols for three-state buffer.



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Draw the full Circuit

→ 1 decoder
4 state buffers for each stage
↳
4 Stages = 16 buffer

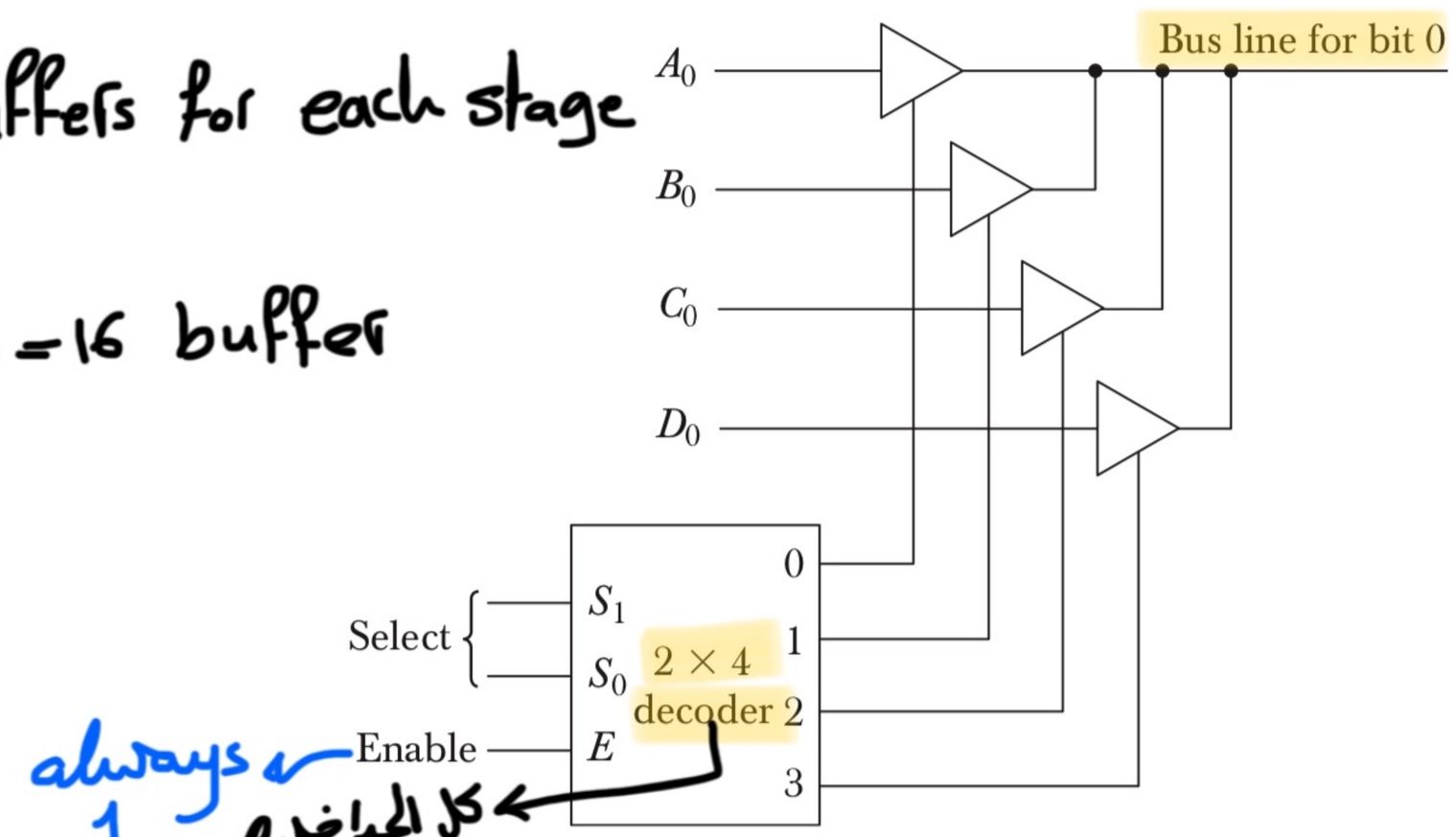

Figure 4-5 Bus line with three state-buffers. → 1 stage

To construct a common bus for four registers of n bits each using three-state buffers, we need n circuits with four buffers in each as shown in Fig. 4-5. Each group of four buffers receives one significant bit from the four registers. Each common output produces one of the lines for the common bus for a total of n lines. Only one decoder is necessary to select between the four registers.

Memory Transfer

memory read

memory write

The operation of a memory unit was described in Sec. 2-7. The transfer of information from a memory word to the outside environment is called a *read* operation. The transfer of new information to be stored into the memory is called a *write* operation. A memory word will be symbolized by the letter M. The particular memory word among the many available is selected by the memory address during the transfer. It is necessary to specify the address of M when writing memory transfer operations. This will be done by enclosing the address in square brackets following the letter M. ⇒ memory location

Special purpose

Consider a memory unit that receives the address from a register, called the address register, symbolized by AR. The data are transferred to another register, called the data register, symbolized by DR. The read operation can be stated as follows:

$$\text{Read:} \quad DR \leftarrow M[AR]$$

This causes a transfer of information into DR from the memory word M selected by the address in AR.

The write operation transfers the content of a data register to a memory word M selected by the address. Assume that the input data are in register

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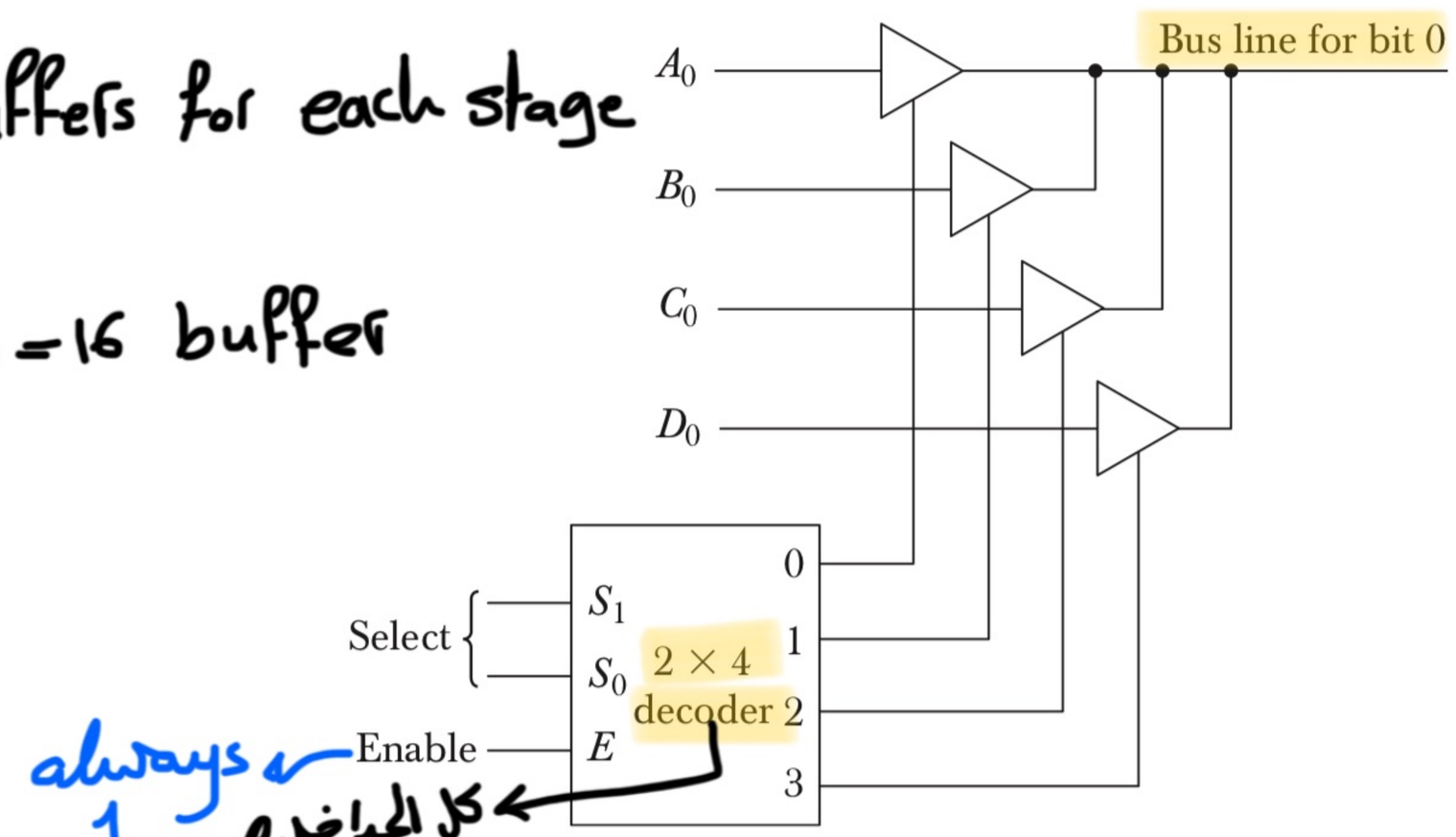
Draw the full Circuit

→ 1 decoder
4 state buffers for each stage
↳
4 Stages = 16 buffer



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$R1$ and the address is in AR. The write operation can be stated symbolically as follows:

$$\text{Write:} \quad M[AR] \leftarrow R1$$

This causes a transfer of information from $R1$ into the memory word M selected by the address in AR.

4-4 Arithmetic Microoperations

A microoperation is an elementary operation performed with the data stored in registers. The microoperations most often encountered in digital computers are classified into four categories:

1. Register transfer microoperations transfer binary information from one register to another.
2. Arithmetic microoperations perform arithmetic operation on numeric data stored in registers.
3. Logic microoperations perform bit manipulation operations on nonnumeric data stored in registers.
4. Shift microoperations perform shift operations on data stored in registers.

The register transfer microoperation was introduced in Sec. 4-2. This type of microoperation does not change the information content when the binary information moves from the source register to the destination register. The other three types of microoperations change the information content during the transfer. In this section we introduce a set of arithmetic microoperations. In the next two sections we present the logic and shift microoperations.

The basic arithmetic microoperations are addition, subtraction, increment, decrement, and shift. Arithmetic shifts are explained later in conjunction with the shift microoperations. The arithmetic microoperation defined by the statement

$$R3 \leftarrow R1 + R2$$

add microoperation

specifies an *add* microoperation. It states that the contents of register $R1$ are added to the contents of register $R2$ and the sum transferred to register $R3$. To implement this statement with hardware we need three registers and the digital component that performs the addition operation. The other basic arithmetic microoperations are listed in Table 4-3. Subtraction is most often

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subtract microoperation

implemented through complementation and addition. Instead of using the minus operator, we can specify the subtraction by the following statement:

$$R3 \leftarrow R1 + \overline{R2} + 1$$

2's Comp. → $\overline{R2}$ ← 1's Comp.

} Sub.

$\overline{R2}$ is the symbol for the 1's complement of $R2$. Adding 1 to the 1's complement produces the 2's complement. Adding the contents of $R1$ to the 2's complement of $R2$ is equivalent to $R1 - R2$.

TABLE 4-3 Arithmetic Microoperations

Symbolic designation	Description
$R3 \leftarrow R1 + R2$	Contents of $R1$ plus $R2$ transferred to $R3$
$R3 \leftarrow R1 - R2$	Contents of $R1$ minus $R2$ transferred to $R3$
$R2 \leftarrow \overline{R2}$	Complement the contents of $R2$ (1's complement)
$R2 \leftarrow \overline{R2} + 1$	2's complement the contents of $R2$ (negate)
$R3 \leftarrow R1 + \overline{R2} + 1$	$R1$ plus the 2's complement of $R2$ (subtraction)
$R1 \leftarrow R1 + 1$	Increment the contents of $R1$ by one
$R1 \leftarrow R1 - 1$	Decrement the contents of $R1$ by one

The increment and decrement microoperations are symbolized by plus-one and minus-one operations, respectively. These microoperations are implemented with a combinational circuit or with a binary up-down counter.

The arithmetic operations of multiply and divide are not listed in Table 4-3. These two operations are valid arithmetic operations but are not included in the basic set of microoperations. The only place where these operations can be considered as microoperations is in a digital system, where they are implemented by means of a combinational circuit. In such a case, the signals that perform these operations propagate through gates, and the result of the operation can be transferred into a destination register by a clock pulse as soon as the output signal propagates through the combinational circuit. In most computers, the multiplication operation is implemented with a sequence of add and shift microoperations. Division is implemented with a sequence of subtract and shift microoperations. To specify the hardware in such a case requires a list of statements that use the basic microoperations of add, subtract, and shift (see Chapter 10).

Binary Adder

To implement the add microoperation with hardware, we need the registers that hold the data and the digital component that performs the arithmetic addition. The digital circuit that forms the arithmetic sum of two bits and a previous carry is called a full-adder (see Fig. 1-17). The digital circuit that

→ 3 inputs

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4 reg. ⇒ 4 full adder
n n

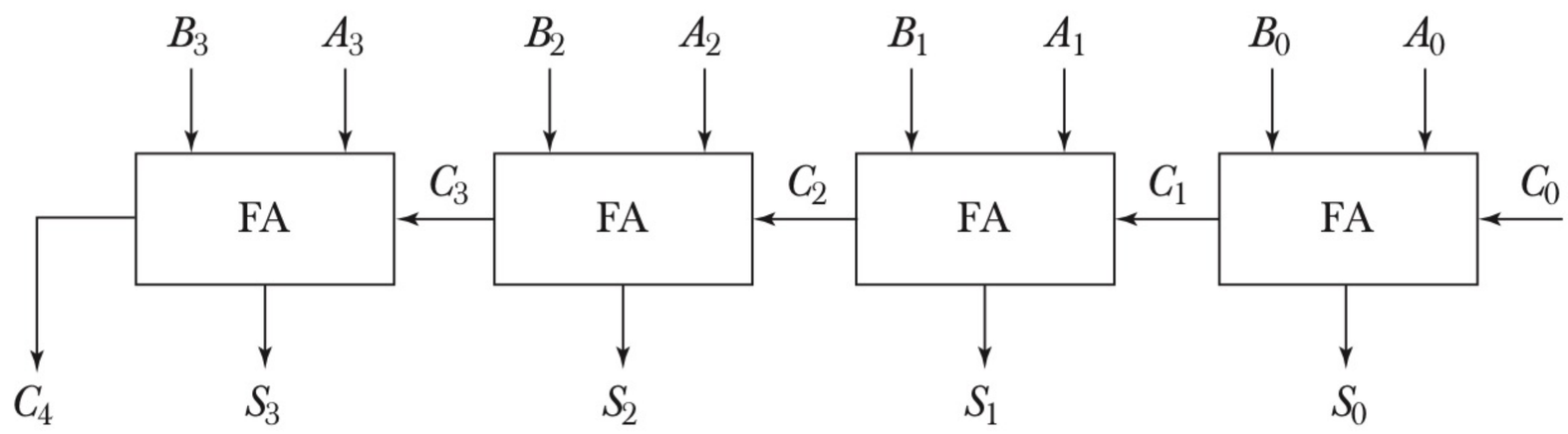

Figure 4-6 4-bit binary adder.

binary adder

full-adder

generates the arithmetic sum of two binary numbers of any length is called a binary adder. The binary adder is constructed with full-adder circuits connected in cascade, with the output carry from one full-adder connected to the input carry of the next full-adder. Figure 4-6 shows the interconnections of four full-adders (FA) to provide a 4-bit binary adder. The augend bits of A and the addend bits of B are designated by subscript numbers from right to left, with subscript 0 denoting the low-order bit. The carries are connected in a chain through the full-adders. The input carry to the binary adder is C_0 and the output carry is C_4. The S outputs of the full-adders generate the required sum bits.

An n-bit binary adder requires n full-adders. The output carry from each full-adder is connected to the input carry of the next-high-order full-adder. The n data bits for the A inputs come from one register (such as $R1$), and the n data bits for the B inputs come from another register (such as $R2$). The sum can be transferred to a third register or to one of the source registers ($R1$ or $R2$), replacing its previous content.

modification →

Binary Adder-Subtractor

The subtraction of binary numbers can be done most conveniently by means of complements as discussed in Sec. 3-2. Remember that the subtraction $A - B$ can be done by taking the 2's complement of B and adding it to A. The 2's complement can be obtained by taking the 1's complement and adding one to the least significant pair of bits. The 1's complement can be implemented with inverters and a one can be added to the sum through the input carry.

adder-subtractor

The addition and subtraction operations can be combined into one common circuit by including an exclusive-OR gate with each full-adder. A 4-bit adder-subtractor circuit is shown in Fig. 4-7. The mode input M controls the operation. When $M = 0$ the circuit is an adder and when $M = 1$ the circuit becomes a subtractor. Each exclusive-OR gate receives input M and one of the inputs of B. When $M = 0$, we have $B \oplus 0 = B$. The full-adders receive the value of B, the input carry is 0, and the circuit performs A plus B. When $M = 1$, we have $B \oplus 1 = B'$ and $C_0 = 1$. The B inputs are all complemented and a 1 is added through the input carry. The circuit performs the operation A plus the

Binary sub ⇒ replace XOR with not → B يصير لي not (عكسية)

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not داخلة ع

0 XOR B = B
1 XOR B = $\overline{B}$

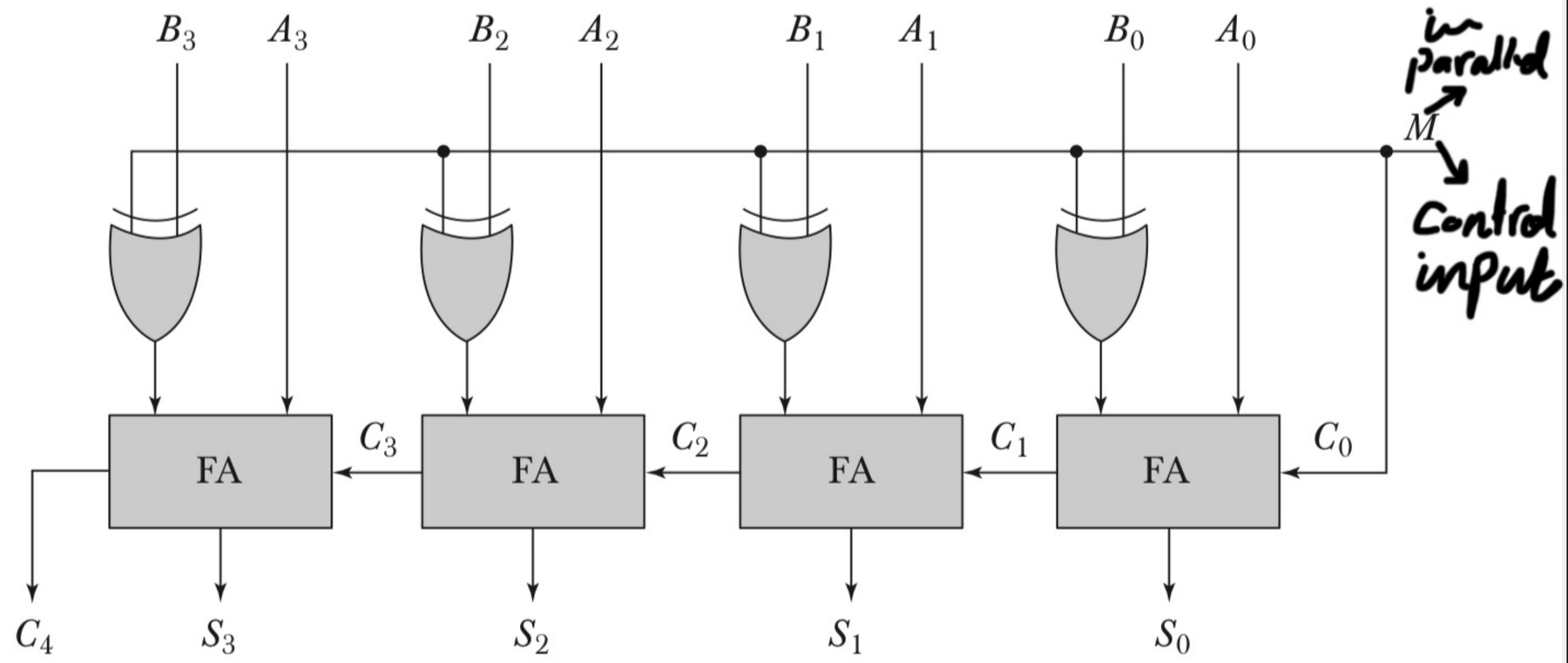

Figure 4-7 4-bit adder-subtractor.

لو بدي اياها بس تطلع بغير XOR ← not بصير بس B داخلة عالـ not

2's complement of B. For unsigned numbers, this gives $A - B$ if $A \geq B$ or the 2's complement of $(B - A)$ if $A < B$. For signed numbers, the result is $A - B$ provided that there is no overflow.

Binary Incrementer

The increment microoperation adds one to a number in a register. For example, if a 4-bit register has a binary value 0110, it will go to 0111 after it is incremented. This microoperation is easily implemented with a binary counter (see Fig. 2-10). Every time the count enable is active, the clock pulse transition increments the content of the register by one. There may be occasions when the increment microoperation must be done with a combinational circuit independent of a particular register. This can be accomplished by means of half-adders (see Fig. 1-16) connected in cascade.

incrementer

The diagram of a 4-bit combinational circuit incrementer is shown in Fig. 4-8. One of the inputs to the least significant half-adder (HA) is connected to logic-1 and the other input is connected to the least significant bit of the number to be incremented. The output carry from one half-adder is connected to one of the inputs of the next-higher-order half-adder. The circuit receives the four bits from A_0 through A_3, adds one to it, and generates the incremented output in S_0 through S_3. The output carry C_4 will be 1 only after incrementing binary 1111. This also causes outputs S_0 through S_3 to go to 0.

The circuit of Fig. 4-8 can be extended to an n-bit binary incrementer by extending the diagram to include n half-adders. The least significant bit must have one input connected to logic-1. The other inputs receive the number to be incremented or the carry from the previous stage.

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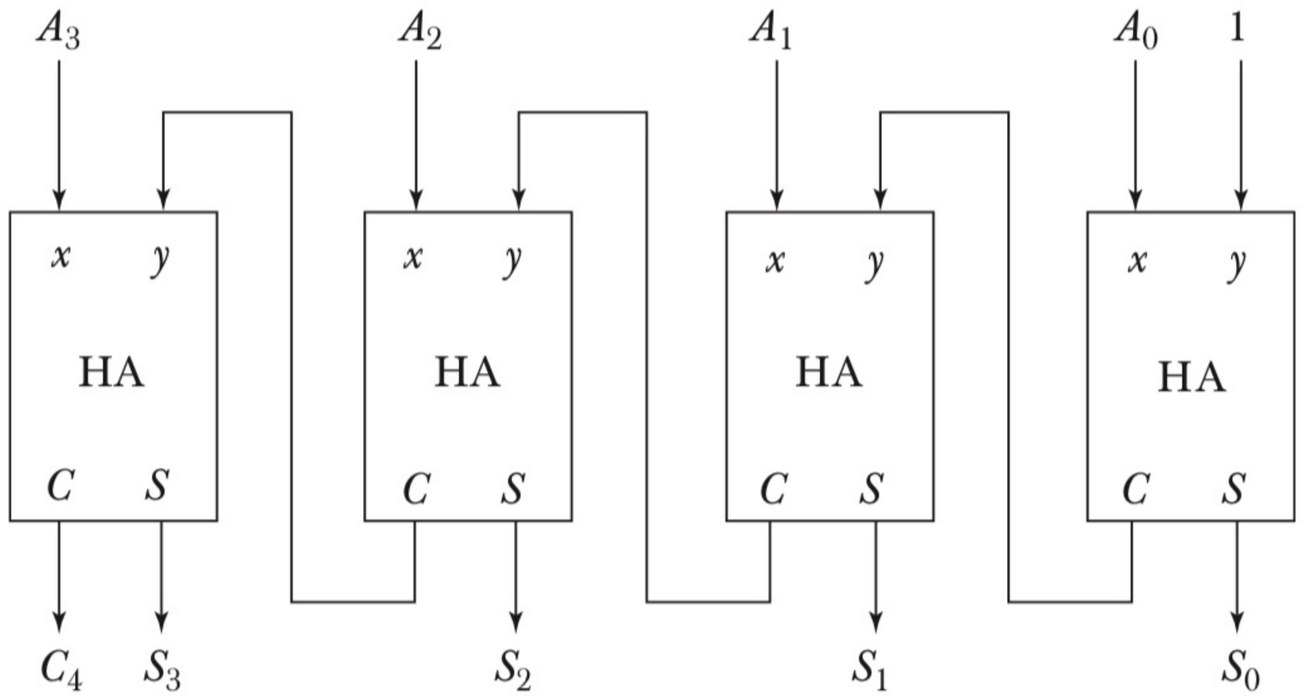

Figure 4-8 4-bit binary incrementer.

↳no need for full adder

Arithmetic Circuit

arithmetic circuit

The arithmetic microoperations listed in Table 4-3 can be implemented in one composite arithmetic circuit. The basic component of an arithmetic circuit is the parallel adder. By controlling the data inputs to the adder, it is possible to obtain different types of arithmetic operations.

The diagram of a 4-bit arithmetic circuit is shown in Fig. 4-9. It has four full-adder circuits that constitute the 4-bit adder and four multiplexers for choosing different operations. There are two 4-bit inputs A and B and a 4-bit output D. The four inputs from A go directly to the X inputs of the binary adder. Each of the four inputs from B are connected to the data inputs of the multiplexers. The multiplexers data inputs also receive the complement of B. The other two data inputs are connected to logic-0 and logic-1. Logic-0 is a fixed voltage value (0 volts for TTL integrated circuits) and the logic-1 signal can be generated through an inverter whose input is 0. The four multiplexers are controlled by two selection inputs, S_1 and S_0. The input carry C_{in} goes to the carry input of the FA in the least significant position. The other carries a connected from one stage to the next.

input carry

The output of the binary adder is calculated from the following arithmetic sum:

$$D = A + Y + C_{in}$$

where A is the 4-bit binary number at the X inputs and Y is the 4-bit binary number at the Y inputs of the binary adder. C_{in} is the input carry, which can be equal to 0 or 1. Note that the symbol + in the equation above denotes an arithmetic plus. By controlling the value of Y with the two selection inputs S_1 and S_0 and making C_{in} equal to 0 or 1, it is possible to generate the eight arithmetic microoperations listed in Table 4-4.

4 bit reg
4 MUX (4×1)
3 inputs
Work in Parallel
For MUX
Carry on the 1st full adder

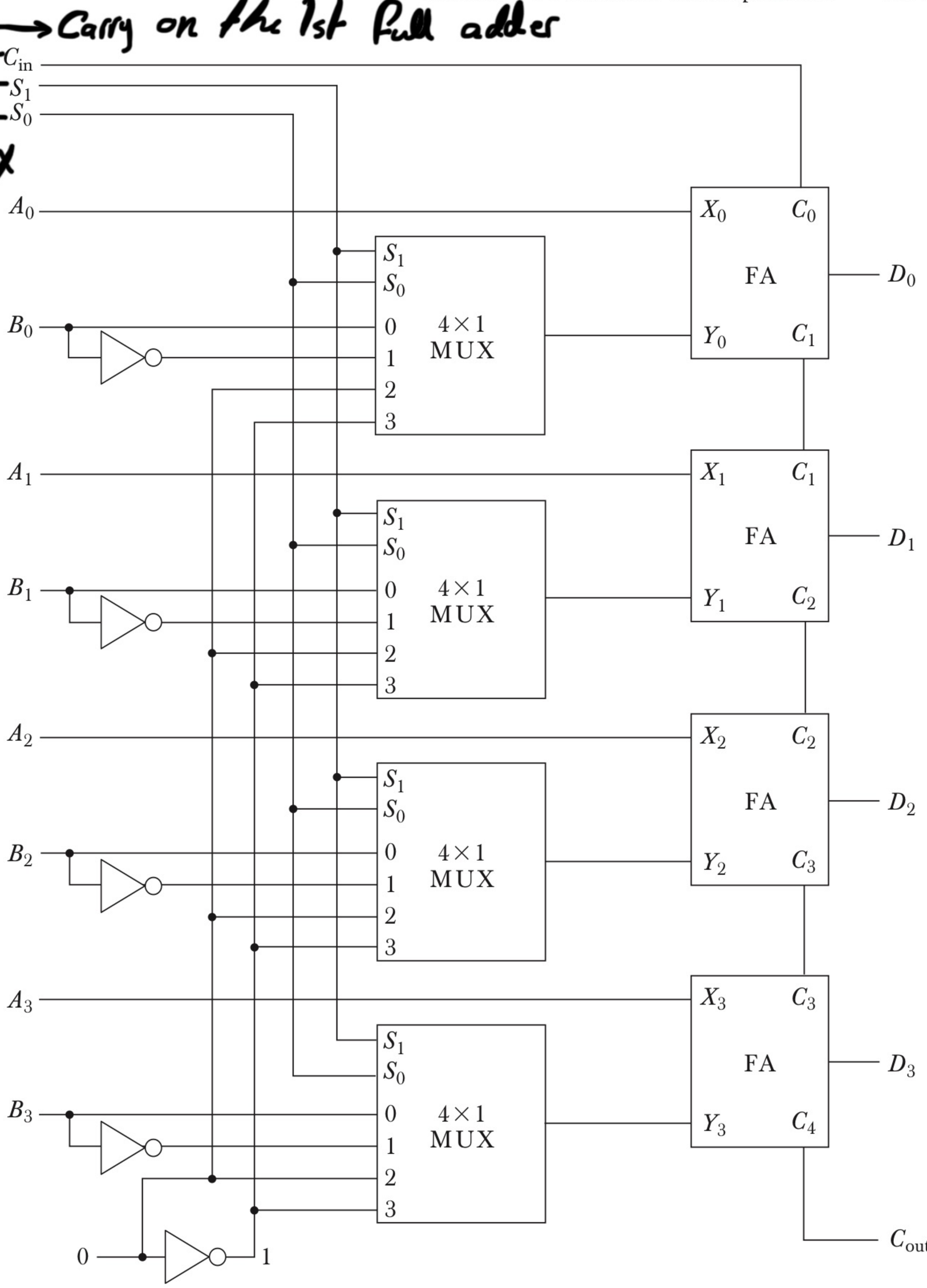

Figure 4-9 4-bit arithmetic circuit.

ALU ← كل الدوائر موجودين فيها

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TABLE 4-4 Arithmetic Circuit Function Table

Select			Input	Output	
S_1	S_0	C_{in}	Y	$D = A + Y + C_{in}$	Microoperation
0	0	0	B	$D = A + B$	Add
0	0	1	B	$D = A + B + 1$	Add with carry
0	1	0	$\overline{B}$	$D = A + \overline{B}$	Subtract with borrow
0	1	1	$\overline{B}$	$D = A + \overline{B} + 1$	Subtract
1	0	0	0	$D = A$	Transfer A
1	0	1	0	$D = A + 1$	Increment A
1	1	0	1	$D = A - 1$	Decrement A
1	1	1	1	$D = A$	Transfer A

addition

When $S_1S_0 = 00$, the value of B is applied to the Y inputs of the adder. If $C_{in} = 0$, the output $D = A + B$. If $C_{in} = 1$, output $D = A + B + 1$. Both cases perform the add microoperation with or without adding the input carry.

subtraction

When $S_1S_0 = 01$, the complement of B is applied to the Y inputs of the adder. If $C_{in} = 1$, then $D = A + \overline{B} + 1$. This produces A plus the 2's complement of B, which is equivalent to a subtraction of $A - B$. When $C_{in} = 0$, then $D = A + \overline{B}$. This is equivalent to a subtract with borrow, that is, $A - B - 1$.

increment

When $S_1S_0 = 10$, the inputs from B are neglected, and instead, all 0's are inserted into the Y inputs. The output becomes $D = A + 0 + C_{in}$. This gives $D = A$ when $C_{in} = 0$ and $D = A + 1$ when $C_{in} = 1$. In the first case we have a direct transfer from input A to output D. In the second case, the value of A is incremented by 1.

decrement

When $S_1S_0 = 11$, all 1's are inserted into the Y inputs of the adder to produce the decrement operation $D = A - 1$ when $C_{in} = 0$. This is because a number with all 1's is equal to the 2's complement of 1 (the 2's complement of binary 0001 is 1111). Adding a number A to the 2's complement of 1 produces $F = A +$ 2's complement of $1 = A - 1$. When $C_{in} = 1$, then $D = A - 1 + 1 = A$, which causes a direct transfer from input A to output D. Note that the microoperation $D = A$ is generated twice, so there are only seven distinct microoperations in the arithmetic circuit.

4-5 Logic Microoperations

Logic microoperations specify binary operations for strings of bits stored in registers. These operations consider each bit of the register separately and treat them as binary variables. For example, the exclusive-OR microoperation with the contents of two registers $R1$ and $R2$ is symbolized by the statement

$$P: \quad R1 \leftarrow R1 \oplus R2$$

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It specifies a logic microoperation to be executed on the individual bits of the registers provided that the control variable $P = 1$. As a numerical example, assume that each register has four bits. Let the content of $R1$ be 1010 and the content of $R2$ be 1100. The exclusive-OR microoperation stated above symbolizes the following logic computation:

```
1010    Content of R1
1100    Content of R2
0110    Content of R1 after P = 1
```

The content of $R1$, after the execution of the microoperation, is equal to the bit-by-bit exclusive-OR operation on pairs of bits in $R2$ and previous values of $R1$. The logic microoperations are seldom used in scientific computations, but they are very useful for bit manipulation of binary data and for making logical decisions.

special symbols

Special symbols will be adopted for the logic microoperations OR, AND, and complement, to distinguish them from the corresponding symbols used to express Boolean functions. The symbol $\vee$ will be used to denote an OR microoperation and the symbol $\wedge$ to denote an AND microoperation. The complement microoperation is the same as the 1's complement and uses a bar on top of the symbol that denotes the register name. By using different symbols, it will be possible to differentiate between a logic microoperation and a control (or Boolean) function. Another reason for adopting two sets of symbols is to be able to distinguish the symbol +, when used to symbolize an arithmetic plus, from a logic OR operation. Although the + symbol has two meanings, it will be possible to distinguish between them by noting where the symbol occurs. When the symbol + occurs in a microoperation, it will denote an arithmetic plus. When it occurs in a control (or Boolean) function, it will denote an OR operation. We will never use it to symbolize an OR microoperation. For example, in the statement

$$P + Q\text{:}\quad R1 \leftarrow R2 + R3,\quad R4 \leftarrow R5 \vee R6$$

the + between P and Q is an OR operation between two binary variables of a control function. The + between $R2$ and $R3$ specifies an add microoperation. The OR microoperation is designated by the symbol $\vee$ between registers $R5$ and $R6$.

List of Logic Microoperations

There are 16 different logic operations that can be performed with two binary variables. They can be determined from all possible truth tables obtained with two binary variables as shown in Table 4-5. In this table, each of the 16 columns F_0 through F_{15} represents a truth table of one possible Boolean

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TABLE 4-5 Truth Tables for 16 Functions of Two Variables

x	y	F_0	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1

function for the two variables x and y. Note that the functions are determined from the 16 binary combinations that can be assigned to F.

The 16 Boolean functions of two variables x and y are expressed in algebraic form in the first column of Table 4-6. The 16 logic microoperations are derived from these functions by replacing variable x by the binary content of register A and variable y by the binary content of register B. It is important to realize that the Boolean functions listed in the first column of Table 4-6 represent a relationship between two binary variables x and y. The logic microoperations listed in the second column represent a relationship between the binary content of two registers A and B. Each bit of the register is treated as a binary variable and the microoperation is performed on the string of bits stored in the registers.

TABLE 4-6 Sixteen Logic Microoperations

Boolean function	Microoperation	Name
$F_0 = 0$	$F \leftarrow 0$	Clear
$F_1 = xy$	$F \leftarrow A \wedge B$	AND
$F_2 = xy'$	$F \leftarrow A \wedge \overline{B}$	
$F_3 = x$	$F \leftarrow A$	Transfer A
$F_4 = x'y$	$F \leftarrow \overline{A} \wedge B$	
$F_5 = y$	$F \leftarrow B$	Transfer B
$F_6 = x \oplus y$	$F \leftarrow A \oplus B$	Exclusive-OR
$F_7 = x + y$	$F \leftarrow A \vee B$	OR
$F_8 = (x + y)'$	$F \leftarrow \overline{A \vee B}$	NOR
$F_9 = (x \oplus y)'$	$F \leftarrow \overline{A \oplus B}$	Exclusive-NOR
$F_{10} = y'$	$F \leftarrow \overline{B}$	Complement B
$F_{11} = x + y'$	$F \leftarrow A \vee \overline{B}$	
$F_{12} = x'$	$F \leftarrow \overline{A}$	Complement A
$F_{13} = x' + y$	$F \leftarrow \overline{A} \vee B$	
$F_{14} = (xy)'$	$F \leftarrow \overline{A \wedge B}$	NAND
$F_{15} = 1$	$F \leftarrow$ all 1's	Set to all 1's

micro operation & boolean تحويل بين
يكون فيه
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اميزه
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Hardware Implementation

The hardware implementation of logic microoperations requires that logic gates be inserted for each bit or pair of bits in the registers to perform the required logic function. Although there are 16 logic microoperations, most computers use only four–AND, OR, XOR (exclusive-OR), and complement–from which all others can be derived.

logic circuit

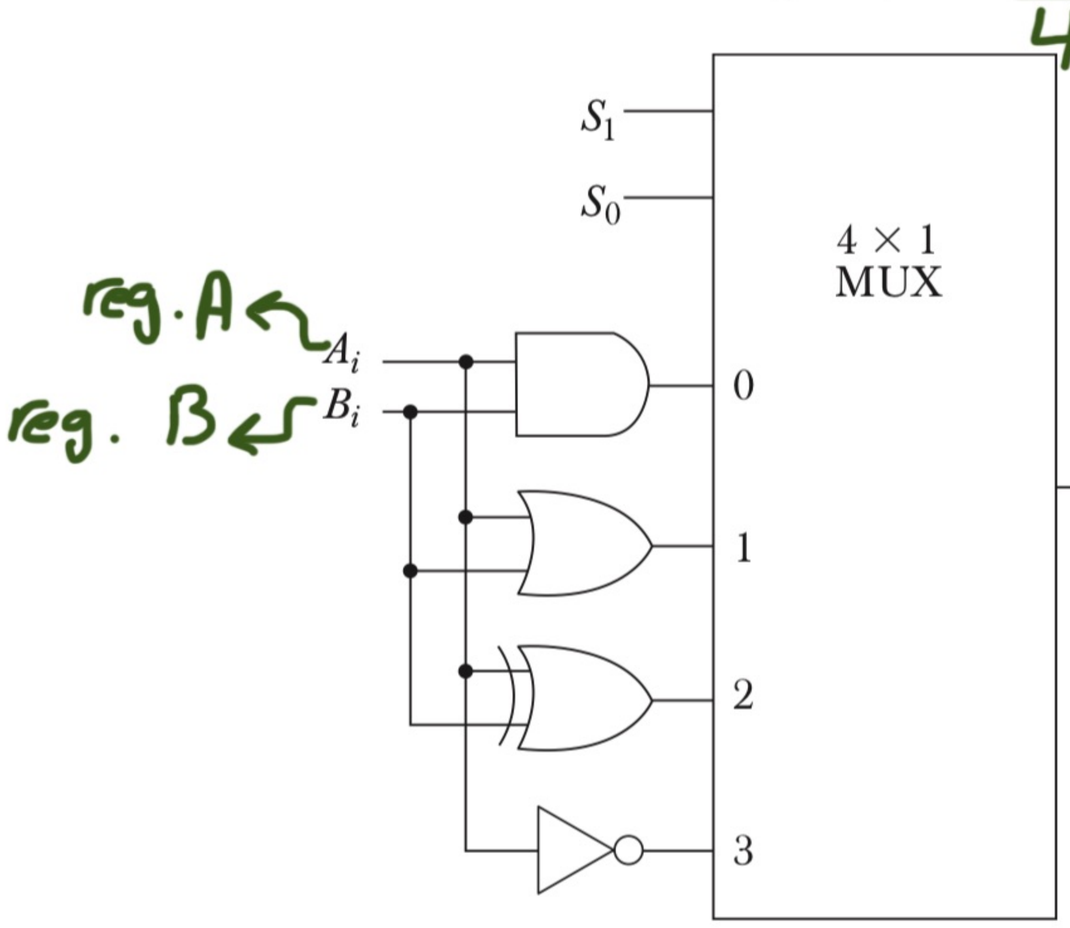
Figure 4-10 shows one stage of a circuit that generates the four basic logic microoperations. It consists of four gates and a multiplexer. Each of the four logic operations is generated through a gate that performs the required logic. The outputs of the gates are applied to the data inputs of the multiplexer. The two selection inputs S_1 and S_0 choose one of the data inputs of the multiplexer and direct its value to the output. The diagram shows one typical stage with subscript i. For a logic circuit with n bits, the diagram must be repeated n times for $i = 0, 1, 2, \ldots, n - 1$. The selection variables are applied to all stages. The function table in Fig. 4-10(b) lists the logic microoperations obtained for each combination of the selection variables.

Some Applications

Logic microoperations are very useful for manipulating individual bits or a portion of a word stored in a register. They can be used to change bit values, delete a group of bits, or insert new bit values into a register. The following examples show how the bits of one register (designated by A) are manipulated

Figure 4-10 One stage of logic circuit.

4 stages , 4 bit reg.



(a) Logic diagram

S_1	S_0	Output	Operation
0	0	$E = A \wedge B$	AND
0	1	$E = A \vee B$	OR
1	0	$E = A \oplus B$	XOR
1	1	$E = \overline{A}$	Complement

(b) Functional table

* اذا بدي اعمل load للـ B بدي اعمل transfer اطلعها في A

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by logic microoperations as a function of the bits of another register (designated by B). In a typical application, register A is a processor register and the bits of register B constitute a logic operand extracted from memory and placed in register B.

selective-set

The *selective-set* operation sets to 1 the bits in register A where there are corresponding 1's in register B. It does not affect bit positions that have 0's in B. The following numerical example clarifies this operation:

```
1010    A before
1100    B (logic operand)
----
1110    A after
```

The two leftmost bits of B are 1's, so the corresponding bits of A are set to 1. One of these two bits was already set and the other has been changed from 0 to 1. The two bits of A with corresponding 0's in B remain unchanged. The example above serves as a truth table since it has all four possible combinations of two binary variables. From the truth table we note that the bits of A after the operation are obtained from the logic-OR operation of bits in B and previous values of A. Therefore, the OR microoperation can be used to selectively set bits of a register.

selective-complement

The *selective-complement* operation complements bits in A where there are corresponding 1's in B. It does not affect bit positions that have 0's in B. For example:

```
1010    A before
1100    B (logic operand)
----
0110    A after
```

Again the two leftmost bits of B are 1's, so the corresponding bits of A are complemented. This example again can serve as a truth table from which one can deduce that the selective-complement operation is just an exclusive-OR microoperation. Therefore, the exclusive-OR microoperation can be used to selectively complement bits of a register.

selective-clear

The *selective-clear* operation clears to 0 the bits in A only where there are corresponding 1's in B. For example:

```
1010    A before
1100    B (logic operand)
----
0010    A after
```

Again the two leftmost bits of B are 1's, so the corresponding bits of A are cleared to 0. One can deduce that the Boolean operation performed on the individual bits is AB'. The corresponding logic microoperation is

$$A \leftarrow A \wedge \overline{B}$$

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The *mask* operation is similar to the selective-clear operation except that the bits of *A* are cleared only where there are corresponding 0's in *B*. The mask operation is an AND micro operation as seen from the following numerical example:

```
1010      A before
1100      B (logic operand)
1000      A after masking
```

The two rightmost bits of *A* are cleared because the corresponding bits of *B* are 0's. The two leftmost bits are left unchanged because the corresponding bits of *B* are 1's. The mask operation is more convenient to use than the selective-clear operation because most computers provide an AND instruction, and few provide an instruction that executes the microoperation for selective-clear.

The *insert* operation inserts a new value into a group of bits. This is done by first masking the bits and then ORing them with the required value. For example, suppose that an *A* register contains eight bits, 0110 1010. To replace the four leftmost bits by the value 1001 we first mask the four unwanted bits:

```
0110  1010      A before
0000  1111      B (mask)
0000  1010      A after masking
```

and then insert the new value:

```
0000  1010      A before
1001  0000      B (insert)
1001  1010      A after insertion
```

The mask operation is an AND microoperation and the insert operation is an OR microoperation.

The *clear* operation compares the words in *A* and *B* and produces an all 0's result if the two numbers are equal. This operation is achieved by an exclusive-OR microoperation as shown by the following example:

```
1010      A
1010      B
0000      A ← A ⊕ B
```

When *A* and *B* are equal, the two corresponding bits are either both 0 or both 1. In either case the exclusive-OR operation produces a 0. The all-0's result is then checked to determine if the two numbers were equal.

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4-6 Shift Microoperations

Shift microoperations are used for serial transfer of data. They are also used in conjunction with arithmetic, logic, and other data-processing operations. The contents of a register can be shifted to the left or the right. At the same time that the bits are shifted, the first flip-flop receives its binary information from the serial input. During a shift-left operation the serial input transfers a bit into the rightmost position. During a shift-right operation the serial input transfers a bit into the leftmost position. The information transferred through the serial input determines the type of shift. There are three types of shifts: logical, circular, and arithmetic.

A *logical* shift is one that transfers 0 through the serial input. We will adopt, the symbols *shl* and *shr* for logical shift-left and shift-right microoperations. For example:

$$R1 \leftarrow \text{shl } R1$$
$$R2 \leftarrow \text{shr } R2$$

are two microoperations that specify a 1-bit shift to the left of the content of register $R1$ and a 1-bit shift to the right of the content of register $R2$. The register symbol must be the same on both sides of the arrow. The bit transferred to the end position through the serial input is assumed to be 0 during a logical shift.

The *circular* shift (also known as a *rotate* operation) circulates the bits of the register around the two ends without loss of information. This is accomplished by connecting the serial output of the shift register to its serial input. We will use the symbols *cil* and *cir* for the circular shift left and right, respectively. The symbolic notation for the shift microoperations is shown in Table 4-7.

TABLE 4-7 Shift Microoperations

Symbolic designation	Description
$R \leftarrow \text{shl } R$	Shift-left register R
$R \leftarrow \text{shr} R$	Shift-right register R
$R \leftarrow \text{cil } R$	Circular shift-left register R
$R \leftarrow \text{cir } R$	Circular shift-right register R
$R \leftarrow \text{ashl } R$	Arithmetic shift-left R
$R \leftarrow \text{ashr } R$	Arithmetic shift-right R

An *arithmetic* shift is a microoperation that shifts a signed binary number to the left or right. An arithmetic shift-left multiplies a signed binary number by 2. An arithmetic shift-right divides the number by 2. Arithmetic shifts must leave the sign bit unchanged because the sign of the number remains the same

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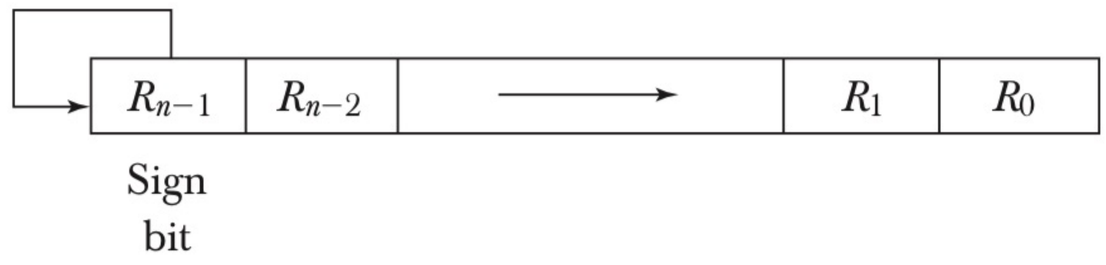

Figure 4-11 Arithmetic shift right.

when it is multiplied or divided by 2. The leftmost bit in a register holds the sign bit, and the remaining bits hold the number. The sign bit is 0 for positive and 1 for negative. Negative numbers are in 2's complement form. Figure 4-11 shows a typical register of n bits. Bit R_{n-1} in the leftmost position holds the sign bit. R_{n-2} is the most significant bit of the number and R_0 is the least significant bit. The arithmetic shift-right leaves the sign bit unchanged and shifts the number (including the sign bit) to the right. Thus R_{n-1} remains the same, R_{n-2} receives the bit from R_{n-1}, and so on for the other bits in the register. The bit in R_0 is lost.

The arithmetic shift-left inserts a 0 into R_0, and shifts all other bits to the left. The initial bit of R_{n-1} is lost and replaced by the bit from R_{n-2}. A sign reversal occurs if the bit in R_{n-1} changes in value after the shift. This happens if the multiplication by 2 causes an overflow. An overflow occurs after an arithmetic shift left if initially, before the shift, R_{n-1} is not equal to R_{n-2}. An overflow flip-flop V_s can be used to detect an arithmetic shift-left overflow.

$$V_s = R_{n-1} \oplus R_{n-2}$$

If $V_s = 0$, there is no overflow, but if $V_s = 1$, there is an overflow and a sign reversal after the shift. V_s must be transferred into the overflow flip-flop with the same clock pulse that shifts the register.

Hardware Implementation

A possible choice for a shift unit would be a bidirectional shift register with parallel load (see Fig. 2-9). Information can be transferred to the register in parallel and then shifted to the right or left. In this type of configuration, a clock pulse is needed for loading the data into the register, and another pulse is needed to initiate the shift. In a processor unit with many registers it is more efficient to implement the shift operation with a combinational circuit. In this way the content of a register that has to be shifted is first placed onto a common bus whose output is connected to the combinational shifter, and the shifted number is then loaded back into the register. This requires only one clock pulse for loading the shifted value into the register.

shifter

A combinational circuit shifter can be constructed with multiplexers as shown in Fig. 4-12. The 4-bit shifter has four data inputs, A_0 through A_3, and four data outputs, H_0 through H_3. There are two serial inputs, one for shift left

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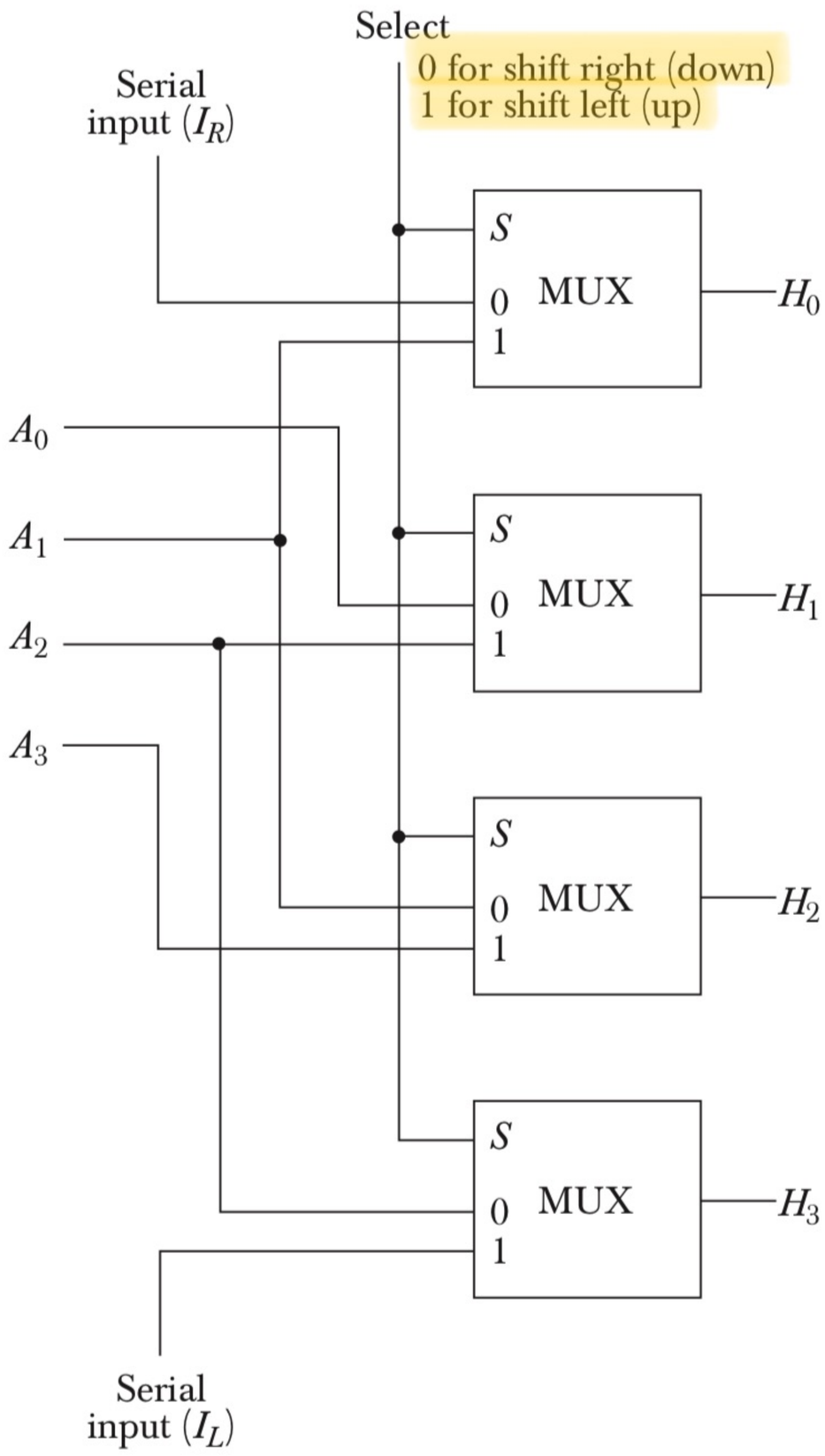

before shift

Functional table

Select	Output			
S	H_0	H_1	H_2	H_2
0	I_R	A_0	A_1	A_2
1	A_1	A_2	A_3	I_L

Figure 4-12 4-bit combinational circuit shifter.

↳ logical shifter

(I_L) and the other for shift right (I_L). When the selection input $S = 0$, the input data are shifted right (down in the diagram). When $S = 1$, the input data are shifted left (up in the diagram). The function table in Fig. 4-12 shows which input goes to each output after the shift. A shifter with n data inputs and outputs requires n multiplexers. The two serial inputs can be controlled by another multiplexer to provide the three possible types of shifts.

4-7 Arithmetic Logic Shift Unit

ALU

ALU

Instead of having individual registers performing the microoperations directly, computer systems employ a number of storage registers connected to a common operational unit called an arithmetic logic unit, abbreviated ALU. To

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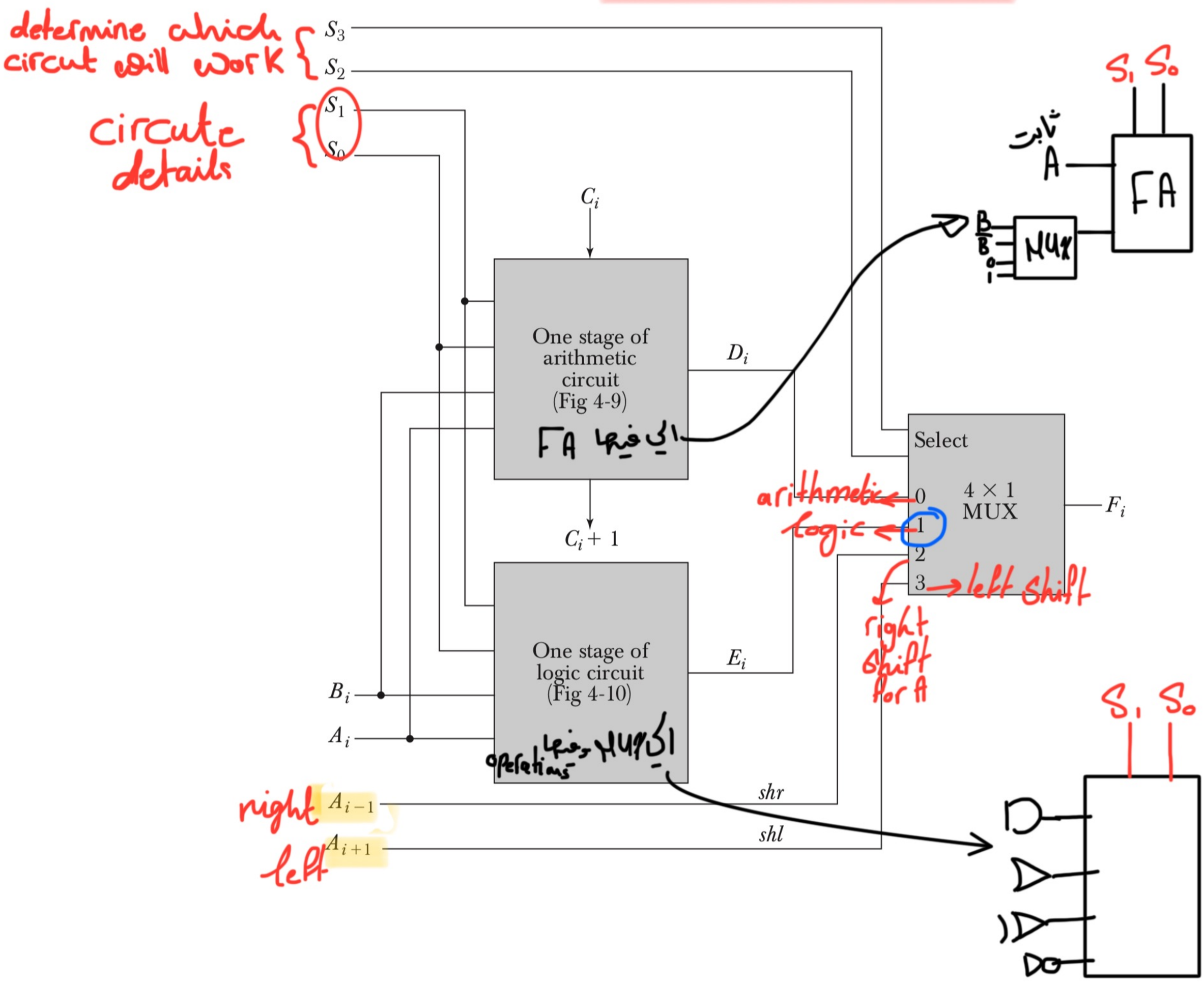
perform a microoperation, the contents of specified registers are placed in the inputs of the common ALU. The ALU performs an operation and the result of the operation is then transferred to a destination register. The ALU is a combinational circuit so that the entire register transfer operation from the source registers through the ALU and into the destination register can be performed during one clock pulse period. The shift microoperations are often performed in a separate unit, but sometimes the shift unit is made part of the overall ALU.

The arithmetic, logic, and shift circuits introduced in previous sections can be combined into one ALU with common selection variables. One stage of an arithmetic logic shift unit is shown in Fig. 4-13. The subscript i designates a typical stage. Inputs A_i and B_i are applied to both the arithmetic and logic

Figure 4-13 One stage of arithmetic logic shift unit.



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units. A particular microoperation is selected with inputs S_1 and S_0. A 4×1 multiplexer at the output chooses between an arithmetic output in E_i and a logic output in H_i. The data in the multiplexer are selected with inputs S_3 and S_2. The other two data inputs to the multiplexer receive inputs A_{i-1} for the shift-right operation and A_{i+1} for the shift-left operation. Note that the diagram shows just one typical stage. The circuit of Fig. 4-13 must be repeated n times for an n-bit ALU. The output carry C_{i+1} of a given arithmetic stage must be connected to the input carry C_i of the next stage in sequence. The input carry to the first stage is the input carry C_{in}, which provides a selection variable for the arithmetic operations.

The circuit whose one stage is specified in Fig. 4-13 provides eight arithmetic operation, four logic operations, and two shift operations. Each operation is selected with the five variables S_3, S_2, S_1, S_0, and C_{in}. The input carry C_{in} is used for selecting an arithmetic operation only.

Table 4-8 lists the 14 operations of the ALU. The first eight are arithmetic operations (see Table 4-4) and are selected with $S_3S_2 = 00$. The next four are logic operations (see Fig. 4-10) and are selected with $S_3S_2 = 01$. The input carry has no effect during the logic operations and is marked with don't-care $\times$'s. The last two operations are shift operations and are selected with $S_3S_2 = 10$ and 11. The other three selection inputs have no effect on the shift.

TABLE 4-8 Function Table for Arithmetic Logic Shift Unit

Operation select						
S_3	S_2	S_1	S_0	C_{in}	Operation	Function
0	0	0	0	0	$F = A$	Transfer A
0	0	0	0	1	$F = A + 1$	Increment A
0	0	0	1	0	$F = A + B$	Addition
0	0	0	1	1	$F = A + B + 1$	Add with carry
0	0	1	0	0	$F = A + \bar{B}$	Subtract with borrow
0	0	1	0	1	$F = A + \bar{B} + 1$	Subtraction
0	0	1	1	0	$F = A - 1$	Decrement A
0	0	1	1	1	$F = A$	Transfer A
0	1	0	0	$\times$	$F = A \wedge B$	AND
0	1	0	1	$\times$	$F = A \vee B$	OR
0	1	1	0	$\times$	$F = A \oplus B$	XOR
0	1	1	1	$\times$	$F = \bar{A}$	Complement A
1	0	$\times$	$\times$	$\times$	$F =$ shr A	Shift right A into F
1	1	$\times$	$\times$	$\times$	$F =$ shl A	Shift left A into F

4-8 Hardware Description Languages

The design of complex digital systems at the gate and flip-flop level is a tedious and time consuming process. Digital systems can be described at the register transfer level by means of hardware description language (HDL). The

CHAPTER FIVE

Basic Computer Organization and Design

IN THIS CHAPTER

5-1 Instruction Codes
5-2 Computer Registers
5-3 Computer Instructions
5-4 Timing and Control
5-5 Instruction Cycle
5-6 Memory-Reference Instructions
5-7 Input–Output and Interrupt
5-8 Complete Computer Description
5-9 Design of Basic Computer
5-10 Design of Accumulator Logic

5-1 Instruction Codes

Program is a sequence of instructions
instruction " " " = microoperations

In this chapter we introduce a basic computer and show how its operation can be specified with register transfer statements. The organization of the computer is defined by its internal registers, the riming and control structure, and the set of instructions that it uses. The design of the computer is then carried out in detail. Although the basic computer presented in this chapter is very small compared to commercial computers, it has the advantage of being simple enough so we can demonstrate the design process without too many complications.

The internal organization of a digital system is defined by the sequence of microoperations it performs on data stored in its registers. The general-purpose digital computer is capable of executing various microoperations and, in addition, can be instructed as to what specific sequence of operations it must perform. The user of a computer can control the process by means of a program. A program is a set of instructions that specify the operations,

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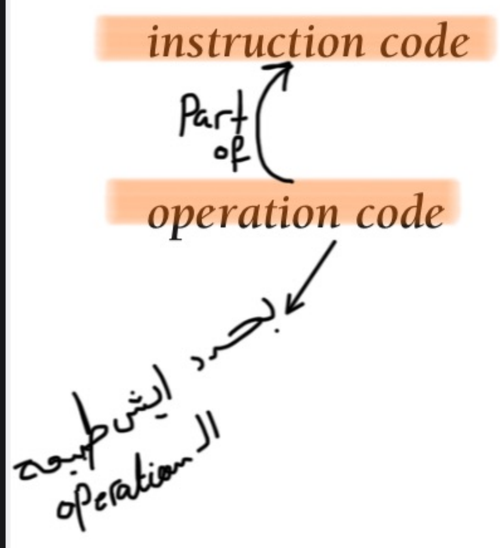
operands, and the sequence by which processing has to occur. The data-processing task may be altered by specifying a new program with different instructions or specifying the same instructions with different data.

A computer instruction is a binary code that specifies a sequence of microoperations for the computer. Instruction codes together with data are stored in memory. The computer reads each instruction from memory and places it in a control register. The control then interprets the binary code of the instruction and proceeds to execute it by issuing a sequence of microoperations. Every computer has its own unique instruction set. The ability to store and execute instructions, the stored program concept, is the most important property of a general-purpose computer.



An instruction code is a group of bits that instruct the computer to perform a specific operation. It is usually divided into parts, each having its own particular interpretation. The most basic part of an instruction code is its operation part. The operation code of an instruction is a group of bits that define such operations as add, subtract, multiply, shift, and complement. The number of bits required for the operation code of an instruction depends on the total number of operations available in the computer. The operation code must consist of at least n bits for a given 2^n (or less) distinct operations. As an illustration, consider a computer with 64 distinct operations, one of them being an ADD operation. The operation code consists of six bits, with a bit configuration 110010 assigned to the ADD operation. When this operation code is decoded in the control unit, the computer issues control signals to read an operand from memory and add the operand to a processor register.

At this point we must recognize the relationship between a computer operation and a microoperation. An operation is part of an instruction stored in computer memory. It is a binary code that tells the computer to perform a specific operation. The control unit receives the instruction from memory and interprets the operation code bits. It then issues a sequence of control signals to initiate microoperations in internal computer registers. For every operation code, the control issues a sequence of microoperations needed for the hardware implementation of the specified operation. For this reason, an operation code is sometimes called a macrooperation because it specifies a set of micro-operations.

The operation part of an instruction code specifies the operation to be performed. This operation must be performed on some data stored in processor registers or in memory. An instruction code must therefore specify not only the operation but also the registers or the memory words where the operands are to be found, as well as the register or memory word where the result is to be stored. Memory words can be specified in instruction codes by their address. Processor registers can be specified by assigning to the instruction another binary code of k bits that specifies one of 2^k registers. There are many variations for arranging the binary code of instructions, and each computer has its own particular instruction code format. Instruction code formats

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are conceived by computer designers who specify the architecture of the computer. In this chapter we choose a particular instruction code to explain the basic organization and design of digital computers.

Stored Program Organization

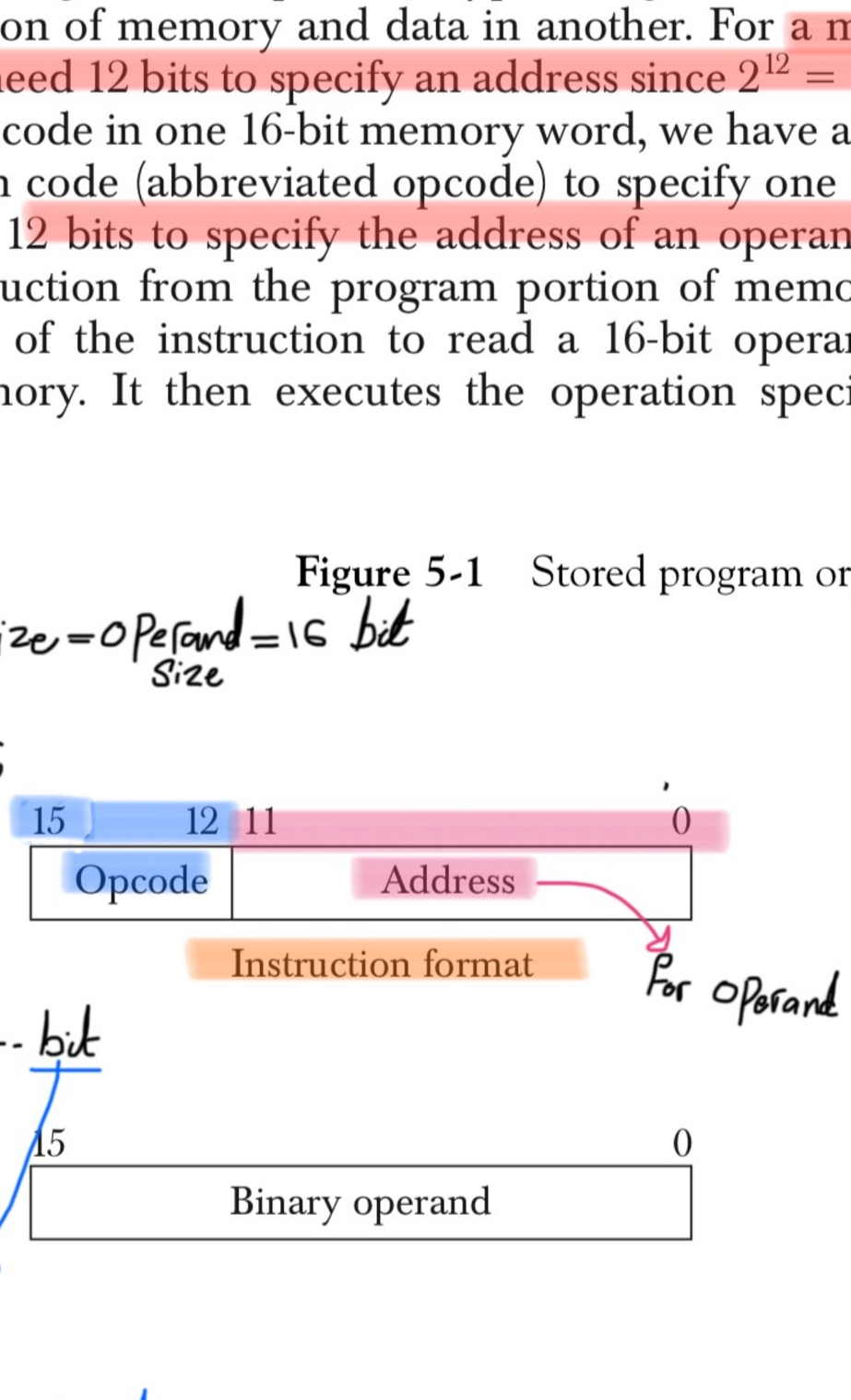
The simplest way to organize a computer is to have one processor register and an instruction code format with two parts. The first part specifies the operation to be performed and the second specifies an address. The memory address tells the control where to find an operand in memory. This operand is read from memory and used as the data to be operated on together with the data stored in the processor register.

Figure 5-1 depicts this type of organization. Instructions are stored in one section of memory and data in another. For a memory unit with 4096 words we need 12 bits to specify an address since $2^{12} = 4096$. If we store each instruction code in one 16-bit memory word, we have available four bits for the operation code (abbreviated opcode) to specify one out of 16 possible operations, and 12 bits to specify the address of an operand. The control reads a 16-bit instruction from the program portion of memory. It uses the 12-bit address part of the instruction to read a 16-bit operand from the data portion of memory. It then executes the operation specified by the operation code.

opcode
unique for each instruction

Figure 5-1 Stored program organization.

location size = instruction size = operand size = 16 bit
numbers in the memory = $2^{12} = 4096$
Max number of unique add.
Max memory size = $2^{12} * 16 = \ldots$ bit
locations · size for 1 location
8 تقسم على byte لتحويل
1024 تقسم على kbyte



Max size of instruction set = 2^4 unique combinations (4 ← opcode)

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accumulator (AC)

register الـ الرئيسي

Computers that have a single-processor register usually assign to it the name accumulator and label it *AC*. The operation is performed with the memory operand and the content of *AC*.

If an operation in an instruction code does not need an operand from memory, the rest of the bits in the instruction can be used for other purposes. For example, operations such as clear *AC*, complement *AC*, and increment *AC* operate on data stored in the *AC* register. They do not need an operand from memory. For these types of operations, the second part of the instruction code (bits 0 through 11) is not needed for specifying a memory address and can be used to specify other operations for the computer.

Indirect Address

immediate instruction

It is sometimes convenient to use the address bits of an instruction code not as an address but as the actual operand. When the second part of an instruction code specifies an operand, the instruction is said to have an immediate operand. When the second part specifies the address of an operand, the instruction is said to have a direct address. This is in contrast to a third possibility called indirect address, where the bits in the second part of the instruction designate an address of a memory word in which the address of the operand is found. One bit of the instruction code can be used to distinguish between a direct and an indirect address.

As an illustration of this configuration, consider the instruction code format shown in Fig. 5-2(a). It consists of a 3-bit operation code, a 12-bit address, and an indirect address mode bit designated by *I*. The mode bit is 0 for a direct address and 1 for an indirect address. A direct address instruction is shown in Fig. 5-2(b). It is placed in address 22 in memory. The *I* bit is 0, so the instruction is recognized as a direct address instruction. The opcode specifies an ADD instruction, and the address part is the binary equivalent of 457. The control finds the operand in memory at address 457 and adds it to the content of *AC*. The instruction in address 35 shown in Fig. 5-2(c) has a mode bit $I = 1$. Therefore, it is recognized as an indirect address instruction. The address part is the binary equivalent of 300. The control goes to address 300 to find the address of the operand. The address of the operand in this case is 1350. The operand found in address 1350 is then added to the content of *AC*. The indirect address instruction needs two references to memory to fetch an operand. The first reference is needed to read the address of the operand; the second is for the operand itself. We define the *effective address* to be the address of the operand in a computation-type instruction or the target address in a branch-type instruction. Thus the effective address in the instruction of Fig. 5-2(b) is 457 and in the instruction of Fig 5-2(c) is 1350.

effective address

The direct and indirect addressing modes are used in the computer presented in this chapter. The memory word that holds the address of the operand in an indirect address instruction is used as a pointer to an array of

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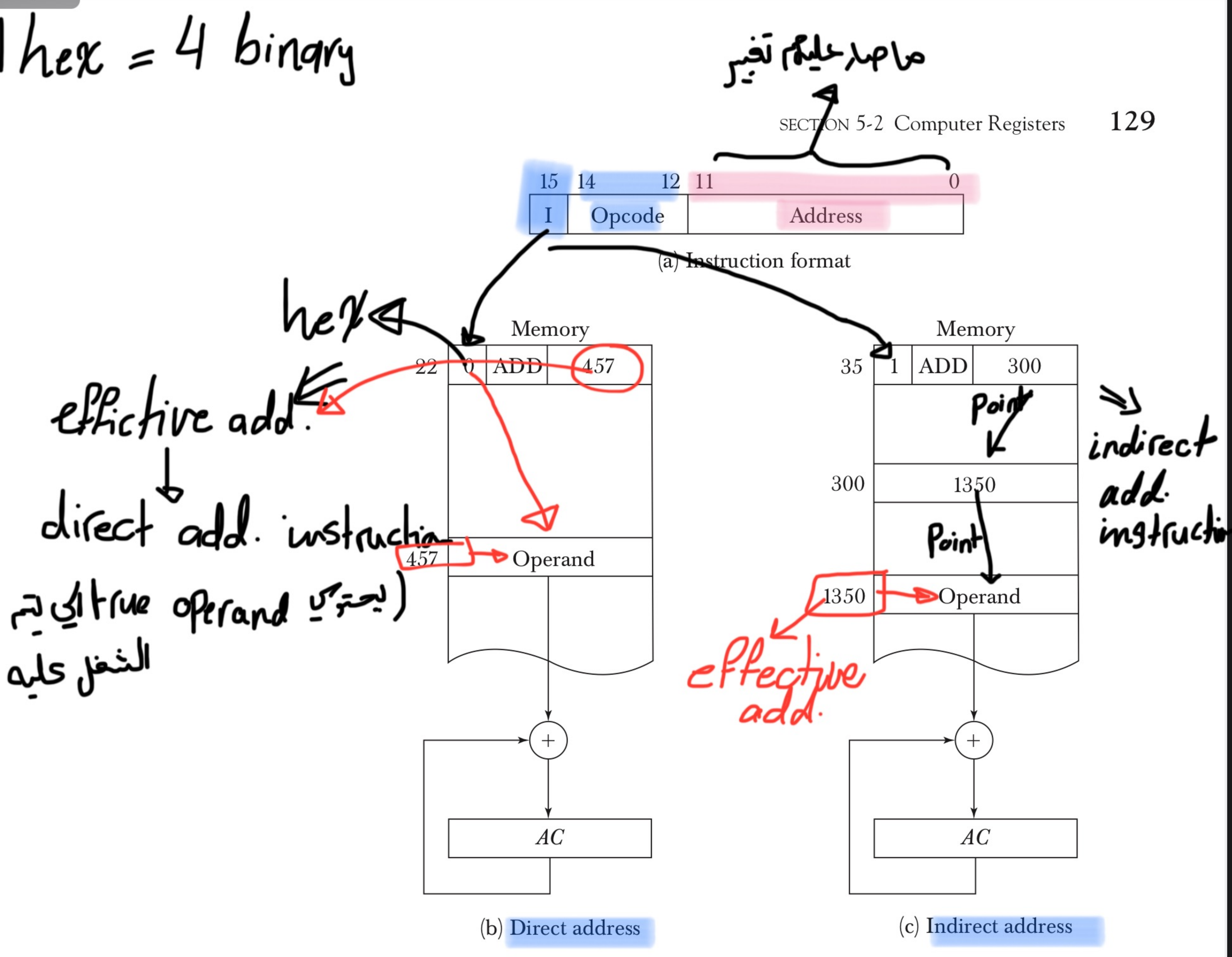

Figure 5-2 Demonstration of direct and indirect address.

data. The pointer could be placed in a processor register instead of memory as done in commercial computers.

5-2 Computer Registers

Computer instructions are normally stored in consecutive memory locations and are executed sequentially one at a time. The control reads an instruction from a specific address in memory and executes it. It then continues by reading the next instruction in sequence and executes it, and so on. This type of instruction sequencing needs a counter to calculate the address of the next instruction after execution of the current instruction is completed. It is also necessary to provide a register in the control unit for storing the instruction code after it is read

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from memory. The computer needs processor registers for manipulating data and a register for holding a memory address. These requirements dictate the register configuration shown in Fig. 5-3. The registers are also listed in Table 5-1 together with a brief description of their function and the number of bits that they contain.

The memory unit has a capacity of 4096 words and each word contains 16 bits. Twelve bits of an instruction word are needed to specify the address of an operand. This leaves three bits for the operation, part of the instruction and a bit to specify a direct or indirect address. The data register (*DR*) holds the operand read from memory. The accumulator (*AC*) register is a general-purpose processing register. The instruction read from memory is placed in the instruction register (*IR*). The temporary register (*TR*) is used for holding temporary data during the processing.

TABLE 5-1 List of Registers for the Basic Computer

Register symbol	Number of bits	Register name	Function
DR	16	Data register	Holds memory operand
AR	12	Address register	Holds address for memory
AC	16	Accumulator	Processor register
IR	16	Instruction register	Holds instruction code
PC	12	Program counter	Holds address of instruction
TR	16	Temporary register	Holds temporary data
INPR	8	Input register	Holds input character
OUTR	8	Output register	Holds output character

program counter (PC)

The memory address register (*AR*) has 12 bits since this is the width of a memory address. The program counter (*PC*) also has 12 bits and it holds the address of the next instruction to be read from memory after the current instruction is executed. The *PC* goes through a counting sequence and causes the computer to read sequential instructions previously stored in memory. Instruction words are read and executed in sequence unless a branch instruction is encountered. A branch instruction calls for a transfer to a nonconsecutive instruction in the program. The address part of a branch instruction is transferred to *PC* to become the address of the next instruction. To read an instruction, the content of *PC* is taken as the address for memory and a memory read cycle is initiated. *PC* is then incremented by one, so it holds the address of the next instruction in sequence.

Two registers are used for input and output. The input register (*INPR*) receives an 8-bit character from an input device. The output register (*OUTR*) holds an 8-bit character for an output device.

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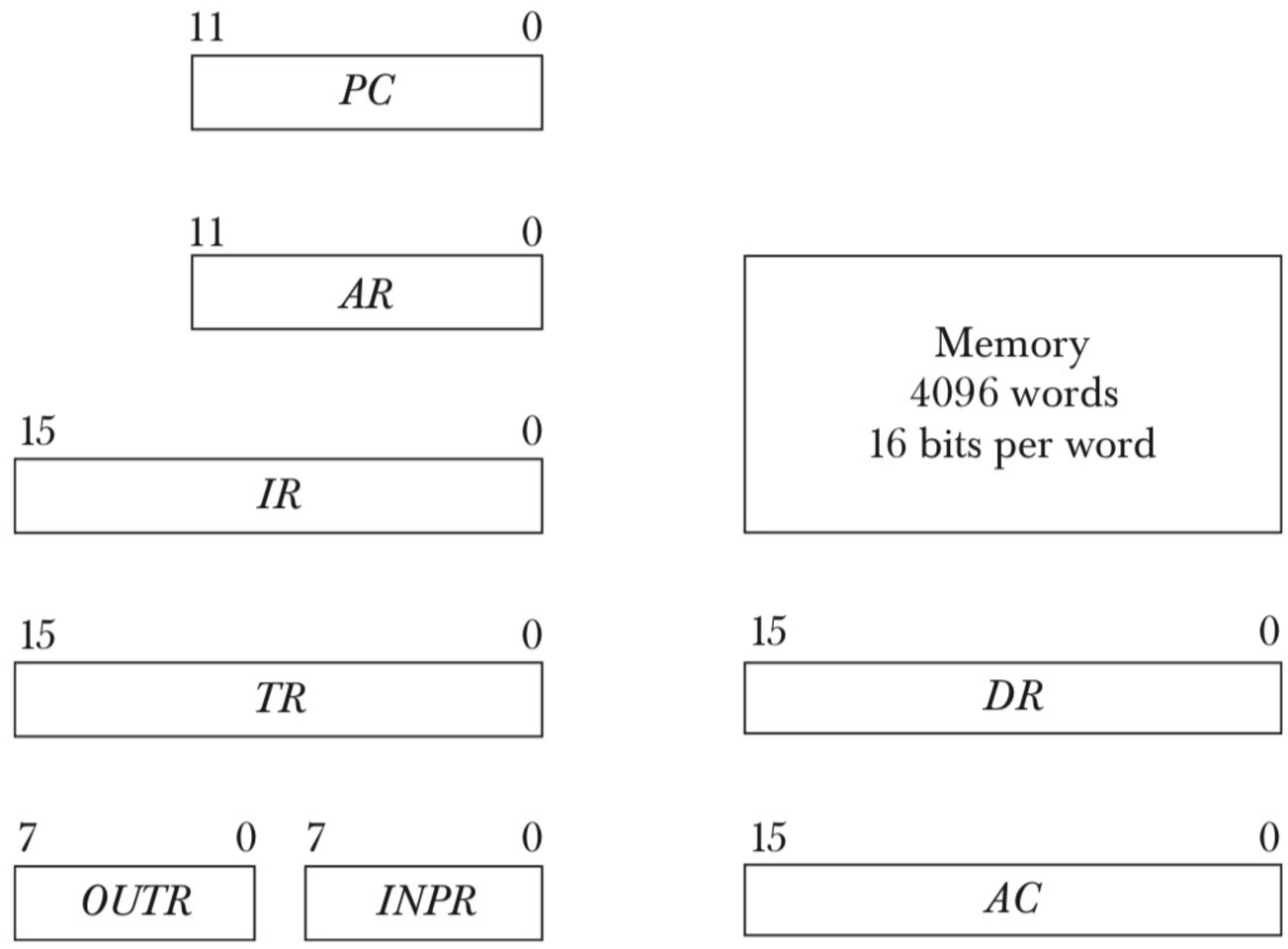


Figure 5-3 Basic computer registers and memory.

Common Bus System

The basic computer has eight registers, a memory unit, and a control unit (to be presented in Sec. 5-4). Paths must be provided to transfer information from one register to another and between memory and registers. The number of wires will be excessive if connections are made between the outputs of each register and the inputs of the other registers. A more efficient scheme for transferring information in a system with many registers is to use a common bus. We have shown in Sec. 4-3 how to construct a bus system using multiplexers or three-state buffer gates. The connection of the registers and memory of the basic computer to a common bus system is shown in Fig. 5-4.

The outputs of seven registers and memory are connected to the common bus. The specific output that is selected for the bus lines at any given time is determined from the binary value of the selection variables S_2, S_1, and S_0. The number along each output shows the decimal equivalent of the required binary selection. For example, the number along the output of *DR* is 3. The 16-bit outputs of *DR* are placed on the bus lines when $S_2S_1S_0 = 011$ since this is the binary value of decimal 3. The lines from the common bus are connected to the inputs of each register and the data inputs of the memory. The particular register whose LD (load) input is enabled receives the data from the bus during the next clock pulse transition. The memory receives the contents of the bus when its write input is activated. The memory places its 16-bit output onto the bus when the read input is activated and $S_2S_1S_0 = 111$.

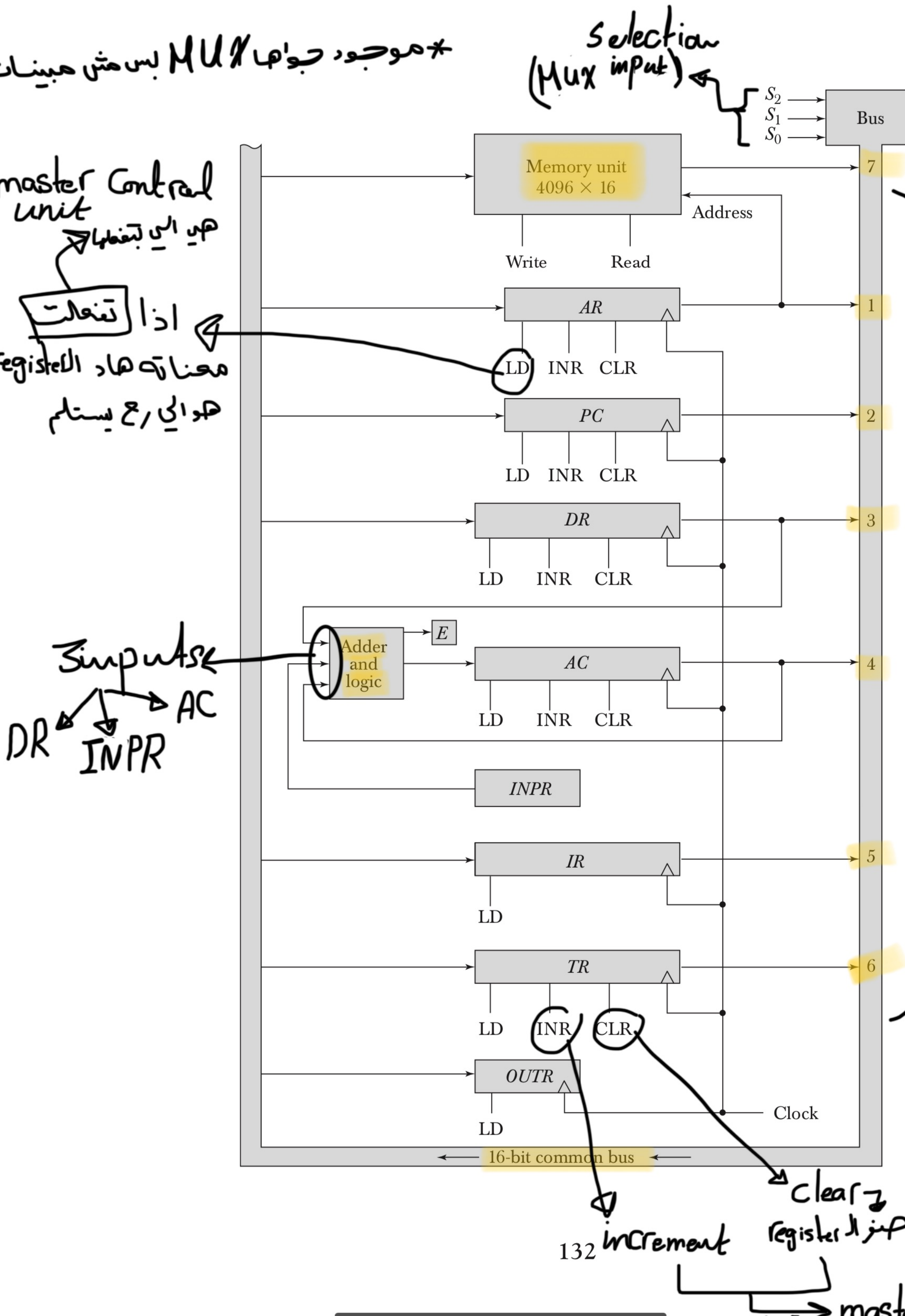
load (LD)

*موجود جواها MUX بس هش هبينات

selection (Mux input)

master Control unit هي اللي بتفعلها

اذا تفعلت registerال هاد معناته هاد هوالي رح يستلم



Code for each register

⇓

Compination of the 3 Selection

Simputse

AC

DR

INPR

clear

register ال جوز

132 increment

master Control

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Four registers, *DR*, *AC*, *IR*, and *TR*, have 16 bits each. Two registers, *AR* and *PC*, have 12 bits each since they hold a memory address. When the contents of *AR* or *PC* are applied to the 16-bit common bus, the four most significant bits are set to 0's. When *AR* or *PC* receive information from the bus, only the 12 least significant bits are transferred into the register.

The input register *INPR* and the output register *OUTR* have 8 bits each and communicate with the eight least significant bits in the bus. *INPR* is connected to provide information to the bus but *OUTR* can only receive information from the bus. This is because *INPR* receives a character from an input device which is then transferred to *AC*. *OUTR* receives a character from *AC* and delivers it to an output device. There is no transfer from *OUTR* to any of the other registers.

The 16 lines of the common bus receive information from six registers and the memory unit. The bus lines are connected to the inputs of six registers and the memory. Five registers have three control inputs: LD (load), INR (increment), and CLR (clear). This type of register is equivalent to a binary counter with parallel load and synchronous clear similar to the one shown in Fig. 2-11. The increment operation is achieved by enabling the count input of the counter. Two registers have only a LD input. This type of register is shown in Fig. 2-7.

memory address

The input data and output data of the memory are connected to the common bus, but the memory address is connected to *AR*. Therefore, *AR* must always be used to specify a memory address. By using a single register for the address, we eliminate the need for an address bus that would have been needed otherwise. The content of any register can be specified for the memory data input during a write operation. Similarly, any register can receive the data from memory after a read operation except *AC*.

The 16 inputs of *AC* come from an adder and logic circuit. This circuit has three sets of inputs. One set of 16-bit inputs come from the outputs of *AC*. They are used to implement register microoperations such as complement *AC* and shift *AC*. Another set of 16-bit inputs come from the data regisler *DR*. The inputs from *DR* and *AC* are used for arithmetic and logic microoperations, such as add *DR* to *AC* or AND *DR* to *AC*. The result of an addition is transferred to *AC* and the end carry-out of the addition is transferred to flip-flop *E* (extended *AC* bit). A third set of 8-bit inputs come from the input register *INPR*. The operation of *INPR* and *OUTR* is explained in Sec. 5-7.

Note that the content of any register can be applied onto the bus and an operation can be performed in the adder and logic circuit during the same clock cycle. The clock transition at the end of the cycle transfers the content of the bus into the designated destination register and the output of the adder and logic circuit into *AC*. For example, the two microoperations

$$DR \leftarrow AC \quad \text{and} \quad AC \leftarrow DR$$

can be executed at the same time. This can be done by placing the content of *AC* on the bus (with $S_2S_1S_0 = 100$), enabling the LD (load) input of *DR*,

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transferring the content of *DR* through the adder and logic circuit into *AC*, and enabling the LD (load) input of *AC*, all during the same clock cycle. The two transfers occur upon the arrival of the clock pulse transition at the end of the clock cycle.

5-3 Computer Instructions

Instruction format

The basic computer has three instruction code formats, as shown in Fig. 5-5. Each format has 16 bits. The operation code (opcode) part of the instruction contains three bits and the meaning of the remaining 13 bits depends on the operation code encountered. A memory-reference instruction uses 12 bits to specify an address and one bit to specify the addressing mode *I*. *I* is equal to 0 for direct address and to 1 for indirect address (see Fig. 5-2). The register-reference instructions are recognized by the operation code 111 with a 0 in the leftmost bit (bit 15) of the instruction. A register-reference instruction specifies an operation on or a test of the *AC* register. An operand from memory is not needed; therefore, the other 12 bits are used to specify the operation or test to be executed. Similarly, an input–output instruction does not need a reference to memory and is recognized by the operation code 111 with a 1 in the left-most bit of the instruction. The remaining 12 bits are used to specify the type of input–output operation or test performed.

The type of instruction is recognized by the computer control from the four bits in positions 12 through 15 of the instruction. If the three opcode bits in positions 12 though 14 are not equal to 111, the instruction is a memory-reference type and the bit in position 15 is taken as the addressing mode *I*. If the 3-bit opcode is equal to 111, control then inspects the bit in position 15. If

Figure 5-5 Basic computer instruction formats.

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this bit is 0, the instruction is a register-reference type. If the bit is 1, the instruction is an input–output type. Note that the bit in position 15 of the instruction code is designated by the symbol *I* but is not used as a mode bit when the operation code is equal to 111.

Only three bits of the instruction are used for the operation code. It may seem that the computer is restricted to a maximum of eight distinct operations. However, since register-reference and input–output instructions use the remaining 12 bits as part of the operation code, the total number of instructions can exceed eight. In fact, the total number of instructions chosen for the basic computer is equal to 25.

The instructions for the computer are listed in Table 5-2. The symbol designation is a three-letter word and represents an abbreviation intended for

TABLE 5-2 Basic Computer Instructions

	Hexadecimal code		
Symbol	$I = 0$	$I = 1$	Description
AND	0xxx	8xxx	AND memory word to *AC*
ADD	lxxx	9xxx	Add memory word to *AC*
LDA	2xxx	Axxx	Load memory word to *AC*
STA	3xxx	Bxxx	Store content of *AC* in memory
BUN	4xxx	Cxxx	Branch unconditionally
BSA	5xxx	Dxxx	Branch and save return address
ISZ	6xxx	Exxx	Increment and skip if zero
CLA	7800		Clear *AC*
CLE	7400		Clear *E*
CMA	7200		Complement *AC*
CME	7100		Complement *E*
CIR	7080		Circulate right *AC* and *E*
CIL	7040		Circulate left *AC* and *E*
INC	7020		Increment *AC*
SPA	7010		Skip next instruction if *AC* positive
SNA	7008		Skip next instruction if *AC* negative
SZA	7004		Skip next instruction if *AC* zero
SZE	7002		Skip next instruction if *E* is 0
HLT	7001		Halt computer
INP	F800		Input character to *AC*
OUT	F400		Output character from *AC*
SKI	F200		Skip on input flag
SKO	F100		Skip on output flag
ION	F080		Interrupt on
IOF	F040		Interrupt off

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user. Programs and data must be transferred into memory and results of computations must be transferred back to the user.

The instructions listed in Table 5-2 constitute a minimum set that provides all the capabilities mentioned above. There is one arithmetic instruction, ADD, and two related instructions, complement *AC*(CMA) and increment *AC*(INC). With these three instructions we can add and subtract binary numbers when negative numbers are in signed-2's complement representation. The circulate instructions, CIR and CIL, can be used for arithmetic shifts as well as any other type of shifts desired. Multiplication and division can be performed using addition, subtraction, and shifting. There are three logic operations: AND, complement *AC*(CMA), and clear *AC*(CLA). The AND and complement provide a NAND operation. It can be shown that with the NAND operation it is possible to implement all the other logic operations with two variables (listed in Table 4-6). Moving information from memory to *AC* is accomplished with the load *AC*(LDA) instruction. Storing information from *AC* into memory is done with the store *AC*(STA) instruction. The branch instructions BUN, BSA, and ISZ, together with the four skip instructions, provide capabilities for program control and checking of status conditions. The input (INP) and output (OUT) instructions cause information to be transferred between the computer and external devices.

Although the set of instructions for the basic computer is complete, it is not efficient because frequently used operations are not performed rapidly. An efficient set of instructions will include such instructions as subtract, multiply, OR, and exclusive-OR. These operations must be programmed in the basic computer. The programs are presented in Chap. 6 together with other programming examples for the basic computer. By using a limited number of instructions it is possible to show the detailed logic design of the computer. A more complete set of instructions would have made the design too complex. In this way we can demonstrate the basic principles of computer organization and design without going into excessive complex details. In Chap. 8 we present a complete list of computer instructions that are included in most commercial computers.

The function of each instruction listed in Table 5-2 and the microoperations needed for their execution are presented in Secs. 5-5 through 5-7 We delay this discussion because we must first consider the control unit and understand its internal organization.

5-4 Timing and Control

clock pulses

The timing for all registers in the basic computer is controlled by a master clock generator. The clock pulses are applied to all flip-flops and registers in the system, including the flip-flops and registers in the control unit. The clock pulses do not change the state of a register unless the register is enabled by a

1 Program is a sequence of instructions

1 instruction " " " " microoperations

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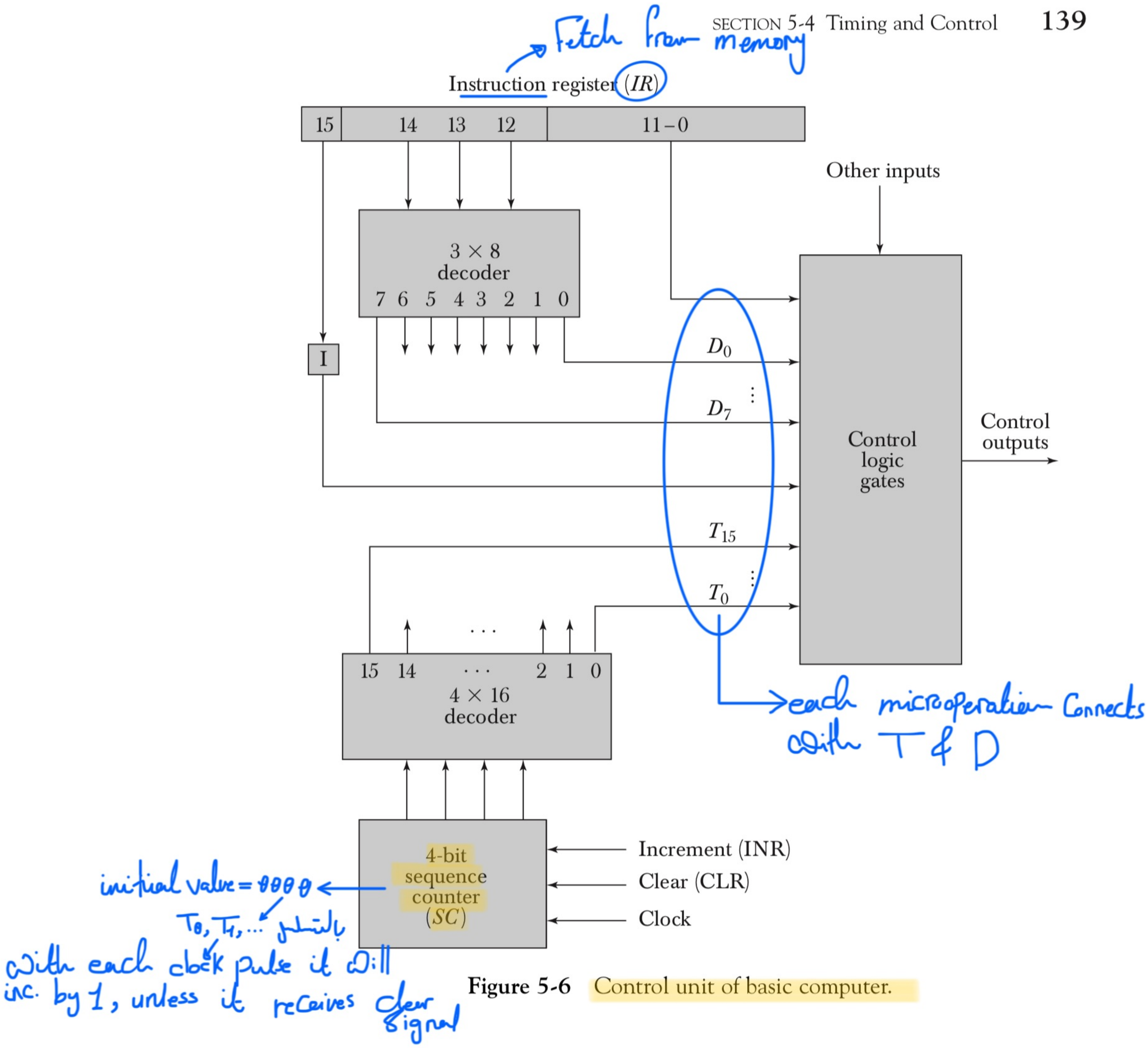

Figure 5-6 Control unit of basic computer.

clears SC to 0, which in turn activates the timing signal T_0 out of the decoder. T_0 is active during one clock cycle. The positive clock transition labeled T_0 in the diagram will trigger only those registers whose control inputs are connected to timing signal T_0. SC is incremented with every positive clock transition, unless its CLR input is active. This produces the sequence of timing signals T_0, T_1, T_2, T_3, T_4, and so on, as shown in the diagram. (Note the relationship between the timing signal and its corresponding positive clock transition.) If SC is not cleared, the timing signals will continue with T_5, T_6, up to T_{15} and back to T_0.

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processor until the memory word is available. To facilitate the presentation, we will assume that a wait period is not necessary in the basic computer.

To fully comprehend the operation of the computer, it is crucial that one understands the timing relationship between the clock transition and the timing signals. For example, the register transfer statement

$$T_0: \quad AR \leftarrow PC$$

specifies a transfer of the content of PC into AR if timing signal T_0 is active. T_0 is active during an entire clock cycle interval. During this time the content of PC is placed onto the bus (with $S_2S_1S_0 = 010$) and the LD (load) input of AR is enabled. The actual transfer does not occur until the end of the clock cycle when the clock goes through a positive transition. This same positive clock transition increments the sequence counter SC from 0000 to 0001. The next clock cycle has T_1 active and T_0 inactive.

5-5 Instruction Cycle

A program residing in the memory unit of the computer consists of a sequence of instructions. The program is executed in the computer by going through a cycle for each instruction. Each instruction cycle in turn is subdivided into a sequence of subcycles or phases. In the basic computer each instruction cycle consists of the following phases:

1. Fetch an instruction from memory.
2. Decode the instruction.
3. Read the effective address from memory if the instruction has an indirect address.
4. Execute the instruction.

Upon the completion of step 4, the control goes back to step 1 to fetch, decode, and execute the next instruction. This process continues indefinitely unless a HALT instruction is encountered.

Fetch and Decode

Initially, the program counter PC is loaded with the address of the first instruction in the program. The sequence counter SC is cleared to 0, providing a decoded timing signal T_0. After each clock pulse, SC is incremented by one, so that the timing signals go through a sequence T_0, T_1, T_2, and so on. The microoperations for the fetch and decode phases can be specified by the following register transfer statements.

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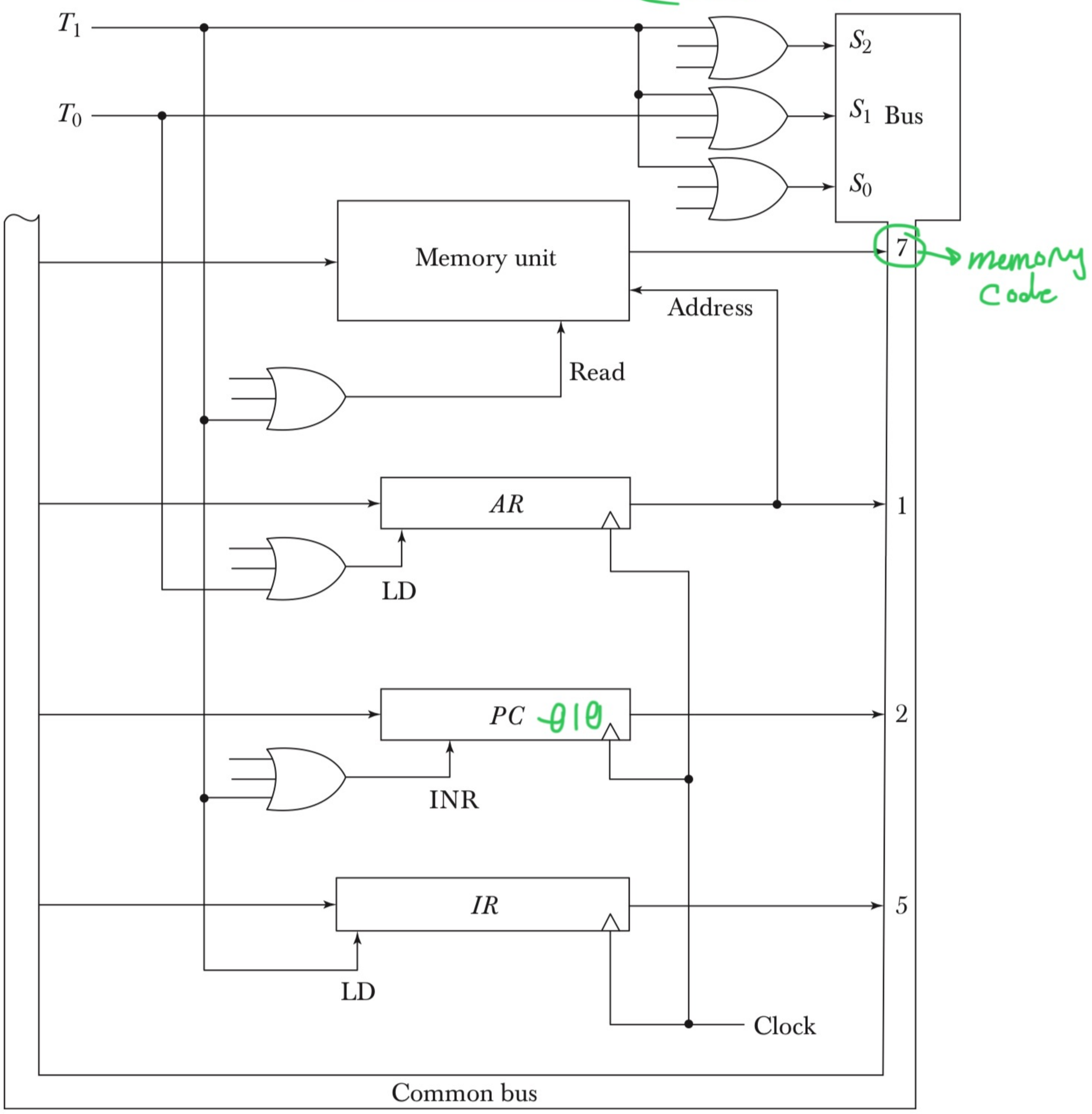
$$T_0: \quad AR \leftarrow PC$$
$$T_1: \quad IR \leftarrow M[AR],\ PC \leftarrow PC + 1$$
$$T_2: \quad D_0, \ldots, D_7 \leftarrow \text{Decode } IR(12\text{–}14),\ AR \leftarrow IR(0\text{–}11),\ I \leftarrow IR(15)$$

Since only *AR* is connected to the address inputs of memory, it is necessary to transfer the address from *PC* to *AR* during the clock transition associated with timing signal T_0. The instruction read from memory is then placed in the instruction register *IR* with the clock transition associated with timing

Figure 5-8 Register transfers for the fetch phase.



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instructions are presented in this section. The memory-reference instructions are explained in the next section. The input-output instructions are included in Sec. 5-7.

Register-Reference Instructions

Register-reference instructions are recognized by the control when $D_7 = 1$ and $I = 0$. These instructions use bits 0 through 11 of the instruction code to specify one of 12 instructions. These 12 bits are available in IR (0–11). They were also transferred to AR during time T_2.

The control functions and microoperations for the register-reference instructions are listed in Table 5-3. These instructions are executed with the

instructions → memory ref. → OPcode in bits 12,13,14 ≠ 1
→ reg. ref → " " " " " " =1 , bit 16 (index 15) = 0 ← الاخير
→ I/O → 4 bits ≥1 = 1

clock transition associated with timing variable T_3. Each control function needs the Boolean relation $D_7 I' T_3$, which we designate for convenience by the symbol r. The control function is distinguished by one of the bits in IR(0–11). By assigning the symbol B_i to bit i of IR, all control functions can be simply denoted by rB_i. For example, the instruction CLA has the hexadecimal code 7800 (see Table 5-2), which gives the binary equivalent 0111 1000 0000 0000. The first bit is a zero and is equivalent to I'. The next three bits constitute the operation code and are recognized from decoder output D_7. Bit 11 in IR is 1 and is recognized from B_{11}. The control function that initiates the microoperation for this instruction is $D_7 I' T_3 B_{11} = rB_{11}$. The execution of a register-reference instruction is completed at time T_3. The sequence counter SC is cleared to 0 and the control goes back to fetch the next instruction with timing signal T_0.

The first seven register-reference instructions perform clear, complement, circular shift, and increment microoperations on the AC or E registers. The next four instructions cause a skip of the next instruction in sequence when a stated condition is satisfied. The skipping of the instruction is achieved by incrementing PC once again (in addition, it is being incremented during the fetch phase at time T_1). The condition control statements must be recognized as part of the control conditions. The AC is positive when the sign bit in AC (15) = 0; it is negative when $AC(15) = 1$. The content of AC is zero ($AC = 0$) if all the flip-flops of the register are zero. The HLT instruction clears a start-stop flip-flop S and stops the sequence counter from counting. To restore the operation of the computer, the start–stop flip-flop must be set manually.

TABLE 5-3 Execution of Register-Reference Instructions

$D_7 I' T_3 = r$ (common to all registcr-reference instructions)
$IR(i) = B_i$ [bit in IR(0–11) that specifies the operation]

	r:	$SC \leftarrow 0$	Clear SC
CLA	rB_{11}:	$AC \leftarrow 0$	Clear AC
CLE	rB_{10}:	$E \leftarrow 0$	Clear E

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5-6 Memory-Reference Instructions

In order to specify the microoperations needed for the execution of each instruction, it is necessary that the function that they are intended to perform be defined precisely. Looking back to Table 5-2, where the instructions are listed, we find that some instructions have an ambiguous description. This is because the explanation of an instruction in words is usually lengthy, and not enough space is available in the table for such a lengthy explanation. We will now show that the function of the memory-reference instructions can be defined precisely by means of register transfer notation.

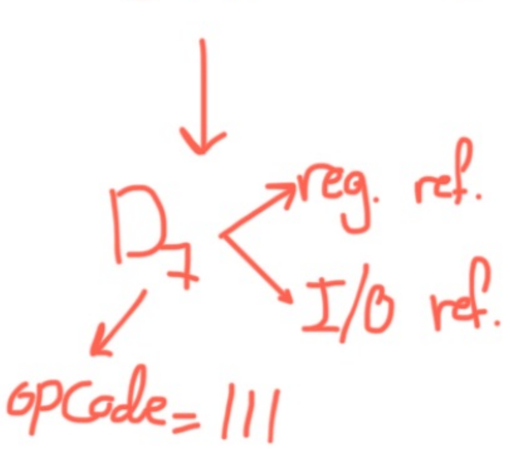
effective address

Table 5-4 lists the seven memory-reference instructions. The decoded output D_i for $i = 0, 1, 2, 3, 4, 5$, and 6 from the operation decoder that belongs to each instruction is included in the table. The effective address of the instruction is in the address register AR and was placed there during timing signal T_2 when $I = 0$, or during timing signal T_3 when $I = 1$. The execution of the memory-reference instructions starts with timing signal T_4. The symbolic description of each instruction is specified in the table in terms of register transfer notation. The actual execution of the instruction in the bus system will require a sequence of microoperations. This is because data stored in memory cannot be processed directly. The data must be read from memory to a register where they can be operated on with logic circuits. We now explain the operation of each instruction and list the control functions and microoperations needed for their execution. A flowchart that summarizes all the microoperations is presented at the end of this section.

TABLE 5-4 Memory-Reference Instructions

Symbol	Operation decoder	Symbolic description
AND	D_0	$AC \leftarrow AC \wedge M[AR]$
ADD	D_1	$AC \leftarrow AC + M[AR]$, $E \leftarrow C_{out}$
LDA	D_2	$AC \leftarrow M[AR]$
STA	D_3	$M[AR] \leftarrow AC$
BUN	D_4	$PC \leftarrow AR$
BSA	D_5	$M[AR] \leftarrow PC$, $PC \leftarrow AR + 1$
ISZ	D_6	$M[AR] \leftarrow M[AR] + 1$, If $M[AR] + 1 = 0$ then $PC \leftarrow PC + 1$

memory ref. تكون مارح D7 ما في ← لو في D7 ←



AND to AC

This is an instruction that performs the AND logic operation on pairs of bits in AC and the memory word specified by the effective address. The result of

the operation is transferred to *AC*. The microoperations that execute this instruction are:

$$D_0T_4: \quad DR \leftarrow M[AR]$$

$$D_0T_5: \quad AC \leftarrow AC \wedge DR, \quad SC \leftarrow 0$$

The control function for this instruction uses the operation decoder D_0 since this output of the decoder is active when the instruction has an AND operation whose binary code value is 000. Two timing signals are needed to execute the instruction. The clock transition associated with timing signal T_4 transfers the operand from memory into *DR*. The clock transition associated with the next timing signal T_5 transfers to *AC* the result of the AND logic operation between the contents of *DR* and *AC*. The same clock transition clears *SC* to 0, transferring control to timing signal T_0 to start a new instruction cycle.

ADD to AC

This instruction adds the content of the memory word specified by the effective address to the value of *AC*. The sum is transferred into *AC* and the output carry C_{out} is transferred to the *E* (extended accumulator) flip-flop. The microoperations needed to execute this instruction are

$$D_1T_4: \quad DR \leftarrow M[AR]$$

$$D_1T_5: \quad AC \leftarrow AC + DR, \quad E \leftarrow C_{out}, \quad SC \leftarrow 0$$

The same two timing signals, T_4 and T_5, are used again but with operation decoder D_1 instead of D_0, which was used for the AND instruction. After the instruction is fetched from memory and decoded, only one output of the operation decoder will be active, and that output determines the sequence of microoperations that the control follows during the execution of a memory-reference instruction.

LDA: Load to AC

This instruction transfers the memory word specified by the effective address to *AC*. The microoperations needed to execute this instruction are

$$D_2T_4: \quad DR \leftarrow M[AR]$$

$$D_2T_5: \quad AC \leftarrow DR, \quad SC \leftarrow 0$$

Looking back at the bus system shown in Fig. 5-4 we note that there is no direct path from the bus into *AC*. The adder and logic circuit receive information

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from *DR* which can be transferred into *AC*. Therefore, it is necessary to read the memory word into *DR* first and then transfer the content of *DR* into *AC*. The reason for not connecting the bus to the inputs of *AC* is the delay encountered in the adder and logic circuit. It is assumed that the time it takes to read from memory and transfer the word through the bus as well as the adder and logic circuit is more than the time of one clock cycle. By not connecting the bus to the inputs of *AC* we can maintain one clock cycle per microoperation.

STA: Store AC

This instruction stores the content of *AC* into the memory word specified by the effective address. Since the output of *AC* is applied to the bus and the data input of memory is connected to the bus, we can execute this instruction with one microoperation:

$$D_3T_4: \quad M[AR] \leftarrow AC, \quad SC \leftarrow 0$$

BUN: Branch Unconditionally

This instruction transfers the program to the instruction specified by the effective address. Remember that *PC* holds the address of the instruction to be read from memory in the next instruction cycle. *PC* is incremented at time T_1 to prepare it for the address of the next instruction in the program sequence. The BUN instruction allows the programmer to specify an instruction out of sequence and we say that the program branches (or jumps) unconditionally. The instruction is executed with one microoperation:

$$D_4T_4: \quad PC \leftarrow AR, \quad SC \leftarrow 0$$

The effective address from *AR* is transferred through the common bus to *PC*. Resetting *SC* to 0 transfers control to T_0. The next instruction, is then fetched and executed from the memory address given by the new value in *PC*.

BSA: Branch and Save Return Address

This instruction is useful for branching to a portion of the program called a subroutine or procedure. When executed, the BSA instruction stores the address of the next instruction in sequence (which is available in *PC*) into a memory location specified by the effective address. The effective address plus one is then transferred to *PC* to serve as the address of the first instruction in the subroutine. This operation was specified in Table 5-4 with the following register transfer:

$$M[AR] \leftarrow PC, \quad PC \leftarrow AR + 1$$

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It is not possible to perform the operation of the BSA instruction in one clock cycle when we use the bus system of the basic computer. To use the memory and the bus properly, the BSA instruction must be executed with a sequence of two microoperations:

$$D_5T_4: \quad M[AR] \leftarrow PC, \quad AR \leftarrow AR + 1$$
$$D_5T_5: \quad PC \leftarrow AR, \quad SC \leftarrow 0$$

Timing signal T_4 initiates a memory write operation, places the content of PC onto the bus, and enables the INR input of AR. The memory write operation is completed and AR is incremented by the time the next clock transition occurs. The bus is used at T_5 to transfer the content of AR to PC.

ISZ: Increment and Skip if Zero

This instruction increments the word specified by the effective address, and if the incremented value is equal to 0, PC is incremented by 1. The programmer usually stores a negative number (in 2's complement) in the memory word. As this negative number is repeatedly incremented by one, it eventually reaches the value of zero. At that time PC is incremented by one in order to skip the next instruction in the program.

Since it is not possible to increment a word inside the memory, it is necessary to read the word into DR, increment DR, and store the word back into memory. This is done with the following sequence of microoperations:

$$D_6T_4: \quad DR \leftarrow M[AR]$$
$$D_6T_5: \quad DR \leftarrow DR + 1$$
$$D_6T_6: \quad M[AR] \leftarrow DR, \quad \text{if } (DR = 0) \text{ then } (PC \leftarrow PC + 1), \quad SC \leftarrow 0$$

Control Flowchart

A flowchart showing all microoperations for the execution of the seven memory-reference instructions is shown in Fig. 5-11. The control functions are indicated on top of each box. The microoperations that are performed during time T_4, T_5, or T_6 depend on the operation code value. This is indicated in the flowchart by six different paths, one of which the control takes after the instruction is decoded. The sequence counter SC is cleared to 0 with the last timing signal in each case. This causes a transfer of control to timing signal T_0 to start the next instruction cycle.

Note that we need only seven timing signals to execute the longest instruction (ISZ). The computer can be designed with a 3-bit sequence counter. The reason for using a 4-bit counter for SC is to provide additional timing signals for other instructions that are presented in the problems section.

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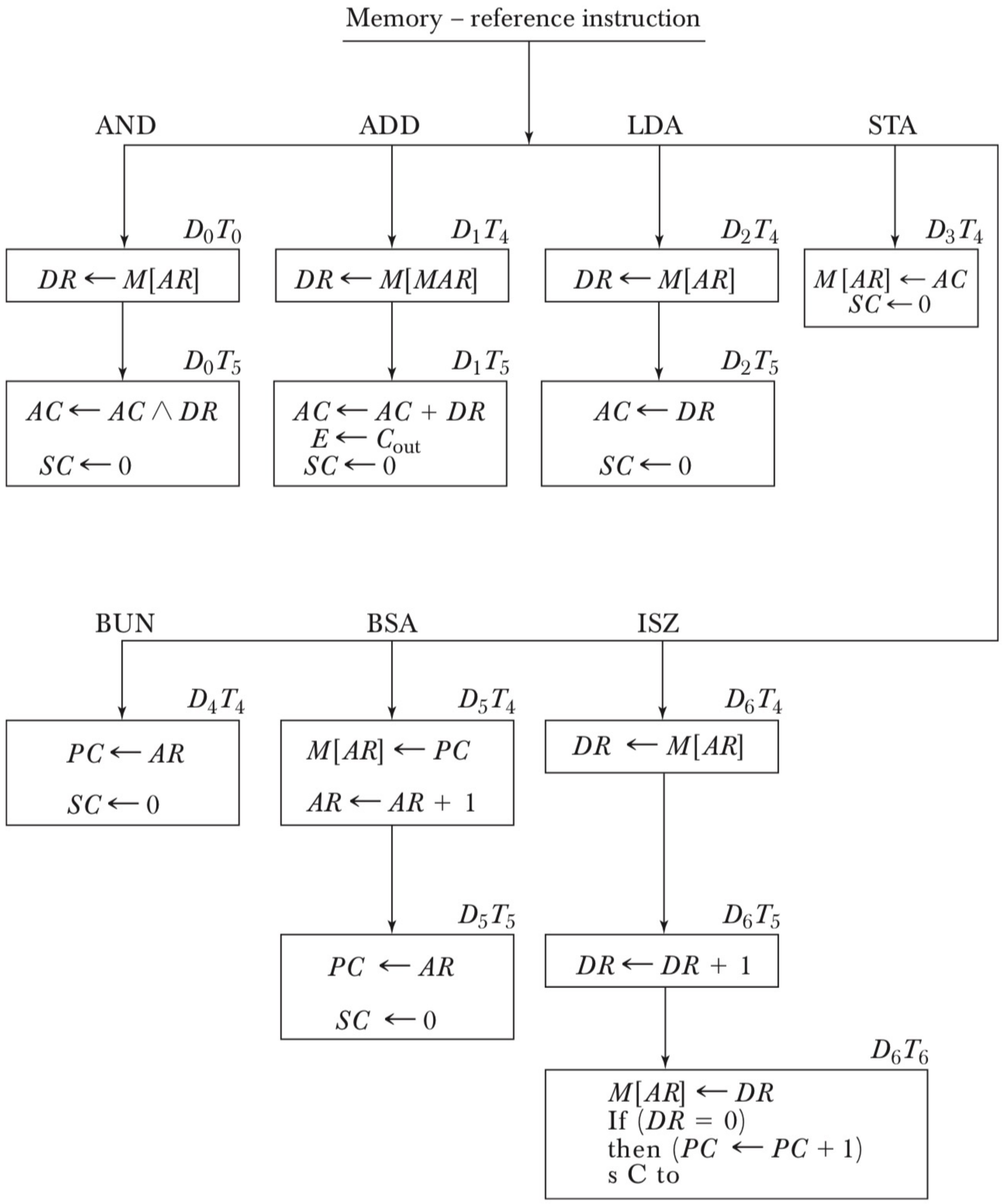

Figure 5-11 Flowchart for memory-reference instructions.

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interface

5-7 Input–Output and Interrupt

A computer can serve no useful purpose unless it communicates with the external environment. Instructions and data stored in memory must come from some input device. Computational results must be transmitted to the user through some output device. Commercial computers include many types

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of input and output devices. To demonstrate the most basic requirements for input and output communication, we will use as an illustration a terminal unit with a keyboard and printer. Input–output organization is discussed further in Chap. 11.

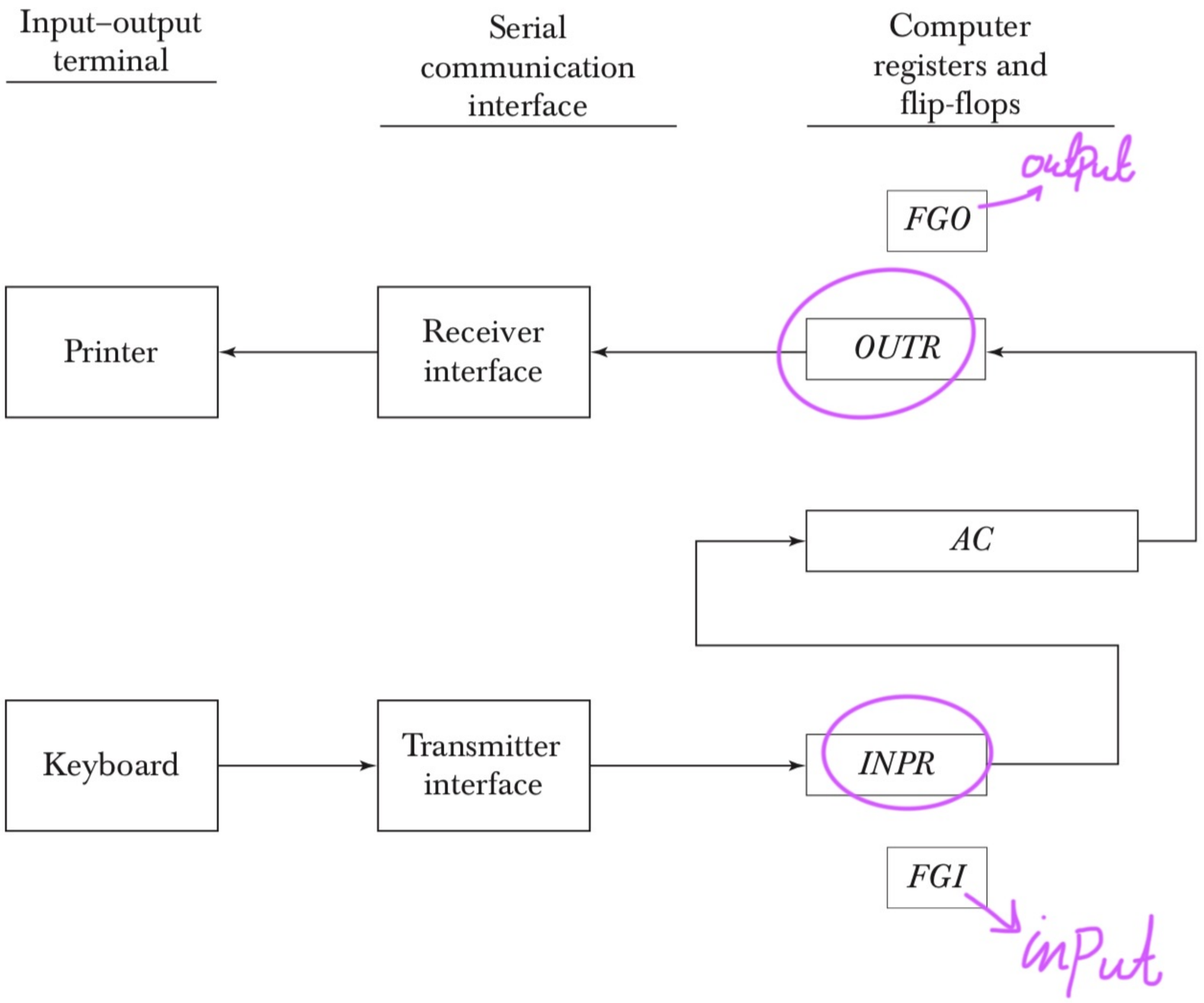
Input–Output Configuration

The terminal sends and receives serial information. Each quantity of information has eight bits of an alphanumeric code. The serial information from the keyboard is shifted into the input register *INPR*. The serial information for the printer is stored in the output register *OUTR*. These two registers communicate with a communication interface serially and with the *AC* in parallel. The input–output configuration is shown in Fig. 5-12. The transmitter interface receives serial information from the keyboard and transmits it to *INPR*. The receiver interface receives information from *OUTR* and sends it to the printer serially. The operation of the serial communication interface is explained in Sec. 11-3.

input register

The input register *INPR* consists of eight bits and holds an alphanumeric input information. The 1-bit input flag *FGI* is a control flip-flop. The flag bit is

Figure 5-12 Input–output configuration.



Flag ← 0 → متاح
1 → مشغول

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set to 1 when new information is available in the input device and is cleared to 0 when the information is accepted by the computer. The flag is needed to synchronize the timing rate difference between the input device and the computer. The process of information transfer is as follows. Initially, the input flag *FGI* is cleared to 0. When a key is struck in the keyboard, an 8-bit alphanumeric code is shifted into *INPR* and the input flag *FGI* is set to 1. As long as the flag is set, the information in *INPR* cannot be changed by striking another key. The computer checks the flag bit; if it is 1, the information from *INPR* is transferred in parallel into *AC* and *FGI* is cleared to 0. Once the flag is cleared, new information can be shifted into *INPR* by striking another key.

output register

The output register *OUTR* works similarly but the direction of information flow is reversed. Initially, the output flag *FGO* is set to 1. The computer checks the flag bit; if it is 1, the information from *AC* is transferred in parallel to *OUTR* and *FGO* is cleared to 0. The output device accepts the coded information, prints the corresponding character, and when the operation is completed, it sets *FGO* to 1. The computer does not load a new character into *OUTR* when *FGO* is 0 because this condition indicates that the output device is in the process of printing the character.

Input–Output Instructions

Input and output instructions are needed for transferring information to and from *AC* register, for checking the flag bits, and for controlling the interrupt facility. Input–output instructions have an operation code 1111 and are recognized by the control when $D_7 = 1$ and $I = 1$. The remaining bits of the instruction specify the particular operation. The control functions and microoperations for the input–output instructions are listed in Table 5-5. These instructions are executed with the clock transition associated with timing signal T_3. Each control function needs a Boolean relation D_7IT_3, which we designate for convenience by the symbol p. The control function is distinguished by one of the bits in $IR(6–11)$. By assigning the symbol B_i to bit i of IR, all control functions can

TABLE 5-5 Input–Output Instructions

$D_7IT_3 = p$ (common to all input–output instructions)
$IR(i) = B_i$ [bit in $IR(6–11)$ that specifies the instruction]

	p:	$SC \leftarrow 0$	Clear *SC*
INP	pB_{11}:	$AC(0–7) \leftarrow INPR$, $FGI \leftarrow 0$	Input character
OUT	pB_{10}:	$OUTR \leftarrow AC(0–7)$, $FGO \leftarrow 0$	Output character
SKI	pB_9:	If $(FGI = 1)$ then $(PC \leftarrow PC + 1)$	Skip on input flag
SKO	pB_8:	If $(FGO = 1)$ then $(PC \leftarrow PC + 1)$	Skip on output flag
ION	pB_7:	$IEN \leftarrow 1$	Interrupt enable on
IOF	pB_6:	$IEN \leftarrow 0$	Interrupt enable off

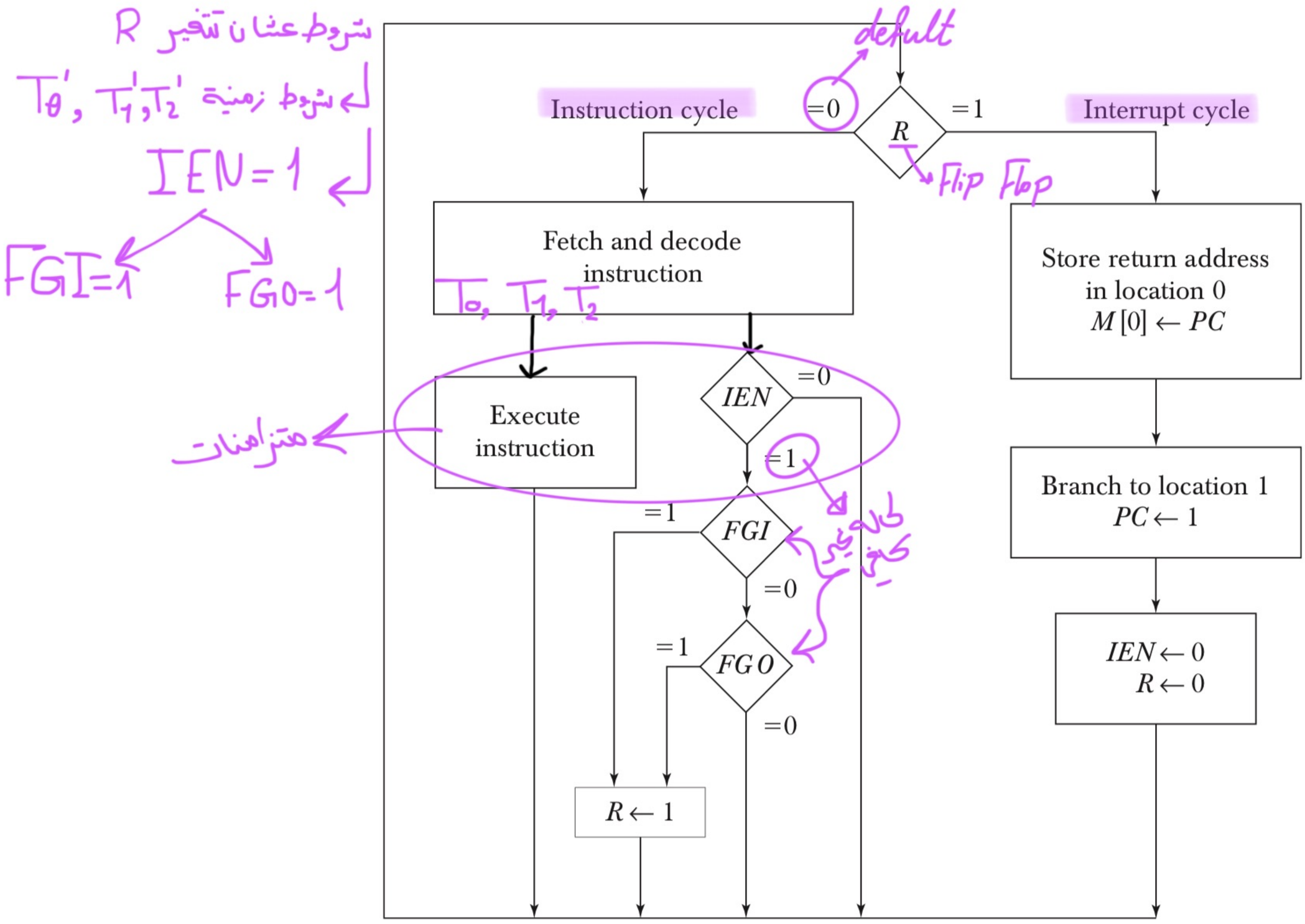

Figure 5-13 Flowchart for interrupt cycle.

The way that the interrupt is handled by the computer can be explained by means of the flowchart of Fig. 5-13. An interrupt flip-flop R is included in the computer. When $R = 0$, the computer goes through an instruction cycle. During the execute phase of the instruction cycle IEN is checked by the control. If it is 0, it indicates that the programmer does not want to use the interrupt, so control continues with the next instruction cycle. If *IEN* is 1, control checks the flag bits. If both flags are 0, it indicates that neither the input nor the output registers are ready for transfer of information. In this case, control continues with the next instruction cycle. If either flag is set to 1 while $IEN = 1$, flip-flop R is set to 1. At the end of the execute phase, control checks the value of R, and if it is equal to 1, it goes to an interrupt cycle instead of an instruction cycle.

interrupt cycle

The interrupt cycle is a hardware implementation of a branch and save return address operation. The return address available in *PC* is stored in a specific location where it can be found later when the program returns to the instruction at which it was interrupted. This location may be a processor

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is read from memory during the fetch phase, control goes to the indirect phase (because $I = 1$) to read the effective address. The effective address is in location 0 and is the return address that was stored there during the previous interrupt cycle. The execution of the indirect BUN instruction results in placing into PC the return address from location 0.

Interrupt Cycle

We are now ready to list the register transfer statements for the interrupt cycle. The interrupt cycle is initiated after the last execute phase if the interrupt flip-flop R is equal to 1. This flip-flop is set to 1 if $IEN = 1$ and either FGI or FGO are equal to 1. This can happen with any clock transition except when timing signals T_0, T_1, or T_2 are active. The condition for setting flip-flop R to 1 can be expressed with the following register transfer statement:

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$$T_0' T_1' T_2' (IEN)(FGI + FGO): \quad R \leftarrow 1$$

AND OR

The symbol + between FGI and FGO in the control function designates a logic OR operation. This is ANDed with IEN and $T_0' T_1' T_2'$.

modified fetch phase

We now modify the fetch and decode phases of the instruction cycle. Instead of using only timing signals T_0, T_1, and T_2 (as shown in Fig. 5-9) we will AND the three timing signals with R' so that the fetch and decode phases will be recognized from the three control functions $R' T_0$, $R' T_1$, and $R' T_2$. The reason for this is that after the instruction is executed and SC is cleared to 0, the control will go through a fetch phase only if $R = 0$. Otherwise, if $R = 1$, the control will go through an interrupt cycle. The interrupt cycle stores the return address (available in PC) into memory location 0, branches to memory location 1, and clears IEN, R, and SC to 0. This can be done with the following sequence of microoperations:

$$\begin{aligned}
RT_0: &\quad AR \leftarrow 0, \quad TR \leftarrow PC \\
RT_1: &\quad M[AR] \leftarrow TR, \quad PC \leftarrow 0 \\
RT_2: &\quad PC \leftarrow PC + 1, \quad IEN \leftarrow 0, \quad R \leftarrow 0, \quad SC \leftarrow 0
\end{aligned}$$

During the first timing signal AR is cleared to 0, and the content of PC is transferred to the temporary register TR. With the second timing signal, the return address is stored in memory at location 0 and PC is cleared to 0. The third timing signal increments PC to 1, clears IEN and R, and control goes back to T_0 by clearing SC to 0. The beginning of the next instruction cycle has the condition $R' T_0$ and the content of PC is equal to 1. The control then goes through an instruction cycle that fetches and executes the BUN instruction in location 1.

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CHAPTER NINE

Pipeline and Vector Processing

IN THIS CHAPTER

9-1 Parallel Processing
9-2 Pipelining
9-3 Arithmetic Pipeline
9-4 Instruction Pipeline
9-5 RISC Pipeline
9-6 Vector Processing
9-7 Array Processors

9-1 Parallel Processing

multiple Func. units ← يشتغلو بنفس الوقت

Parallel processing is a term used to denote a large class of techniques that are used to provide simultaneous data-processing tasks for the purpose of increasing the computational speed of a computer system. Instead of processing each instruction sequentially as in a conventional computer, a parallel processing system is able to perform concurrent data processing to achieve faster execution time. For example, while an instruction is being executed in the ALU, the next instruction can be read from memory. The system may have two or more ALUs and be able to execute two or more instructions at the same time. Furthermore, the system may have two or more processors operating concurrently. The purpose of parallel processing is to speed up the computer processing capability and increase its throughput, that is, the amount of processing that can be accomplished during a given interval of time. The amount of hardware increases with parallel processing, and with it, the cost of the system increases. However, technological developments have reduced hardware costs to the point where parallel processing techniques are economically feasible.

throughput

Parallel processing can be viewed from various levels of complexity. At the lowest level, we distinguish between parallel and serial operations by the type of registers used. Shift registers operate in serial fashion one bit at a time,

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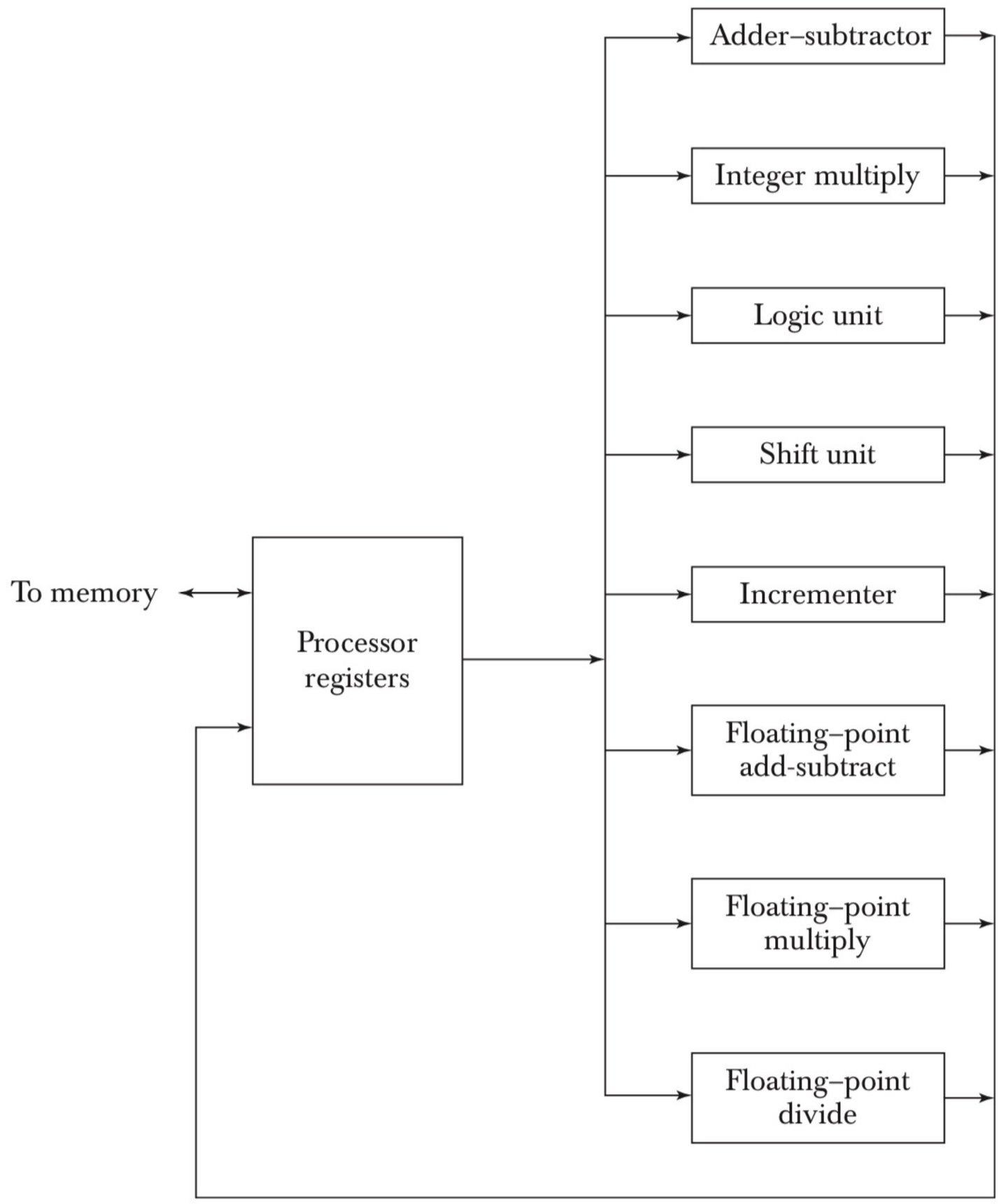
while registers with parallel load operate with all the bits of the word simultaneously. Parallel processing at a higher level of complexity can be achieved by having a multiplicity of functional units that perform identical or different operations simultaneously. Parallel processing is established by distributing the data among the multiple functional units. For example, the arithmetic, logic, and shift operations can be separated into three units and the operands diverted to each unit under the supervision of a control unit.

multiple functional units

Figure 9-1 shows one possible way of separating the execution unit into eight functional units operating in parallel. The operands in the registers are applied to one of the units depending on the operation specified by the

Figure 9-1 Processor with multiple functional units.



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instruction associated with the operands. The operation performed in each functional unit is indicated in each block of the diagram. The adder and integer multiplier perform the arithmetic operations with integer numbers. The floating-point operations are separated into three circuits operating in parallel. The logic, shift, and increment operations can be performed concurrently on different data. All units are independent of each other, so one number can be shifted while another number is being incremented. A multifunctional organization is usually associated with a complex control unit to coordinate all the activities among the various components.

There are a variety of ways that parallel processing can be classified. It can be considered from the internal organization of the processors, from the interconnection structure between processors, or from the flow of information through the system. One classification introduced by M. J. Flynn considers the organization of a computer system by the number of instructions and data items that are manipulated simultaneously. The normal operation of a computer is to fetch instructions from memory and execute them in the processor. The sequence of instructions read from memory constitutes an *instruction stream*. The operations performed on the data in the processor constitutes a *data stream*. Parallel processing may occur in the instruction stream, in the data stream, or in both. Flynn's classification divides computers into four major groups as follows:

Single instruction stream, single data stream (SISD)
Single instruction stream, multiple data stream (SIMD)
Multiple instruction stream, single data stream (MISD)
Multiple instruction stream, multiple data stream (MIMD)

SISD represents the organization of a single computer containing a control unit, a processor unit, and a memory unit. Instructions are executed sequentially and the system may or may not have internal parallel processing capabilities. Parallel processing in this case may be achieved by means of multiple functional units or by pipeline processing.

SIMD

SIMD represents an organization that includes many processing units under the supervision of a common control unit. All processors receive the same instruction from the control unit but operate on different items of data. The shared memory unit must contain multiple modules so that it can communicate with all the processors simultaneously. MISD structure is only of theoretical interest since no practical system has been constructed using this organization. MIMD organization refers to a computer system capable of processing several programs at the same time. Most multiprocessor and multicomputer systems can be classified in this category.

MIMD

Flynn's classification depends on the distinction between the performance of the control unit and the data-processing unit. It emphasizes the behavioral

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characteristics of the computer system rather than its operational and structural interconnections. One type of parallel processing that does not fit Flynn's classification is pipelining. The only two categories used from this classification are SIMD array processors discussed in Sec. 9-7, and MIMD multiprocessors presented in Chap. 13.

Parallel processing computers are required to meet the demands of large scale computations in many scientific, engineering, military, medical, artificial intelligence, and basic research areas. The following are some representative applications of parallel processing computers: Numerical weather forecasting, computational aerodynamics, finite-element analysis, remote-sensing applications, genetic engineering, computer-asseted tomography, and weapon research and defence.

In this chapter we consider parallel processing under the following main topics:

1. Pipeline processing
2. Vector processing
3. Array processors

Pipeline processing is an implementation technique where arithmetic suboperations or the phases of a computer instruction cycle overlap in execution. Vector processing deals with computations involving large vectors and matrices. Array processors perform computations on large arrays of data.

arithmetic / instruction

9-2 Pipelining → divide the task into multiple tasks

Pipelining is a technique of decomposing a sequential process into suboperations, with each subprocess being executed in a special dedicated segment that operates concurrently with all other segments. A pipeline can be visualized as a collection of processing segments through which binary information flows. Each segment performs partial processing dictated by the way the task is partitioned. The result obtained from the computation in each segment is transferred to the next segment in the pipeline. The final result is obtained after the data have passed through all segments. The name "pipeline" implies a flow of information analogous to an industrial assembly line. It is characteristic of pipelines that several computations can be in progress in distinct segments at the same time. The overlapping of computation is made possible by associating a register with each segment in the pipeline. The registers provide isolation between each segment so that each can operate on distinct data simultaneously.

Perhaps the simplest way of viewing the pipeline structure is to imagine that each segment consists of an input register followed by a combinational circuit. The register holds the data and the combinational circuit performs the suboperation in the particular segment. The output of the combinational circuit in a given segment is applied to the input register of the next segment. A clock is applied to all registers after enough time has elapsed to perform all

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segment activity. In this way the information flows through the pipeline one step at a time.

an example

The pipeline organization will be demonstrated by means of a simple example. Suppose that we want to perform the combined multiply and add operations with a stream of numbers.

$$A_i * B_i + C_i \quad \text{for } i = 1, 2, 3, \ldots, 7$$

Each suboperation is to be implemented in a segment within a pipeline. Each segment has one or two registers and a combinational circuit as shown in Fig. 9-2. $R1$ through $R5$ are registers that receive new data with every clock pulse. The multiplier and adder are combinational circuits. The suboperations performed in each segment of the pipeline are as follows:

$R1 \leftarrow A_i, \quad R2 \leftarrow B_i$	Input A_i and B_i
$R3 \leftarrow R1 * R2, \quad R4 \leftarrow C_i$	Multiply and input C_i
$R5 \leftarrow R3 + R4$	Add C_i to product

The five registers are loaded with new data every clock pulse. The effect of each clock is shown in Table 9-1. The first clock pulse transfers A_1 and B_1 into

Figure 9-2 Example or pipeline processing.

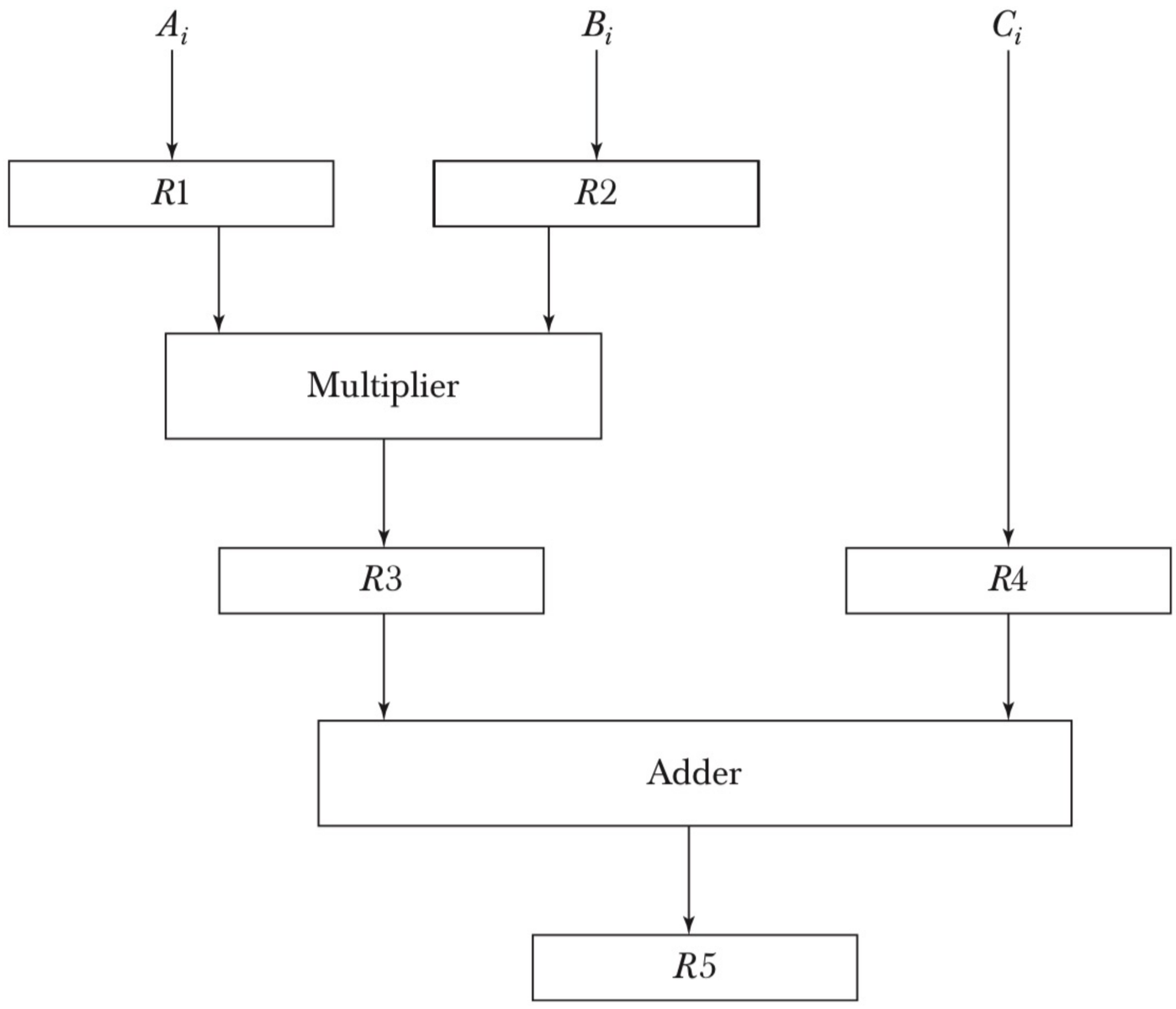

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TABLE 9-1 Content of Registers in Pipeline Example

Clock Pulse Number	Segment 1		Segment 2		Segment 3
	*R*1	*R*2	*R*3	*R*4	*R*5
1	A_1	B_1	–	–	–
2	A_2	B_2	$A_1 * B_1$	C_1	–
3	A_3	B_3	$A_2 * B_2$	C_2	$A_1 * B_1 + C_1$
4	A_4	B_4	$A_3 * B_3$	C_3	$A_2 * B_2 + C_2$
5	A_5	B_5	$A_4 * B_4$	C_4	$A_3 * B_3 + C_3$
6	A_6	B_6	$A_5 * B_5$	C_5	$A_4 * B_4 + C_4$
7	A_7	B_7	$A_6 * B_6$	C_6	$A_5 * B_5 + C_5$
8	–	–	$A_7 * B_7$	C_7	$A_6 * B_6 + C_6$
9	–	–	–	–	$A_7 * B_7 + C_7$

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*R*1 and *R*2. The second clock pulse transfers the product of *R*1 and *R*2 into *R*3 and C_1 into *R*4. The same clock pulse transfers A_2 and B_2 into *R*1 and *R*2. The third clock pulse operates on all three segments simultaneously. It places A_3 and B_3 into *R*1 and *R*2, transfers the product of *R*1 and *R*2 into *R*3, transfers C_2 into *R*4, and places the sum of *R*3 and *R*4 into *R*5. It takes three clock pulses to fill up the pipe and retrieve the first output from *R*5. From there on, each clock produces a new output and moves the data one step down the pipeline. This happens as long as new input data flow into the system. When no more input data are available, the clock must continue until the last output emerges out of the pipeline.

General Considerations

Any operation that can be decomposed into a sequence of suboperations of about the same complexity can be implemented by a pipeline processor. The technique is efficient for those applications that need to repeat the same task many times with different sets of data. The general structure of a four-segment pipeline is illustrated in Fig. 9-3. The operands pass through all four segments in a fixed sequence. Each segment consists of a combinational circuit S_i that performs a suboperation over the data stream flowing through the pipe. The segments are separated by registers R_i that hold the intermediate results between the stages. Information flows between adjacent stages under the control of a common clock applied to all the registers simultaneously. We define

task

a *task* as the total operation performed going through all the segments in the pipeline.

space-time diagram

The behavior of a pipeline can be illustrated with a *space-time* diagram. This is a diagram that shows the segment utilization as a function of time. The space-time diagram of a four-segment pipeline is demonstrated in Fig. 9-4. The horizontal axis displays the time in clock cycles and the vertical axis gives

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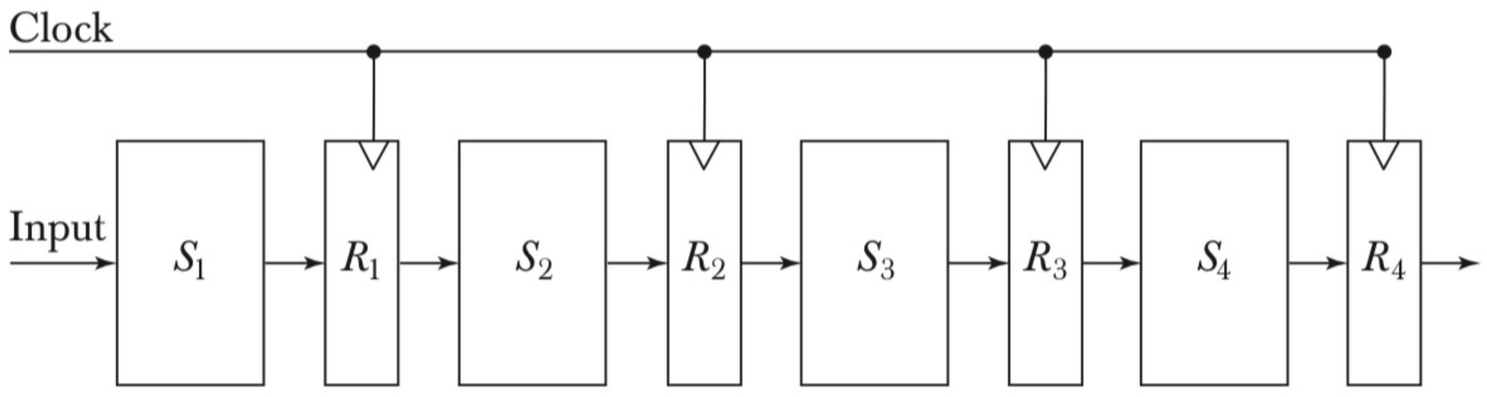

Figure 9-3 Four-segment pipeline.

the segment number. The diagram shows six tasks T_1 through T_6 executed in four segments. Initially, task T_1 is handled by segment 1. After the first clock, segment 2 is busy with T_1, while segment 1 is busy with task T_2. Continuing in this manner, the first task T_1 is completed after the fourth clock cycle. From then on, the pipe completes a task every clock cycle. No matter how many segments there are in the system, once the pipeline is full, it takes only one clock period to obtain an output.

Now consider the case where a k-segment pipeline with a clock cycle time t_p is used to execute n tasks. The first task T_1 requires a time equal to kt_p to complete its operation since there are k segments in the pipe. The remaining $n - 1$ tasks emerge from the pipe at the rate of one task per clock cycle and they will be completed after a time equal to $(n - 1)t_p$. Therefore, to complete n tasks using a k-segment pipeline requires $k + (n - 1)$ clock cycles. For example, the diagram of Fig. 9-4 shows four segments and six tasks. The time required to complete all the operations is $4 + (6 - 1) = 9$ clock cycles, as indicated in the diagram.

speedup

Next consider a nonpipeline unit that performs the same operation and takes a time equal to t_n to complete each task. The total time required for n tasks is nt_n. The speedup of a pipeline processing over an equivalent nonpipeline processing is defined by the ratio

$$S = \frac{nt_n}{(k + n - 1)t_p}$$

Figure 9-4 Space-time diagram for pipeline.

	1	2	3	4	5	6	7	8	9	→ Clock cycles
Segment: 1	T_1	T_2	T_3	T_4	T_5	T_6				
2		T_1	T_2	T_3	T_4	T_5	T_6			
3			T_1	T_2	T_3	T_4	T_5	T_6		
4				T_1	T_2	T_3	T_4	T_5	T_6	

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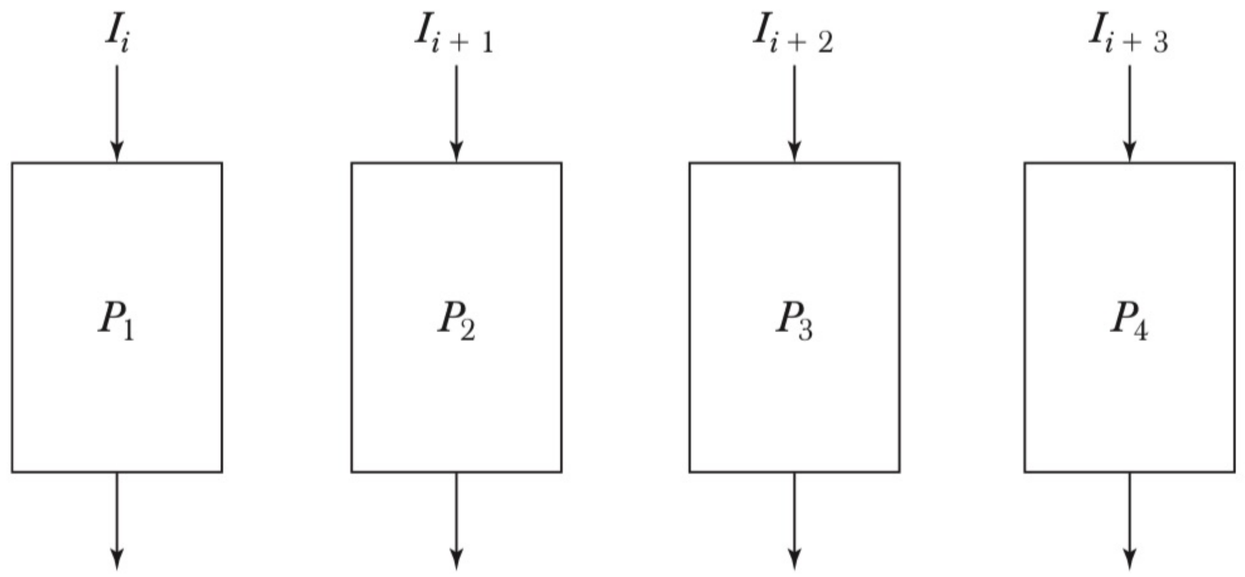

Figure 9-5 Multiple functional units in parallel.

segments to waste time while waiting for the next clock. Moreover, it is not always correct to assume that a nonpipe circuit has the same time delay as that of an equivalent pipeline circuit. Many of the intermediate registers will not be needed in a single-unit circuit, which can usually be constructed entirely as a combinational circuit. Nevertheless, the pipeline technique provids a faster operation over a purely serial sequence even though the maximum theoretical speed is never fully achieved.

There are two areas of computer design where the pipeline organization is applicable. An *arithmetic pipeline* divides an arithmetic operation into suboperations for execution in the pipeline segments. An *instruction pipeline* operates on a stream of instructions by overlapping the fetch, decode, and execute phases of the instruction cycle. The two types of pipelines are explained in the following sections.

9-3 Arithmetic Pipeline

Pipeline arithmetic units are usually found in very high speed computers. They are used to implement floating-point operations, multiplication of fixed-point numbers, and similar computations encountered in scientific problems. A pipeline multiplier is essentially an array multiplier as described in Fig. 10-10, with special adders designed to minimize the carry propagation time through the partial products. Floating-point operations are easily decomposed into suboperations as demonstrated in Sec. 10-5. We will now show an example of a pipeline unit for floating-point addition and subtraction.

The inputs to the floating-point adder pipeline are two normalized floating-point binary numbers.

$$X = A \times 2^a$$
$$Y = B \times 2^b$$

* Same data type ⇒ Same Complexity

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A and B are two fractions that represent the mantissas and a and b are the exponents. The floating-point addition and subtraction can be performed in four segments, as shown in Fig. 9-6. The registers labeled R are placed between the segments to store intermediate results. The suboperations that are performed in the four segments are:

1. Compare the exponents.
2. Align the mantissas.
3. Add or subtract the mantissas.
4. Normalize the result.

This follows the procedure outlined in the flowchart of Fig. 10-15 but with some variations that are used to reduce the execution time of the suboperations. The exponents are compared by subtracting them to determine their difference. The larger exponent is chosen as the exponent of the result. The exponent difference determines how many times the mantissa associated with the smaller exponent must be shifted to the right. This produces an alignment of the two mantissas. It should be noted that the shift must be designed as a combinational circuit to reduce the shift time. The two mantissas are added or subtracted in segment 3. The result is normalized in segment 4. When an overflow occurs, the mantissa of the sum or difference is shifted right and the exponent incremented by one. If an underflow occurs, the number of leading zeros in the mantissa determines the number of left shifts in the mantissa and the number that must be subtracted from the exponent.

The following numerical example may clarify the suboperations performed in each segment. For simplicity, we use decimal numbers, although Fig. 9-6 refers to binary numbers. Consider the two normalized floating-point numbers:

$$X = 0.9504 \times 10^3$$
$$Y = 0.8200 \times 10^2$$

The two exponents are subtracted in the first segment to obtain $3 - 2 = 1$. The larger exponent 3 is chosen as the exponent of the result. The next segment shifts the mantissa of Y to the right to obtain

$$X = 0.9504 \times 10^3$$
$$Y = 0.0820 \times 10^3$$

This aligns the two mantissas under the same exponent. The addition of the two mantissas in segment 3 produces the sum

$$Z = 1.0324 \times 10^3$$

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The sum is adjusted by normalizing the result so that it has a fraction with a nonzero first digit. This is done by shifting the mantissa once to the right and incrementing the exponent by one to obtain the normalized sum.

$$Z = 0.10324 \times 10^4$$

The comparator, shifter, adder-subtractor, incrementer, and decrementer in the floating-point pipeline are implemented with combinational circuits. Suppose that the time delays of the four segments are $t_1 = 60$ ns, $t_2 = 70$ ns, $t_3 = 100$ ns, $t_4 = 80$ ns, and the interface registers have a delay of $t_r = 10$ ns. The clock cycle is chosen to be $t_p = t_3 + t_r = 110$ ns. An equivalent non-pipeline floatingpoint adder-subtractor will have a delay time $t_n = t_1 + t_2 + t_3 + t_4 + t_r = 320$ ns. In this case the pipelined adder has a speedup of $320/110 = 2.9$ over the nonpipelined adder.

9-4 Instruction Pipeline → more effective than non-pipeline

Pipeline processing can occur not only in the data stream but in the instruction stream as well. An instruction pipeline reads consecutive instructions from memory while previous instructions are being executed in other segments. This causes the instruction fetch and execute phases to overlap and perform simultaneous operations. One possible digression associated with such a scheme is that an instruction may cause a branch out of sequence. In that case the pipeline must be emptied and all the instructions that have been read from memory after the branch instruction must be discarded.

Consider a computer with an instruction fetch unit and an instruction execution unit designed to provide a two-segment pipeline. The instruction fetch segment can be implemented by means of a first-in, first-out (FIFO) buffer. This is a type of unit that forms a queue rather than a stack. Whenever the execution unit is not using memory, the control increments the program counter and uses its address value to read consecutive instructions from memory. The instructions are inserted into the FIFO buffer so that they can be executed on a first-in, first-out basis. Thus an instruction stream can be placed in a queue, waiting for decoding and processing by the execution segment. The instruction stream queuing mechanism provides an efficient way for reducing the average access time to memory for reading instructions. Whenever there is space in the FIFO buffer, the control unit initiates the next instruction fetch phase. The buffer acts as a queue from which control then extracts the instructions for the execution unit.

instruction cycle

Computers with complex instructions require other phases in addition to the fetch and execute to process an instruction completely. In the most general case, the computer needs to process each instruction with the following sequence of steps.

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1. Fetch the instruction from memory.
2. Decode the instruction.
3. Calculate the effective address.
4. Fetch the operands from memory.
5. Execute the instruction.
6. Store the result in the proper place.

There are certain difficulties that will prevent the instruction pipeline from operating at its maximum rate. Different segments may take different times to operate on the incoming information. Some segments are skipped for certain operations. For example, a register mode instruction does not need an effective address calculation. Two or more segments may require memory access at the same time, causing one segment to wait until another is finished with the memory. Memory access conflicts are sometimes resolved by using two memory buses for accessing instructions and data in separate modules. In this way, an instruction word and a data word can be read simultaneously from two different modules.

The design of an instruction pipeline will be most efficient if the instruction cycle is divided into segments of equal duration. The time that each step takes to fulfill its function depends on the instruction and the way it is executed.

Example: Four-Segment Instruction Pipeline

Assume that the decoding of the instruction can be combined with the calculation of the effective address into one segment. Assume further that most of the instructions place the result into a processor register so that the instruction execution and storing of the result can be combined into one segment. This reduces the instruction pipeline into four segments.

Figure 9-7 shows how the instruction cycle in the CPU can be processed with a four-segment pipeline. While an instruction is being executed in segment 4, the next instruction in sequence is busy fetching an operand from memory in segment 3. The effective address may be calculated in a separate arithmetic circuit for the third instruction, and whenever the memory is available, the fourth and all subsequent instructions can be fetched and placed in an instruction FIFO. Thus up to four suboperations in the instruction cycle can overlap and up to four different instructions can be in progress of being processed at the same time.

Once in a while, an instruction in the sequence may be a program control type that causes a branch out of normal sequence. In that case the pending operations in the last two segments are completed and all information stored in the instruction buffer is deleted. The pipeline then restarts from the new address stored in the program counter. Similarly, an interrupt request, when acknowledged, will cause the pipeline to empty and start again from a new address value.

الـSegment اكي دخلت بكون مش دوريها

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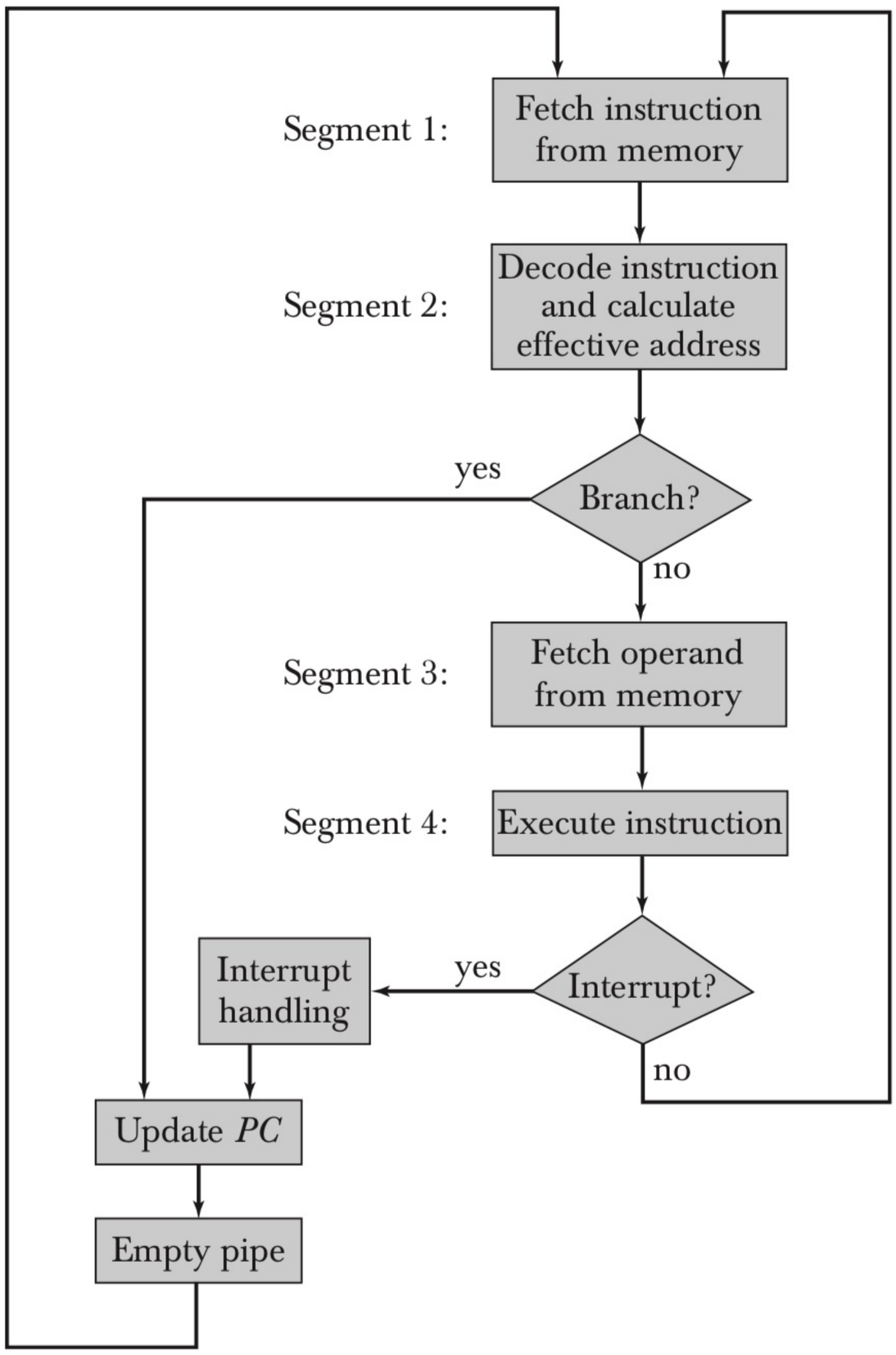

Figure 9-7 Four-segment CPU pipeline.

Figure 9-8 shows the operation of the instruction pipeline. The time in the horizontal axis is divided into steps of equal duration. The four segments are represented in the diagram with an abbreviated symbol.

1. FI is the segment that fetches an instruction.
2. DA is the segment that decodes the instruction and calculates the effective address.
3. FO is the segment that fetches the operand.
4. EX is the segment that executes the instruction.

It is assumed that the processor has separate instruction and data memories so that the operation in FI and FO can proceed at the same time. In the absence

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Step:		1	2	3	4	5	6	7	8	9	10	11	12	13
Instruction:	1	FI	DA	FO	EX									
	2		FI	DA	FO	EX								
(Branch)	3			FI	DA	FO	EX							
	4				FI	–	–	FI	DA	FO	EX			
	5					–	–	–	FI	DA	FO	EX		
	6									FI	DA	FO	EX	
	7										FI	DA	FO	EX

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Figure 9-8 Timing of instruction pipeline.

of a branch instruction, each segment operates on different instructions. Thus, in step 4, instruction 1 is being executed in segment EX; the operand for instruction 2 is being fetched in segment FO; instruction 3 is being decoded in segment DA; and instruction 4 is being fetched from memory in segment FI.

Assume now that instruction 3 is a branch instruction. As soon as this instruction is decoded in segment DA in step 4, the transfer from FI to DA of the other instructions is halted until the branch instruction is executed in step 6. If the branch is taken, a new instruction is fetched in step 7. If the branch is not taken, the instruction fetched previously in step 4 can be used. The pipeline then continues until a new branch instruction is encountered.

Another delay may occur in the pipeline if the EX segment needs to store the result of the operation in the data memory while the FO segment needs to fetch an operand. In that case, segment FO must wait until segment EX has finished its operation.

pipeline conflicts

In general, there are three major difficulties that cause the instruction pipeline to deviate from its normal operation.

→ arithmetic pipeline مش موجودين بالـ
→ Causes delay

1. *Resource conflicts* caused by access to memory by two segments at the same time. Most of these conflicts can be resolved by using separate instruction and data memories.
2. *Data dependency* conflicts arise when an instruction depends on the result of a previous instruction, but this result is not yet available.
3. *Branch difficulties* arise from branch and other instructions that change the value of *PC*.

Data Dependency

A difficulty that may caused a degradation of performance in an instruction pipeline is due to possible collision of data or address. A collision occurs when

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cycle 6 is from address 350, which is the instruction at the branch address.

9-6 Vector Processing

There is a class of computational problems that are beyond the capabilities of a conventional computer. These problems are characterized by the fact that they require a vast number of computations that will take a conventional computer days or even weeks to complete. In many science and engineering

Pipeline can't handle it ← (big data) مشاكل علمية
استخدام شخصي ← لها مثل تنبؤ بالزلازل ـ استخدام عسكري

Clock cycles:	1	2	3	4	5	6	7	8	9	10
1. Load	I	A	E							
2. Increment		I	A	E						
3. Add			I	A	E					
4. Subtract				I	A	E				
5. Branch to X					I	A	E			
6. No-operation						I	A	E		
7. No-operation							I	A	E	
8. Instruction in X								I	A	E

(a) Using no-operation instructions

Clock cycles:	1	2	3	4	5	6	7	8
1. Load	I	A	E					
2. Increment		I	A	E				
3. Branch to X			I	A	E			
4. Add				I	A	E		
5. Subtract					I	A	E	
6. Instruction in X						I	A	E

(b) Rearranging the instructions

Figure 9-10 Example of delayed branch.

applications, the problems can be formulated in terms of vectors and matrices that lend themselves to vector processing.

Computers with vector processing capabilities are in demand in specialized applications. The following are representative application areas where

applications

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Clock cycles:	1	2	3	4	5	6	7	8	9	10
1. Load	I	A	E							
2. Increment		I	A	E						
3. Add			I	A	E					
4. Subtract				I	A	E				
5. Branch to *X*					I	A	E			
6. No-operation						I	A	E		
7. No-operation							I	A	E	
8. Instruction in *X*								I	A	E

(a) Using no-operation instructions

Clock cycles:	1	2	3	4	5	6	7	8
1. Load	I	A	E					
2. Increment		I	A	E				
3. Branch to *X*			I	A	E			
4. Add				I	A	E		
5. Subtract					I	A	E	
6. Instruction in *X*						I	A	E

(b) Rearranging the instructions

Figure 9-10 Example of delayed branch.

applications, the problems can be formulated in terms of vectors and matrices that lend themselves to vector processing.

applications

Computers with vector processing capabilities are in demand in specialized applications. The following are representative application areas where vector processing is of the utmost importance.

- Long-range weather forecasting
- Petroleum explorations
- Seismic data analysis
- Medical diagnosis
- Aerodynamics and space flight simulations

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- Artificial intelligence and expert systems
- Mapping the human genome
- Image processing

Without sophisticated computers, many of the required computations cannot be completed within a reasonable amount of time. To achieve the required level of high performance it is necessary to utilize the fastest and most reliable hardware and apply innovative procedures from vector and parallel processing techniques.

Vector Operations

Many scientific problems require arithmetic operations on large arrays of numbers. These numbers are usually formulated as vectors and matrices of floating-point numbers. A vector is an ordered set of a one-dimensional array of data items. A vector V of length n is represented as a row vector by $V = [V_1\ V_2\ V_3\ \ldots\ V_n]$. It may be represented as a column vector if the data items are listed in a column. A conventional sequential computer is capable of processing operands one at a time. Consequently, operations on vectors must be broken down into single computations with subscripted variables. The element V_i of vector V is written as $V(I)$ and the index I refers to a memory address or register where the number is stored. To examine the difference between a conventional scalar processor and a vector processor, consider the following Fortran DO loop:

```
      DO 20 I = 1, 100
20    C(I) = B(I) + A(I)
```

iterations

This is a program for adding two vectors A and B of length 100 to produce a vector C. This is implemented in machine language by the following sequence of operations.

```
      Initialize I = 0
20    Read A(I)
      Read B(I)
      Store C(I) = A(I) + B(I)
      Increment I = I + 1
      If I ≤ 100 go to 20
      Continue
```

This constitutes a program loop that reads a pair of operands from arrays A and B and performs a floating-point addition. The loop control variable is then updated and the steps repeat 100 times.

A computer capable of vector processing eliminates the overhead associated with the time it takes to fetch and execute the instructions in the program

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بتخلي عن الـ loop

loop. It allows operations to be specified with a single vector instruction of the form

$$C(1:100) = A(1:100) + B(1:100)$$

The vector instruction includes the initial address of the operands, the length of the vectors, and the operation to be performed, all in one composite instruction. The addition is done with a pipelined floating-point adder similar to the one shown in Fig. 9-6.

A possible instruction format for a vector instruction is shown in Fig. 9-11. This is essentially a three-address instruction with three fields specifying the base address of the operands and an additional field that gives the length of the data items in the vectors. This assumes that the vector operands reside in memory. It is also possible to design the processor with a large number of registers and store all operands in registers prior to the addition operation. In that case the base address and length in the vector instruction specify a group of CPU registers.

Matrix Multiplication

Matrix multiplication is one of the most computational intensive operations performed in computers with vector processors. The multiplication of two $n \times n$ matrices consists of n^2 inner products or n^3 multiply–add operations. An $n \times m$ matrix of numbers has n rows and m columns and may be considered as constituting a set of n row vectors or a set of m column vectors. Consider, for example, the multiplication of two 3×3 matrices A and B.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

The product matrix C is a 3×3 matrix whose elements are related to the elements of A and B by the inner product:

$$c_{ij} = \sum_{k=1}^{3} a_{ik} \times b_{kj}$$

For example, the number in the first row and first column of matrix C is calculated by letting $i = 1, j = 1$, to obtain

$$c_{11} = a_{11}b_{11} + a_{12}\,b_{21} + a_{13}\,b_{31}$$

Figure 9-11 Instruction format for vector processor.

Operation code	Base address source 1	Base address source 2	Base address destination	Vector length

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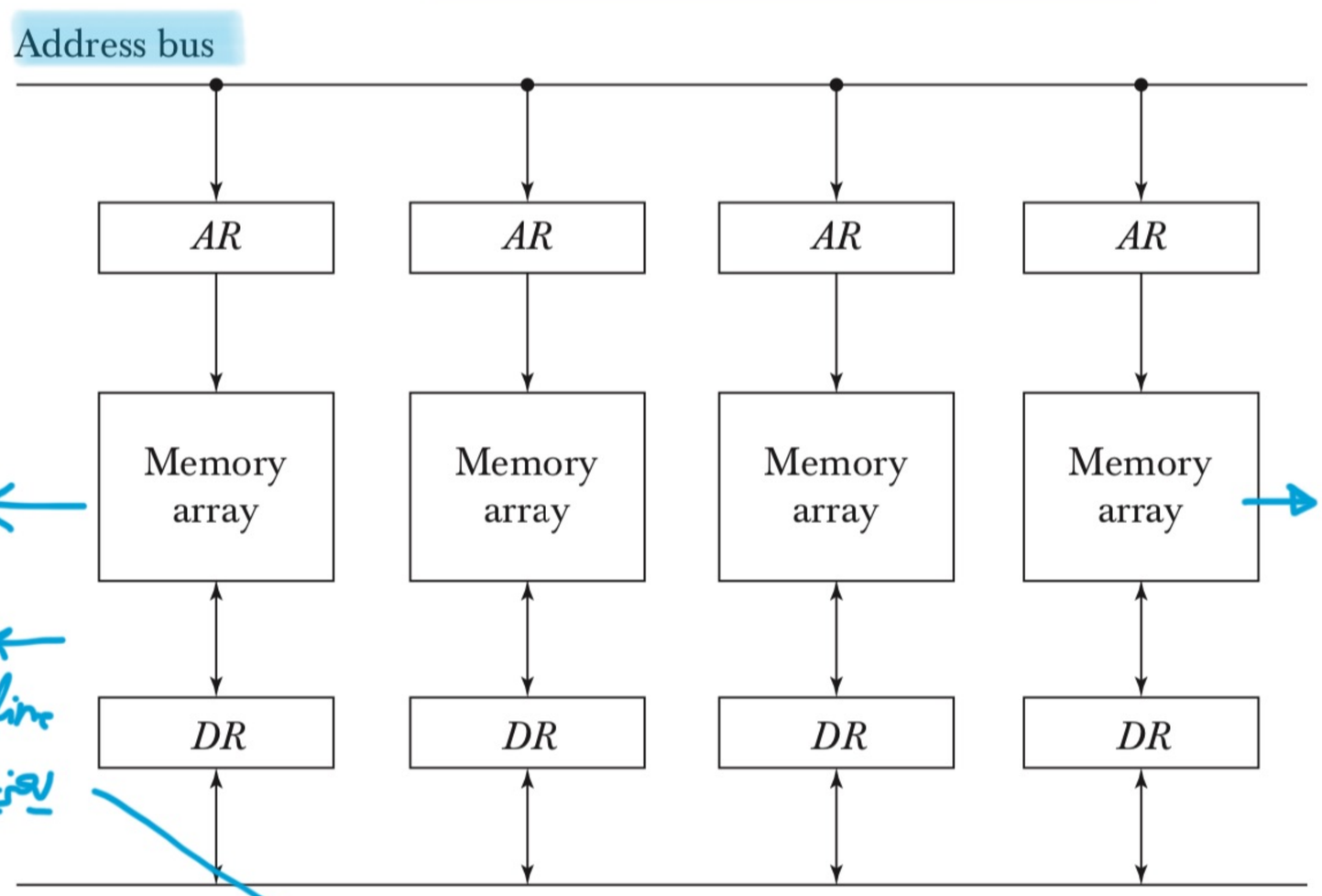
When there are no more product terms to be added, the system inserts four zeros into the multiplier pipeline. The adder pipeline will then have one partial product in each of its four segments, corresponding to the four sums listed in the four rows in the above equation. The four partial sums are then added to form the final sum.

Memory Interleaving

Pipeline and vector processors often require simultaneous access to memory from two or more sources. An instruction pipeline may require the fetching of an instruction and an operand at the same time from two different segments. Similarly, an arithmetic pipeline usually requires two or more operands to enter the pipeline at the same time. Instead of using two memory buses for simultaneous access, the memory can be partitioned into a number of modules connected to a common memory address and data buses. A memory module is a memory array together with its own address and data registers. Figure 9-13 shows a memory unit with four modules. Each memory array has its own address register *AR* and data register *DR*. The address registers receive information from a common address bus and the data registers communicate with a bidirectional data bus. The two least significant bits of the address can be used to distinguish between the four modules. The modular system permits one module to initiate a memory access while other modules are in the process of reading or writing a word and each module can honor a memory request independent of the state of the other modules.

Figure 9-13 Multiple module memory organization.



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قلنا احتمالية حدوث Conflict ← زيادة الـ Performance وتقليل الفترات الزمنية المهدورة في الـ Pipeline & Vector processors

The advantage of a modular memory is that it allows the use of a technique called *interleaving*. In an interleaved memory, different sets of addresses are assigned to different memory modules. For example, in a two-module memory system, the even addresses may be in one module and the odd addresses in the other. When the number of modules is a power of 2, the least significant bits of the address select a memory module and the remaining bits designate the specific location to be accessed within the selected module.

A modular memory is useful in systems with pipeline and vector processing. A vector processor that uses an n-way interleaved memory can fetch n operands from n different modules. By staggering the memory access, the effective memory cycle time can be reduced by a factor close to the number of modules. A CPU with instruction pipeline can take advantage of multiple memory modules so that each segment in the pipeline can access memory independent of memory access from other segments.

Solve the Problem ← Caused by multiple access (conflict) → delay

Superscalar Processors

A superscalar processor architecture has a form of parallelism on a single chip allowing the system as a whole to run much faster than it would otherwise be able to at a given clock speed. A superscalar architecture fetches, executes, and returns results from more than one instruction during a single pipeline stage. A scalar processor processes one data item at a time. In a vector processor, by contrast, a single instruction operates simultaneously on multiple data items. The difference is analogous to the difference between scalar and vector arithmetic. A superscalar processor is sort of a mixture of the two. Each instruction processes one data item, but there are multiple processing units so that multiple instructions can be processing separate data items at the same time.

A superscalar processor normally has an execution rate in excess of one instruction per machine cycle. But just processing multiple instructions at the same time does not make an architecture superscalar. Simple pipelining, where a processor may be loading an instruction while doing arithmetic for the previous one and storing the results from the one before that (thus executing three instructions at the same time) is not superscalar processing. In a superscalar processor, there are several functional units of the same type, along with additional circuitry to dispatch instructions to the units. For instance, most superscalar designs include more than one arithmetic and logic unit. The dispatcher reads instructions from memory and decides which ones can be run in parallel, dispatching them to the two units. Seymour Cray's CDC 6600 from 1965 is often mentioned as the first superscalar design. The Intel i960CA (1988) and the AMD 29000-series 29050 (1990) microprocessors were the first commercial single-chip superscalar microprocessors. The RS6000 from IBM was released in 1990 and was the world's first superscalar RISC microprocessor. Intel followed in 1993 with the Pentium, which with its two ALUs brought the x86 world into the superscalar era.

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Supercomputers

A commercial computer with vector instructions and pipelined floating-point arithmetic operations is referred to as a *supercomputer.* Supercomputers are very powerful, high-performance machines used mostly for scientific computations. To speed up the operation, the components are packed tightly together to minimize the distance that the electronic signals have to travel. Supercomputers also use special techniques for removing the heat from circuits to prevent them from burning up because of their close proximity.

The instruction set of supercomputers contains the standard data transfer, data manipulation, and program control instructions of conventional computers. This is augmented by instructions that process vectors and combinations of scalars and vectors. A supercomputer is a computer system best known for its high computational speed, fast and large memory systems, and the extensive use of parallel processing. It is equipped with multiple functional units and each unit has its own pipeline configuration. Although the supercomputer is capable of general-purpose applications found in all other computers, it is specifically optimized for the type of numerical calculations involving vectors and matrices of floating-point numbers.

Supercomputers are not suitable for normal everyday processing of a typical computer installation. They are limited in their use to a number of scientific applications, such as numerical weather forecasting, seismic wave analysis, and space research. They have limited use and limited market because of their high price.

A measure used to evaluate computers in their ability to perform a given number of floating-point operations per second is referred to as *flops.* The term *megaflops* is used to denote million flops and *gigaflops* to denote billion flops. A typical supercomputer has a basic cycle time of 4 to 20 ns. If the processor can calculate a floating-point operation through a pipeline each cycle time, it will have the ability to perform 50 to 250 megaflops. This rate would be sustained from the time the first answer is produced and does not include the initial setup time of the pipeline.

The first supercomputer developed in 1976 is the Cray-1 supercomputer. It uses vector processing with 12 distinct functional units in parallel. Each functional unit is segmented to process the incoming data through a pipeline. All the functional units can operate concurrently with operands stored in the large number of registers (over 150) in the CPU. A floating-point operation can be performed on two sets of 64-bit operands during one clock cycle of 12.5 ns. This gives a rate of 80 megaflops during the time that the data are processed through the pipeline. It has a memory capacity of 4 million 64-bit words. The memory is divided into 16 banks, with each bank having a 50-ns access time. This means that when all 16 banks are accessed simultaneously, the memory transfer rate is 320 million words per second. Cray research extended its supercomputer to a multiprocessor configuration called Cray X-MP and Cray Y-MP. The new Cray-2 supercomputer is 12 times more powerful than the Cray-1 in vector processing mode.

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Another early model supercomputer is the Fujitsu VP-200. It has a scalar processor and a vector processor that can operate concurrently. Like the Cray supercomputers, a large number of registers and multiple functional units are used to enable register-to-register vector operations. There are four execution pipelines in the vector processor, and when operating simultaneously, they can achieve up to 300 megaflops. The main memory has 32 million words connected to the vector registers through load and store pipelines. The VP-200 has 83 vector instructions and 195 scalar instructions. The newer VP-2600 uses a clock cycle of 3.2 ns and claims a peak performance of 5 gigaflops.

9-7 Array Processors

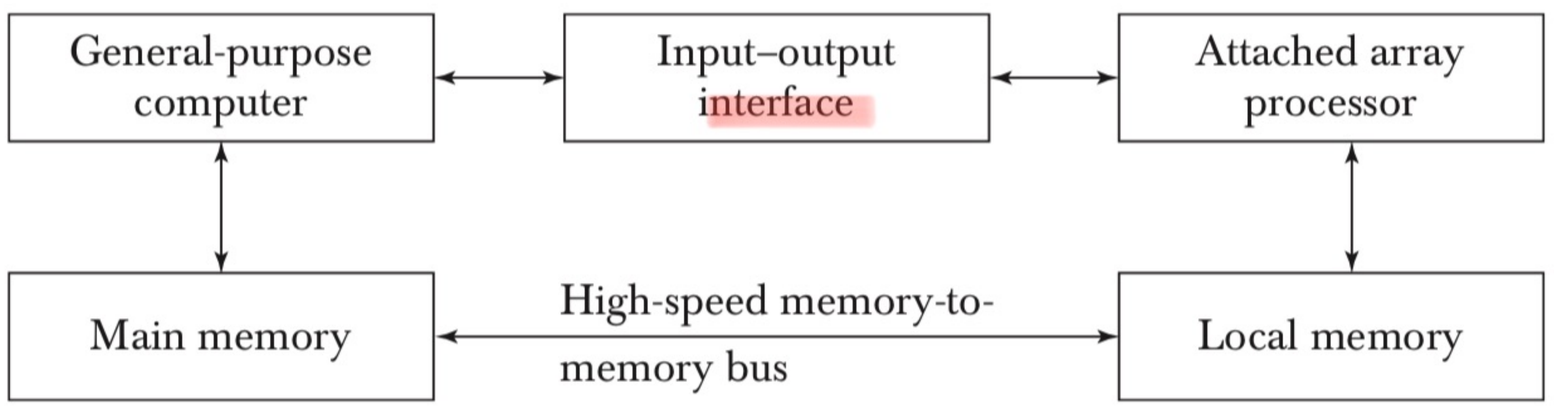
An array processor is a processor that performs computations on large arrays of data. The term is used to refer to two different types of processors. An *attached array processor* is an auxiliary processor attached to a general-purpose computer. It is intended to improve the performance of the host computer in specific numerical computation tasks. An *SIMD array processor* is a processor that has a single-instruction multiple-data organization. It manipulates vector instructions by means of multiple functional units responding to a common instruction. Although both types of array processors manipulate vectors, their internal organization is different.

Attached Array Processor

An attached array processor is designed as a peripheral for a conventional host computer, and its purpose is to enhance the performance of the computer by providing vector processing for complex scientific applications. It achieves high performance by means of parallel processing with multiple functional units. It includes an arithmetic unit containing one or more pipelined floating-point adders and multipliers. The array processor can be programmed by the user to accommodate a variety of complex arithmetic problems.

Figure 9-14 shows the interconnection of an attached array processor to a host computer. The host computer is a general-purpose commercial computer

Figure 9-14 Attached array processor with host computer.



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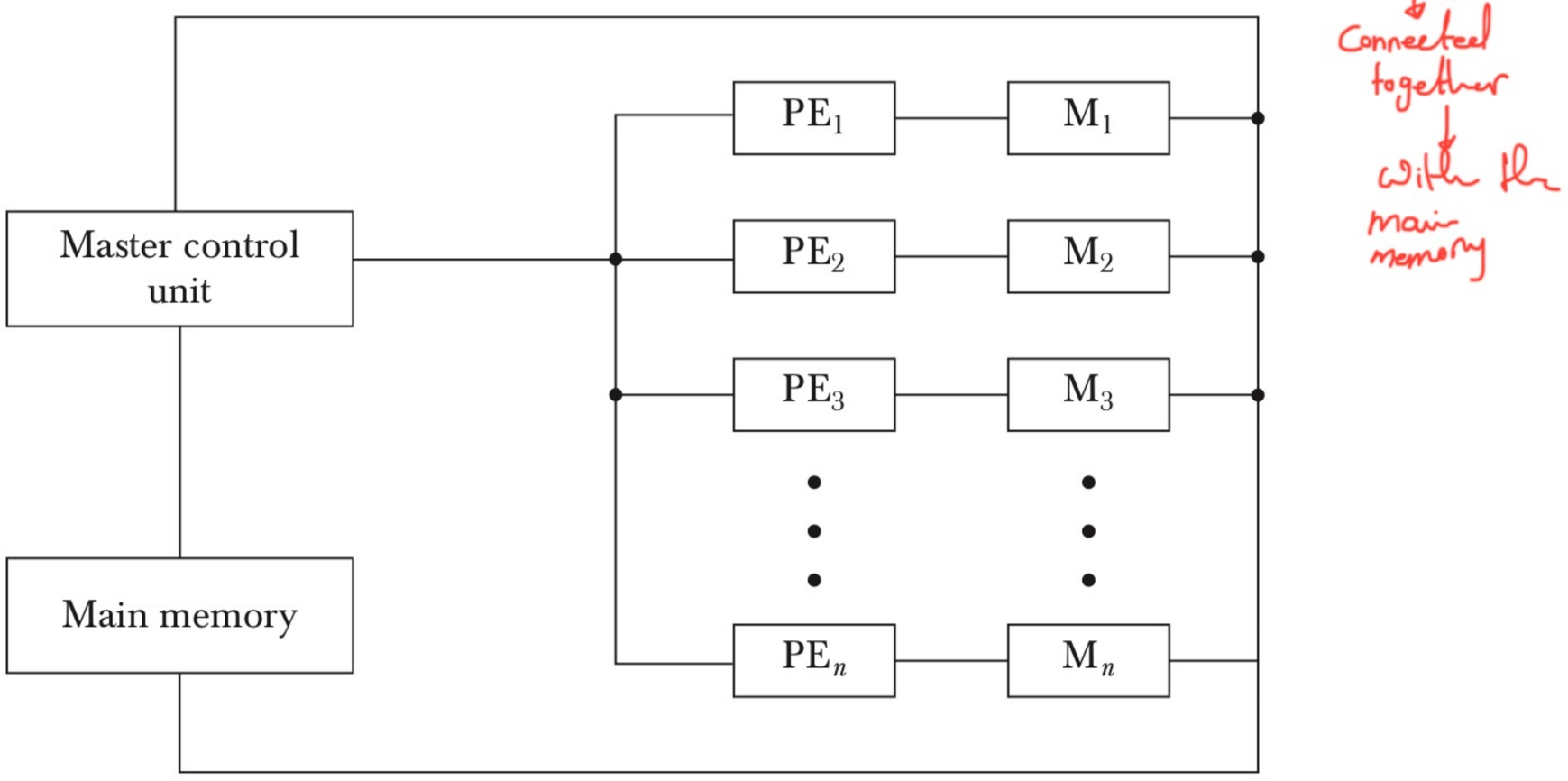
and the attached processor is a back-end machine driven by the host computer. The array processor is connected through an input–output controller to the computer and the computer treats it like an external interface. The data for the attached processor are transferred from main memory to a local memory through a high-speed bus. The general-purpose computer without the attached processor serves the users that need conventional data processing. The system with the attached processor satisfies the needs for complex arithmetic applications.

Some manufacturers of attached array processors offer a model that can be connected to a variety of different host computers. For example, when attached to a VAX 11 computer, the FSP-164/MAX from Floating-Point Systems increases the computing power of the VAX to 100 megaflops. The objective of the attached array processor is to provide vector manipulation capabilities to a conventional computer at a fraction of the cost of supercomputers.

SIMD Array Processor → highest performance

An SIMD array processor is a computer with multiple processing units operating in parallel. The processing units are synchronized to perform the same operation under the control of a common control unit, thus providing a single instruction stream, multiple data stream (SIMD) organization. A general block diagram of an array processor is shown in Fig. 9-15. It contains a set of identical processing elements (PEs), each having a local memory *M*. Each processor element includes an ALU, a floating-point arithmetic unit, and

يقسم الprocessor إلى Processing elements بتنقسم العملية عليهم
Could be operated in Parallel
each one has its own memory
Connected together
with the main memory



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CHAPTER ELEVEN

Input–Output Organization

اول التشابتر فش معلومات مهمة

IN THIS CHAPTER

11-1 Peripheral Devices
11-2 Input–Output Interface
11-3 Asynchronous Data Transfer
11-4 Modes of Transfer
11-5 Priority Interrupt
11-6 Direct Memory Access
11-7 Input–Output Processor
11-8 Serial Communication

11-1 Peripheral Devices any device (keyboard, printer, scanner, ...)

I/O

The input–output subsystem of a computer, referred to as I/O, provides an efficient mode of communication between the central system and the outside environment. Programs and data must be entered into computer memory for processing and results obtained from computations must be recorded or displayed for the user. A computer serves no useful purpose without the ability to receive information from an outside source and to transmit results in a meaningful form.

The most familiar means of entering information into a computer is through a typewriter-like keyboard that allows a person to enter alphanumeric information directly. Every time a key is depressed, the terminal sends a binary coded character to the computer. The fastest possible speed for entering information this way depends on the person's typing speed. On the other hand, the central processing unit is an extremely fast device capable of performing operations at very high speed. When input information is transferred to the processor via a slow keyboard, the processor will be idle most of the time while waiting for the information to arrive. To use a computer efficiently, a large

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amount of programs and data must be prepared in advance and transmitted into a storage medium such as magnetic tapes or disks. The information in the disk is then transferred into computer memory at a rapid rate. Results of programs are also transferred into a high-speed storage, such as disks, from which they can be transferred later into a printer to provide a printed output of results.

Devices that are under the direct control of the computer are said to be connected on-line. These devices are designed to read information into or out of the memory unit upon command from the CPU and are considered to be part of the total computer system. Input or output devices attached to the computer are also called *peripherals*. There are three types of peripherals such as input, output, and input–output peripherals. These peripherals may be analog or digital and serial or parallel. Among the most common peripherals are keyboards, display units, and printers. Peripherals that provide auxiliary storage for the system are magnetic disks and tapes. Peripherals are electromechanical and electromagnetic devices of some complexity. Only a very brief discussion of their function will be given here without going into detail of their internal construction.

peripheral

monitor and keyboard

Video monitors are the most commonly used peripherals. They consist of a keyboard as the input device and a display unit as the output device. There are different types of video monitors, but the most popular use a cathode ray tube (CRT). The CRT contains an electronic gun that sends an electronic beam to a phosphorescent screen in front of the tube. The beam can be deflected horizontally and vertically. To produce a pattern on the screen, a grid inside the CRT receives a variable voltage that causes the beam to hit the screen and make it glow at selected spots. Horizontal and vertical signals deflect the beam and make it sweep across the tube, causing the visual pattern to appear on the screen. A characteristic feature of display devices is a cursor that marks the position in the screen where the next character will be inserted. The cursor can be moved to any position in the screen, to a single character, the beginning of a word, or to any line. Edit keys add or delete information based on the cursor position. The display terminal can operate in a single-character mode where all characters entered on the screen through the keyboard are transmitted to the computer simultaneously. In the block mode, the edited text is first stored in a local memory inside the terminal. The text is transferred to the computer as a block of data.

printer

Printers provide a permanent record on paper of computer output data or text. There are three basic types of character printers: daisywheel, dot matrix, and laser printers. The daisywheel printer contains a wheel with the characters placed along the circumference. To print a character, the wheel rotates to the proper position and an energized magnet then presses the letter against the ribbon. The dot matrix printer contains a set of dots along the printing mechanism. For example, a 5×7 dot matrix printer that prints 80 characters per line has seven horizontal lines, each consisting of $5 \times 80 = 400$ dots. Each dot can be printed or not, depending on the specific characters that are printed on the line. The laser printer uses a rotating photographic drum

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that is used to imprint the character images. The pattern is then transferred onto paper in the same manner as a copying machine.

magnetic tape

Magnetic tapes are used mostly for storing files of data: for example, a company's payroll record. Access is sequential and consists of records that can be accessed one after another as the tape moves along a stationary read–write mechanism. It is one of the cheapest and slowest methods for storage and has the advantage that tapes can be removed when not in use.

magnetic disk

Magnetic disks have high-speed rotational surfaces coated with magnetic material. Access is achieved by moving a read–write mechanism to a track in the magnetized surface. Disks are used mostly for bulk storage of programs and data. Tapes and disks are discussed further in Sec. 12-1 in conjunction with their role as auxiliary memory.

Other input and output devices encountered in computer systems are digital incremental plotters, optical and magnetic character readers, analog-to-digital converters, and various data acquisition equipment. Not all input comes from people, and not all output is intended for people. Computers are used to control various processes in real time, such as machine tooling, assembly line procedures, and chemical and industrial processes. For such applications, a method must be provided for sensing status conditions in the process and sending control signals to the process being controlled.

The input–output organization of a computer is a function of the size of the computer and the devices connected to it. The difference between a small and a large system is mostly dependent on the amount of hardware the computer has available for communicating with peripheral units and the number of peripherals connected to the system. Since each peripheral behaves differently from any other, it would be prohibitive to dwell on the detailed interconnections needed between the computer and each peripheral. Certain techniques common to most peripherals are presented in this chapter.

ASCII Alphanumeric Characters

Input and output devices that communicate with people and the computer are usually involved in the transfer of alphanumeric information to and from the device and the computer. The standard binary code for the alphanumeric

ASCII

characters is ASCII (American Standard Code for Information Interchange). It uses seven bits to code 128 characters as shown in Table 11-1. The seven bits of the code are designated by b_1 through b_7, with b_7 being the most significant bit. The letter A, for example, is represented in ASCII as 1000001 (column 100, row 0001). The ASCII code contains 94 characters that can be printed and 34 nonprinting characters used for various control functions. The printing characters consist of the 26 uppercase letters A through Z, the 26 lowercase letters, the 10 numerals 0 through 9, and 32 special printable characters such as %, * , and $.

The 34 control characters are designated in the ASCII table with abbreviated names. They are listed again below the table with their functional names. The control characters are used for routing data and arranging the printed text

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UNICode → more symbols (languages)

record separator (RS) and file separator (FS). The communication control characters are useful during the transmission of text between remote terminals. Examples of communication control characters are STX (start of text) and ETX (end of text), which are used to frame a text message when transmitted through a communication medium.

byte

ASCII is a 7-bit code, but most computers manipulate an 8-bit quantity as a single unit called a *byte*. Therefore, ASCII characters most often are stored one per byte. The extra bit is sometimes used for other purposes, depending on the application. For example, some printers recognize 8-bit ASCII characters with the most significant bit set to 0. Additional 128 8-bit characters with the most significant bit set to 1 are used for other symbols, such as the Greek alphabet or italic type font. When used in data communication, the eighth bit may be employed to indicate the parity of the binary-coded character.

11-2 Input–Output Interface

عبارة عن circuit للتخلب على الـ differences بين البيئة الداخلية والخارجية للحاسوب

Input–output interface provides a method for transferring information between internal storage and external I/O devices. Peripherals connected to a computer need special communication links for interfacing them with the central processing unit. The purpose of the communication link is to resolve the differences that exist between the central computer and each peripheral. The major differences are:

1. Peripherals are electromechanical and electromagnetic devices and their manner of operation is different from the operation of the CPU and memory, which are electronic devices. Therefore, a conversion of signal values may be required.
2. The data transfer rate of peripherals is usually slower than the transfer rate of the CPU, and consequently, a synchronization mechanism may be needed.
3. Data codes and formats in peripherals differ from the word format in the CPU and memory.
4. The operating modes of peripherals are different from each other and each must be controlled so as not to disturb the operation of other peripherals connected to the CPU.

interface

To resolve these differences, computer systems include special hardware components between the CPU and peripherals to supervise and synchronize all input and output transfers. These components are called *interface* units because they interface between the processor bus and the peripheral device. The word "Interface" is a general term for the point of contact between two parts of a system. In digital computer system the interface is refered as a complementary set of signal connection points between two parts of a system. Therefore, "to interface" means to attach two or more components or systems,

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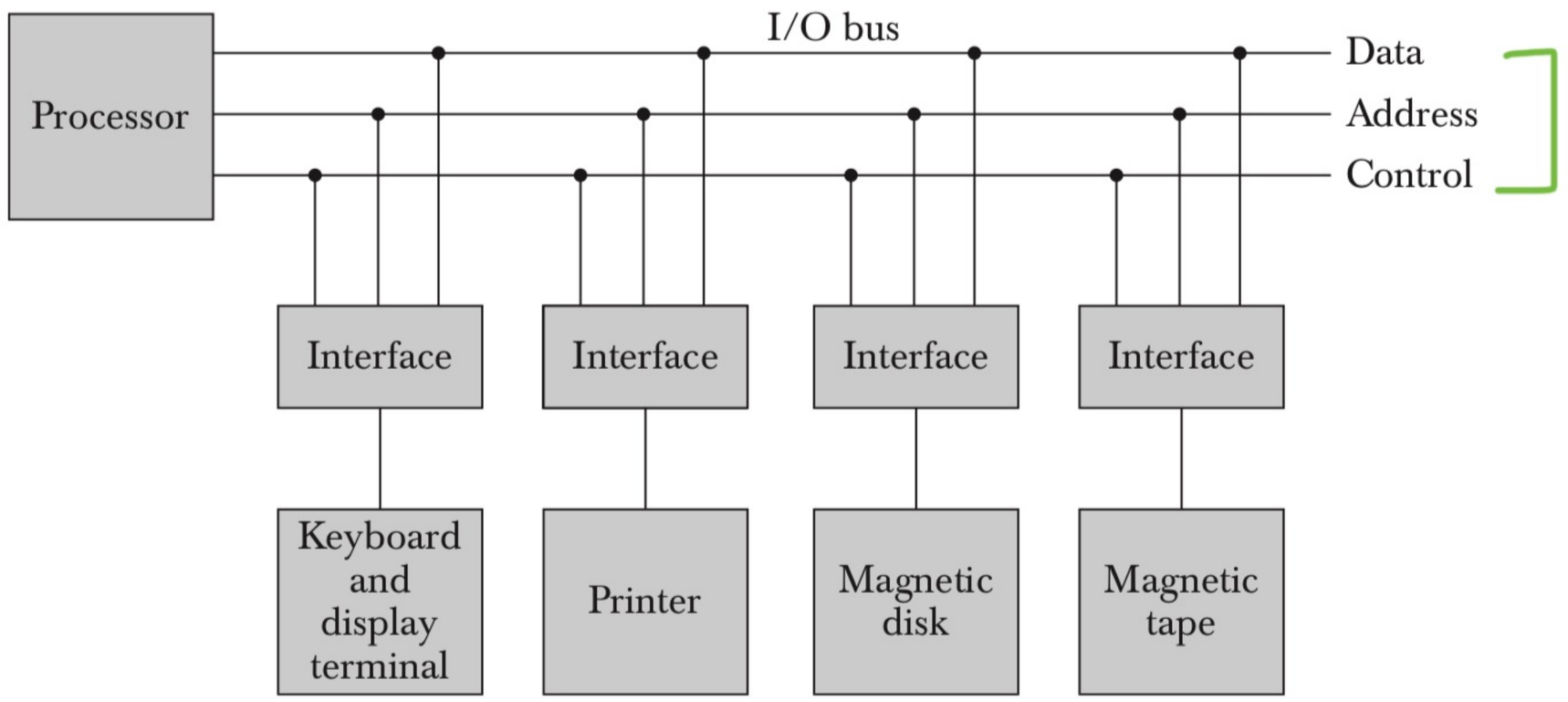
via their respective interface points for data exchanges between them. Two main types of interface are CPU interface that corresponds to the system bus and input–output interface that depends on the nature of input–output device. To attach an input–output device to CPU and input–output interface, circuit is placed between the device and the system bus. This circuit is meant for matching the signal formats and timing characteristics of the CPU interface to those of the input–output device interface. The main function of input–output interface circuit are data conversion, synchronization and device selection. Data conversion refers to conversion between digital and analog signals, and conversion between serial and parallel data formats. Synchronation refers to matching of operating speeds of CPU and other peripherals. Device selection refers to the selection of I/O device by CPU in a queue manner. In addition, each device may have its own controller that supervises the operations of the particular mechanism in the peripheral.

I/O Bus and Interface Modules

A typical communication link between the processor and several peripherals is shown in Fig. 11-1. The I/O bus consists of data lines, address lines, and control lines. The magnetic disk, printer, and terminal are employed in practically any general-purpose computer. The magnetic tape is used in some computers for backup storage. Each peripheral device has associated with it an interface unit. Each interface decodes the address and control received from the I/O bus, interprets them for the peripheral, and provides signals for the peripheral controller. It also synchronizes the data flow and supervises the transfer between peripheral and processor. Each peripheral has its own controller that operates the particular electromechanical device. For example, the printer controller controls the paper motion, the print timing, and the selection of printing characters. A controller may be housed separately or may be physically integrated with the peripheral.

Figure 11-1 Connection of I/O bus to input–output devices.



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The I/O bus from the processor is attached to all peripheral interfaces. To communicate with a particular device, the processor places a device address on the address lines. Each interface attached to the I/O bus contains an address decoder that monitors the address lines. When the interface detects its own address, it activates the path between the bus lines and the device that it controls. All peripherals whose address does not correspond to the address in the bus are disabled by their interface.

I/O command

At the same time that the address is made available in the address lines, the processor provides a function code in the control lines. The interface selected responds to the function code and proceeds to execute it. The function code is referred to as an I/O command and is in essence an instruction that is executed in the interface and its attached peripheral unit. The interpretation of the command depends on the peripheral that the processor is addressing. There are four types of commands that an interface may receive. They are classified as control, status, data output, and data input.

control command

A *control command* is issued to activate the peripheral and to inform it what to do. For example, a magnetic tape unit may be instructed to backspace the tape by one record, to rewind the tape, or to start the tape moving in the forward direction. The particular control command issued depends on the peripheral, and each peripheral receives its own distinguished sequence of control commands, depending on its mode of operation.

status

A *status command* is used to test various status conditions in the interface and the peripheral. For example, the computer may wish to check the status of the peripheral before a transfer is initiated. During the transfer, one or more errors may occur which are detected by the interface. These errors are designated by setting bits in a status register that the processor can read at certain intervals.

output data

A *data output command* causes the interface to respond by transferring data from the bus into one of its registers. Consider an example with a tape unit. The computer starts the tape moving by issuing a control command. The processor then monitors the status of the tape by means of a status command. When the tape is in the correct position, the processor issues a data output command. The interface responds to the address and command and transfers the information from the data lines in the bus to its buffer register. The interface then communicates with the tape controller and sends the data to be stored on tape.

input data

The *data input command* is the opposite of the data output. In this case the interface receives an item of data from the peripheral and places it in its buffer register. The processor checks if data are available by means of a status command and then issues a data input command. The interface places the data on the data lines, where they are accepted by the processor.

I/O versus Memory Bus

In addition to communicating with I/O, the processor must communicate with the memory unit. Like the I/O bus, the memory bus contains data,

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address, and read/write control lines. There are three ways that computer buses can be used to communicate with memory and I/O:

1. Use two separate buses, one for memory and the other for I/O.
2. Use one common bus for both memory and I/O but have separate control lines for each.
3. Use one common bus for memory and I/O with common control lines.

IOP

In the first method, the computer has independent sets of data, address, and control buses, one for accessing memory and the other for I/O. This is done in computers that provide a separate I/O processor (IOP) in addition to the central processing unit (CPU). The memory communicates with both the CPU and the IOP through a memory bus. The IOP communicates also with the input and output devices through a separate I/O bus with its own address, data and control lines. The purpose of the IOP is to provide an independent pathway for the transfer of information between external devices and internal memory. The I/O processor is sometimes called a data channel. In Sec. 11-7 we discuss the function of the IOP in more detail.

Isolated versus Memory-Mapped I/O

Many computers use one common bus to transfer information between memory or I/O and the CPU. The distinction between a memory transfer and I/O transfer is made through separate read and write lines. The CPU specifies whether the address on the address lines is for a memory word or for an interface register by enabling one of two possible read.or write lines. The *I/O read* and *I/O write* control lines are enabled during an I/O transfer. The *memory read* and *memory write* control lines are enabled during a memory transfer. This configuration isolates all I/O interface addresses from the addresses assigned to memory and is referred to as the *isolated I/O method* for assigning addresses in a common bus.

isolated I/O

2 lines isolated control Commands for read & write

In the isolated I/O configuration, the CPU has distinct input and output instructions, and each of these instructions is associated with the address of an interface register. When the CPU fetches and decodes the operation code of an input or output instruction, it places the address associated with the instruction into the common address lines. At the same time, it enables the I/O read (for input) or I/O write (for output) control line. This informs the external components that are attached to the common bus that the address in the address lines is for an interface register and not for a memory word. On the other hand, when the CPU is fetching an instruction or an operand from memory, it places the memory address on the address lines and enables the memory read or memory write control line. This informs the external components that the address is for a memory word and not for an I/O interface.

The isolated I/O method isolates memory and I/O addresses so that memory address values are not affected by interface address assignment since each has its own address space. The other alternative is to use the same

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address space for both memory and I/O. This is the case in computers that employ only one set of read and write signals and do not distinguish between memory and I/O addresses. This configuration is referred to as *memory-mapped I/O*. The computer treats an interface register as being part of the memory system. The assigned addresses for interface registers cannot be used for memory words, which reduces the memory address range available.

memory-mapped

1 line for both read & write

In a memory-mapped I/O organization there are no specific input or output instructions. The CPU can manipulate I/O data residing in interface registers with the same instructions that are used to manipulate memory words. Each interface is organized as a set of registers that respond to read and write requests in the normal address space. Typically, a segment of the total address space is reserved for interface registers, but in general, they can be located at any address as long as there is not also a memory word that responds to the same address.

Computers with memory-mapped I/O can use memory-type instructions to access I/O data. It allows the computer to use the same instructions for either input–output transfers or for memory transfers. The advantage is that the load and store instructions used for reading and writing from memory can be used to input and output data from I/O registers. In a typical computer, there are more memory-reference instructions than I/O instructions. With memory-mapped I/O all instructions that refer to memory are also available for I/O.

I/O port

Example of I/O Interface

input output

An example of an I/O interface unit is shown in block diagram form in Fig. 11-2. It consists of two data registers called *ports*, a control register, a status register, bus buffers, and timing and control circuits. The interface communicates with the CPU through the data bus. The chip select and register select inputs determine the address assigned to the interface. The I/O read and write are two control lines that specify an input or output, respectively. The four registers communicate directly with the I/O device attached to the interface.

The I/O data to and from the device can be transferred into either port *A* or port *B*. The interface may operate with an output device or with an input device, or with a device that requires both input and output. If the interface is connected to a printer, it will only output data, and if it services a character reader, it will only input data. A magnetic disk unit transfers data in both directions but not at the same time, so the interface can use bidirectional lines. A command is passed to the I/O device by sending a word to the appropriate interface register. In a system like this, the function code in the I/O bus is not needed because control is sent to the control register, status information is received from the status register, and data are transferred to and from ports *A* and *B* registers. Thus the transfer of data, control, and status information is always via the common data bus. The distinction between data, control, or status information is determined from the particular interface register with which the CPU communicates.

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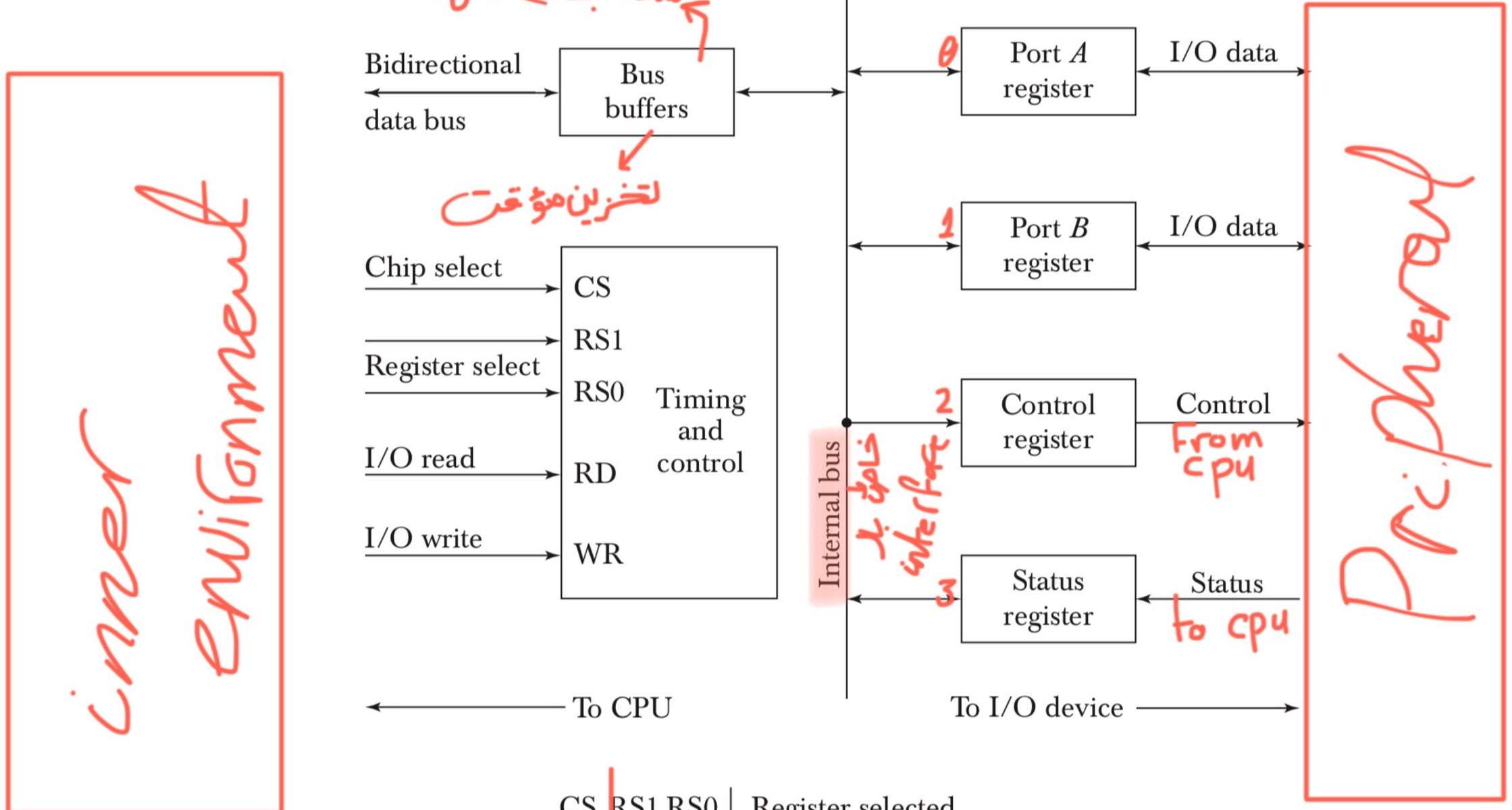

CS	RS1	RS0	Register selected
0	×	×	None: data bus in high–impedance
1	0	0	Port *A* register
1	0	1	Port *B* register
1	1	0	Control register
1	1	1	Status register

Figure 11-2 Example of I/O interface unit.

The control register receives control information from the CPU. By loading appropriate bits into the control register, the interface and the I/O device attached to it can be placed in a variety of operating modes. For example, port *A* may be defined as an input port and port *B* as an output port. A magnetic tape unit may be instructed to rewind the tape or to start the tape moving in the forward direction. The bits in the status register are used for status conditions and for recording errors that may occur during the data transfer. For example, a status bit may indicate that port *A* has received a new data item from the I/O device. Another bit in the status register may indicate that a parity error has occurred during the transfer.

The interface registers communicate with the CPU through the bidirectional data bus. The address bus selects the interface unit through the chip

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select and the two register select inputs. A circuit must be provided externally (usually, a decoder) to detect the address assigned to the interface registers. This circuit enables the chip select (*CS*) input when the interface is selected by the address bus. The two register select inputs *RS*1 and *RS*0 are usually connected to the two least significant lines of the address bus. These two inputs select one of the four registers in the interface as specified in the table accompanying the diagram. The content of the selected register is transfer into the CPU via the data bus when the I/O read signal is enabled. The CPU transfers binary information into the selected register via the data bus when the I/O write input is enabled.

11-3 Asynchronous Data Transfer

The internal operations in a digital system are synchronized by means of clock pulses supplied by a common pulse generator. Clock pulses are applied to all registers within a unit and all data transfers among internal registers occur simultaneously during the occurrence of a clock pulse. Two units, such as a CPU and an I/O interface, are designed independently of each other. If the registers in the interface share a common clock with the CPU registers, the transfer between the two units is said to be synchronous. In most cases, the internal timing in each unit is independent from the other in that each uses its own private clock for internal registers. In that case, the two units are said to be asynchronous to each other. This approach is widely used in most computer systems.

Asynchronous data transfer between two independent units requires that control signals be transmitted between the communicating units to indicate the time at which data is being transmitted. One way of achieving this is by means of a *strobe* pulse supplied by one of the units to indicate to the other unit when the transfer has to occur. Another method commonly used is to accompany each data item being transferred with a control signal that indicates the presence of data in the bus. The unit receiving the data item responds with another control signal to acknowledge receipt of the data. This type of agreement between two independent units is referred to as *handshaking*.

The strobe pulse method and the handshaking method of asynchronous data transfer are not restricted to I/O transfers. In fact, they are used extensively on numerous occasions requiring the transfer of data between two independent units. In the general case we consider the transmitting unit as the source and the receiving unit as the destination. For example, the CPU is the source unit during an output or a write transfer and it is the destination unit during an input or a read transfer. It is customary to specify the asynchronous transfer between two independent units by means of a timing diagram that shows the timing relationship that must exist between the control signals and the data in the buses. The sequence of control during an asynchronous transfer depends on whether the transfer is initiated by the source or by the destination unit.

timing diagram

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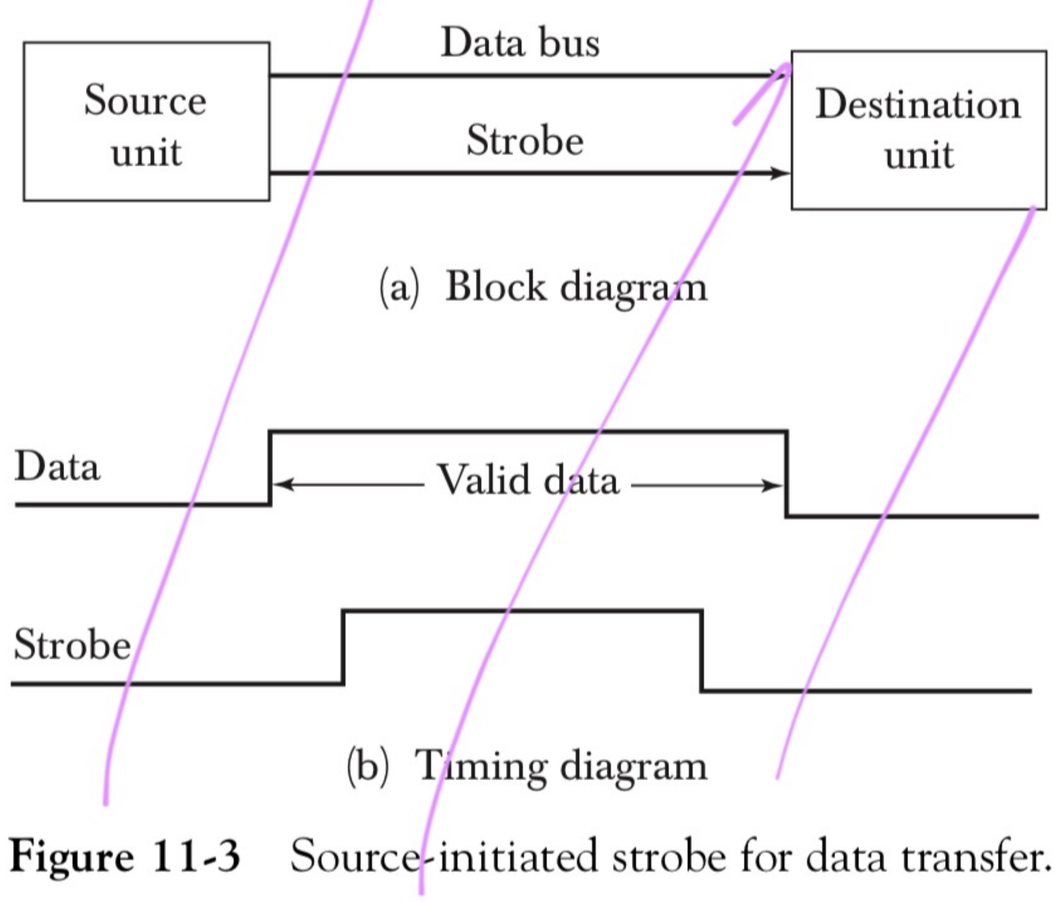

Figure 11-3 Source-initiated strobe for data transfer.

Strobe Control

The strobe control method of asynchronous data transfer employs a single control line to time each transfer. The strobe may be activated by either the source or the destination unit. Figure 11-3(a) shows a source-initiated transfer. The data bus carries the binary information rom source unit to the destination unit. Typically, the bus has multiple lines to transfer an entire byte or word. The strobe is a single line that informs the destination unit when a valid data word is available in the bus.

As shown in the timing diagram of Fig. 11-3(b), the source unit first places the data on the data bus. After a brief delay to ensure that the data settle to a steady value, the source activates the strobe pulse. The information on the data bus and the strobe signal remain in the active state for a sufficient time period to allow the destination unit to receive the data. Often, the destination unit uses the falling edge of the strobe pulse to transfer the contents of the data bus into one of its internal registers. The source removes the data from the bus a brief period after it disables its strobe pulse. Actually, the source does not have to change the information in the data bus. The fact that the strobe signal is disabled indicates that the data bus does not contain valid data. New valid data will be available only after the strobe is enabled again.

Figure 11-4 shows a data transfer initiated by the destination unit. In this case the destination unit activates the strobe pulse, informing the source to provide the data. The source unit responds by placing the requested binary information on the data bus. The data must be valid and remain in the bus long enough for the destination unit to accept it. The falling edge of the strobe pulse can be used again to trigger a destination register. The destination unit then disables the strobe. The source removes the data from the bus after a predetermined time interval.

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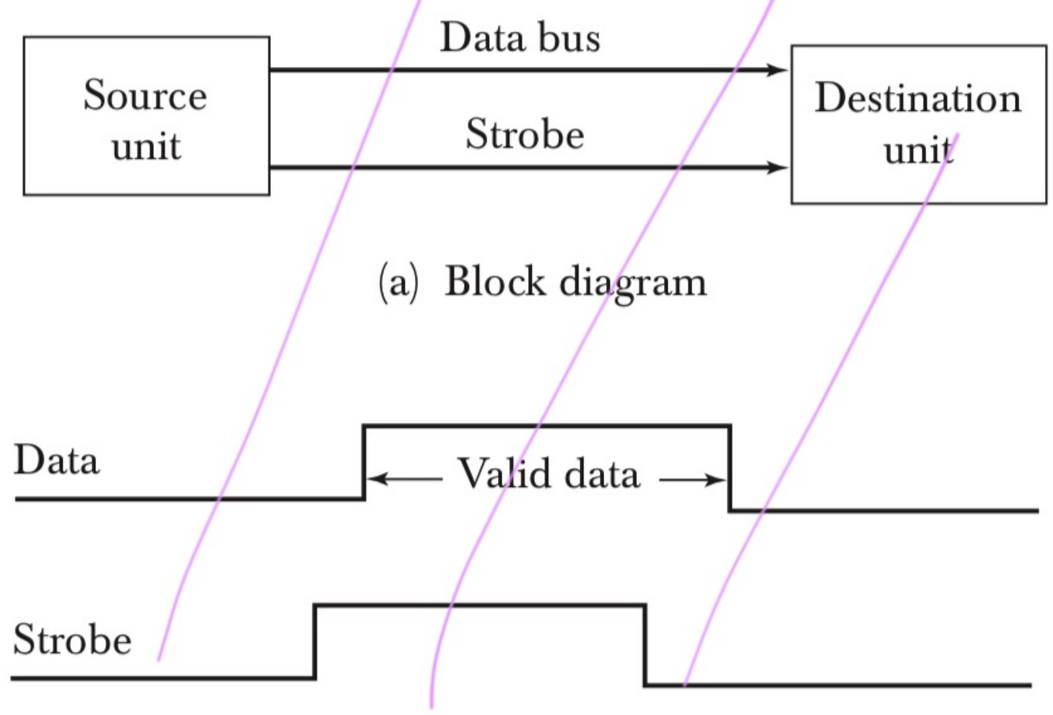

Figure 11-4 Destination-initiated strobe for data transfer.

In many computers the strobe pulse is actually controlled by the clock pulses in the CPU. The CPU is always in control of the buses and informs the external units how to transfer data. For example, the strobe of Fig. 11-3 could be a memory-write control signal from the CPU to a memory unit. The source, being the CPU, places a word on the data bus and informs the memory unit, which is the destination, that this is a write operation. Similarly, the strobe of Fig. 11-4 could be a memory-read control signal from the CPU to a memory unit. The destination, the CPU, initiates the read operation to inform the memory, which is the source, to place a selected word into the data bus.

The transfer of data between the CPU and an interface unit is similar to the strobe transfer just described. Data transfer between an interface and an I/O device is commonly controlled by a set of handshaking lines.

Handshaking

The disadvantage of the strobe method is that the source unit that initiates the transfer has no way of knowing whether the destination unit has actually received the data item that was placed in the bus. Similarly, a destination unit that initiates the transfer has no way of knowing whether the source unit has actually placed the data on the bus. The handshake method solves this problem by introducing a second control signal that provides a reply to the unit that initiates the transfer. The basic principle of the two-wire handshaking method of data transfer is as follows. One control line is in the same direction as the data flow in the bus from the source to the destination. It is used by the source unit to inform the destination unit whether there are valid data in the bus. The other control line is in the other direction from the destination to the source. It is used by the destination unit to inform the source whether it can accept data. The sequence of control during the transfer depends on the unit that initiates the transfer.

two-wire control

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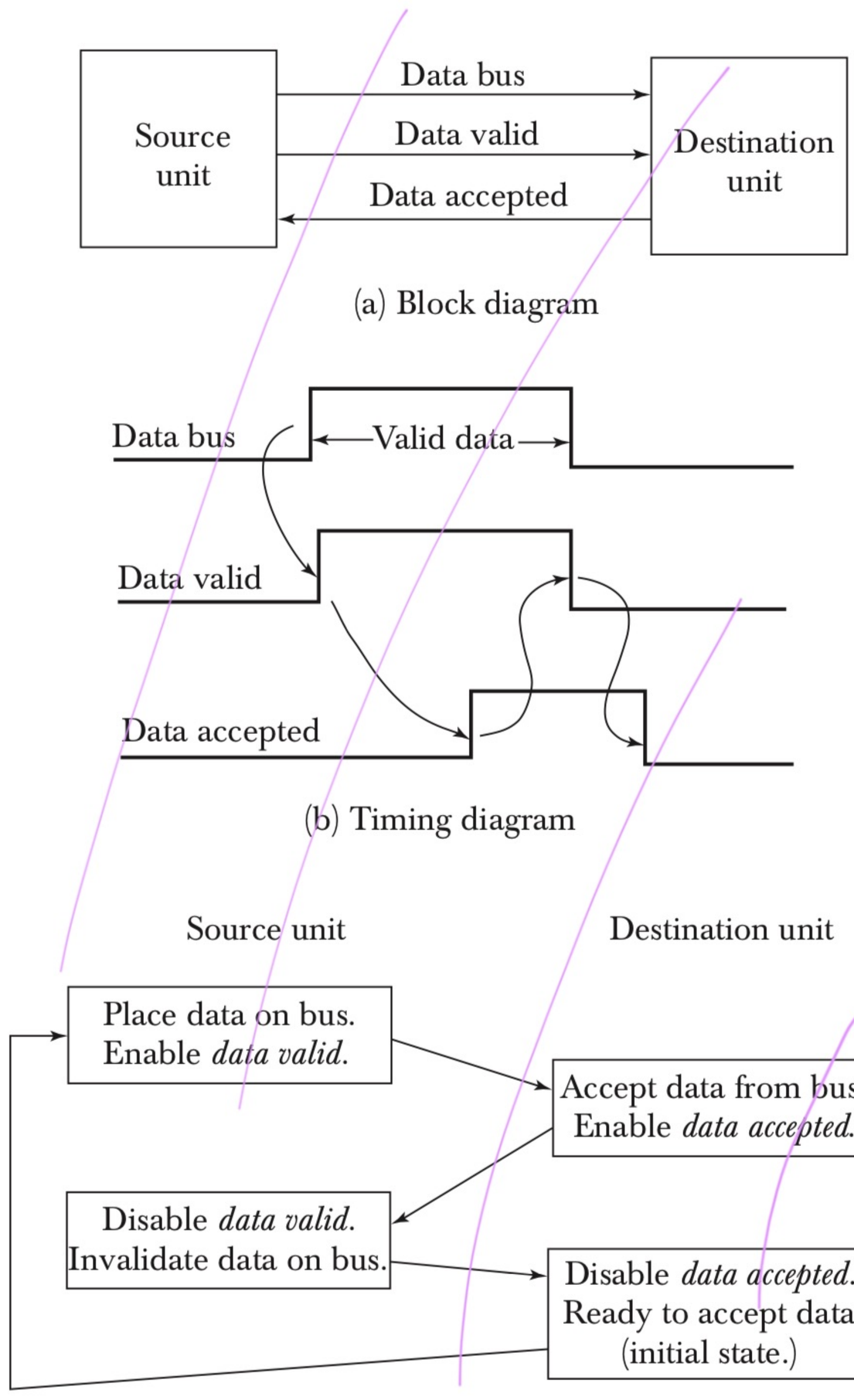

Figure 11-5 Source-initiated transfer using handshaking.

Figure 11-5 shows the data transfer procedure when initiated by the source. The two handshaking lines are *data valid*, which is generated by the source unit, and *data accepted*, generated by the destination unit. The timing diagram shows the exchange of signals between the two units. The sequence of events listed in part (c) shows the four possible states that the system can be at any given time. The source unit initiates the transfer by placing the data on the bus and enabling its *data valid* signal. The *data accepted* signal is activated by the destination unit after it accepts the data from the bus. The source unit then disables its *data valid* signal, which invalidates the data on the bus. The destination unit then disables its *data accepted* signal and the system goes into

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in this case does not place data on the bus until after it receives the *ready for data* signal from the destination unit. From there on, the handshaking procedure follows the same pattern as in the source-initiated case. Note that the sequence of events in both cases would be identical if we consider the *ready for data* signal as the complement of *data accepted*. In fact, the only difference between the source-initiated and the destination-initiated transfer is in their choice of initial state.

The handshaking scheme provides a high degree of flexibility and reliability because the successful completion of a data transfer relies on active participation by both units. If one unit is faulty, the data transfer will not be completed. Such an error can be detected by means of a *timeout* mechanism, which produces an alarm if the data transfer is not completed within a predetermined time. The timeout is implemented by means of an internal clock that starts counting time when the unit enables one of its handshaking control signals. If the return handshake signal does not respond within a given time period, the unit assumes that an error has occurred. The timeout signal can be used to interrupt the processor and hence execute a service routine that takes appropriate error recovery action.

timeout

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Asynchronous Serial Transfer

The transfer of data between two units may be done in parallel or serial. In parallel data transmission, each bit of the message has its own path and the total message is transmitted at the same time. This means that an n-bit message must be transmitted through n separate conductor paths. In serial data transmission, each bit in the message is sent in sequence one at a time. This method requires the use of one pair of conductors or one conductor and.a common ground. Parallel transmission is faster but requires many wires. It is used for short distances and where speed is important. Serial transmission is slower but is less expensive since it requires only one pair of conductors.

synchronous

Serial transmission can be synchronous or asynchronous. In synchronous transmission, the two units share a common clock frequency and bits are transmitted continuously at the rate dictated by the clock pulses. In long-distant serial transmission, each unit is driven by a separate clock of the same frequency. Synchronization signals are transmitted periodically between the two units to keep their clocks in step with each other. In asynchronous transmission, binary information is sent only when it is available and the line remains idle when there is no information to be transmitted. This is in contrast to synchronous transmission, where bits must be transmitted continuously to keep the clock frequency in both units synchronized with each other. Synchronous serial transmission is discussed further in Sec. 11-8.

asynchronous

A serial asynchronous data transmission technique used in many interactive terminals employs special bits that are inserted at both ends of the character code. With this technique, each character consists of three parts: a start bit, the character bits, and stop bits. The convention is that the transmitter